

Harakosan Technical Documentation: User's Guide

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Summary

This User's Guide provides operators instructions on how to operate the Harakosan Z72-2000-MV wind turbine.



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Permanent performance in wind

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1. Abstract

This User's Guide provides operators instructions on how to operate the Harakosan Z72 wind turbine.

After providing some general information in Chapter 3, safety instructions are presented in Chapter 4. It is essential to appreciate these instructions before attempting any work on the turbine. A general overview of the Z72 wind turbine is presented in Chapter 5, followed by a hardware description and a functional description in Chapter 6. Powering on the turbine for the first time is described in Chapter 7. The basic operation of the turbine is done with the Local Operator Panel. Chapter 8 is devoted to this panel, and explains how to use the touch screen and the pushbutton panel comprising the Local Operator Panel. People responsible for service will find in Chapter 9 how the control system of the turbine can be used for their activities.

Chapter 10 provides hints to solve problems, based on the events shown on the touch screen. Eventually, Chapter 11 describes procedures for configuration of the control system.

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3. General information

3.1. Project information and configuration data

This User's Guide is applicable to the following wind turbine park:

Type of application	Harakosan Z72 3 bladed variable speed 2MW wind turbine with Medium Voltage generator and liquid cooled converter
Blades	GFRP, 34.0 m, Umoe, Norway
Generator	60 pole Permanent Magnet Synchronous Machine, ABB Finland
Main bearing	IMO
Hub height	64.0 m
Power converter	ABB ACS1000, ABB Turgi, Switzerland, providing 3.5 kV to the grid connection equipment.
Grid connection	3.5 kV, 60 Hz for main power connection 400 VAC, 60 Hz for auxiliary power is required for the control system
Control system	Bachmann Electronic / Harakosan According to: • Z72-2000-2B-TD-18-01-R03 Z72 Control System block diagram • Z72-2000-2B-TD-18-04-R03 Nacelle control box 540A001 electrical diagram • Z72-2000-2B-TD-18-10-R03 Rotor control box 541A001 electrical diagram • Z72-2000-2B-TD-18-14-R03 Main turbine control system 542A001 electrical diagram
Main turbine PLC application software version	V3.0
Rotor PLC station application software version	V1.0
Local Operator Panel software version	V2.1
Overspeed sensor configuration	V1.0
Accelerometer configuration	V1.0
Converter software version	16101
SCADA	Garrad Hassan SCADA V2.1

3.2. Contacting Harakosan

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If you require more information on specific aspects of the turbine, its operation or its maintenance, please contact Harakosan at service@harakosan.nl .

For urgent questions regarding the installation, you can call Harakosan at +31-321-291 000

Internet: www.harakosan.nl

4. Safety instructions

4.1. General

NOTE:

This chapter includes the safety instructions that must be complied with when operating or servicing the Z72 wind turbine. The contents of this chapter must be read carefully before attempting any work on or with the unit.

Safety instructions related to work on or with the power converter are found in [ABB]

It is the owners' responsibility to ensure that each person involved in the operation or maintenance of the turbine, has received the appropriate training or instructions and has thoroughly read and clearly understood the safety instructions in this chapter.

4.1.1. Intended audience

This User's Guide is intended for the following two target groups:

- Operators of the turbine
- Maintenance personnel of the turbine

This manual is not intended for commissioning staff.

Operation:

The operator is mainly responsible for the operation of the wind turbine (starting and stopping the turbine) using the Local Operator Panel mounted on the ABB Wind Power Converter. An operator is able to diagnose faults, and can decide if it is safe to reset faults and to restart the turbine.

Maintenance:

The maintenance personnel is mainly responsible for:

- Periodical checks of the turbine
- Maintenance tasks

Maintenance people need to know how to stop the turbine for service, put the turbine into Service mode, use the service features of the control system, and how to restart the turbine after service.

4.1.2. Required qualification

Personnel must be fully trained and acquainted with the turbine and its control system. Operators of the turbine are not expected to have prior experience of wind turbines. However, the operator must know how to operate the turbine and must understand the meaning of the events on the Local Operator Panel located on the converter door to decide on the appropriate actions to take.

WARNING:

USE OF THIS EQUIPMENT BY UNTRAINED OR UNQUALIFIED PERSONNEL COULD RESULT IN DAMAGE TO THE WIND TURBINE, POSSIBLE LOSS OF LIFE OR PERSONAL INJURY. MAKE SURE PERSONNEL USING OR WORKING ON THIS TURBINE ARE CORRECTLY TRAINED.

4.2. General safety precautions

Always obey the following general safety precautions:

- Obey all cautions or warnings given in the procedures.
- Always take great care when operating the wind turbine or doing maintenance.
- Always use sufficiently trained personnel and the right equipment to operate and maintain the machine.
- Always work at or near the turbine with at least two people, carrying mobile phones or walkie-talkies (buddy principle).
- Only work on the wind turbine if you are in a healthy condition and in a fit state
- Do not carry out any work at wind speeds above 20 m/s (gale force 8).
- Do not work in the turbine or in the immediate vicinity during a thunderstorm or an impending thunderstorm.
- Do not enter the immediate vicinity of the turbine when ice is present on the blades, because there is a danger of falling ice, especially when the ice is melting. Switch off the wind turbine when ice is present on the blades.
- Always be aware that objects may fall down; a light bolt that falls from a great height can become a dangerous object. Always wear approved safety helmets.
- Always close door hatches in the tower and the nacelle after passing through them.
- Always turn off the power supply before doing work on the electrical installation.
- Always lock the rotor when working on the rotor or rotor blades.
- Do not open control system cabinets unless you have been authorized to do so.
- Never bypass or override machine safety devices.
- Never jumper or force input/output signals while operating the turbine.
- Do not change parameter settings unless completely sure of the consequences of that change.
- Do not leave tools or cleaning rags behind.

4.3. Climbing instructions

Before climbing, check the following points:

- Always wear an approved certified climbing harness and fall protection device when climbing the wind turbine. In any other case you must use a life line and a climbing belt.
- Wear appropriate shoes, preferably with rubber soles.
- Always make sure you have both hands free for climbing. If you have tools with you, you can carry these in a back pack. Large tools can be winched up.
- Always climb at a nice steady speed. It is recommended that you use the resting platforms to take an occasional rest. When using the resting platform, always stay hooked onto the climbing cable or fall protection.

Procedure for climbing:

- 1) Open the tower door and engage it in an opened position.
- 2) Prevent getting locked in by hanging the padlock back in position.
- 3) Check that the climbing harness shows no sign of wear.
- 4) Put on the climbing harness.
- 5) Check that the fall protection is working properly: when you pull the hook up, the fall protection should be able to move up and down along the climbing cable; when you pull the hook down, the fall protection should not be able to move along the climbing cable.
- 6) Hook the climbing harness onto the hook of the fall protection device. Screw the lock of the fall protection hand tight.
- 7) Start climbing, resting on the resting platforms if necessary. Do not detach the climbing harness.
- 8) When you arrive in the nacelle, always close the hatch before releasing the climbing harness from the fall protection device.

CAUTION

SOMETIMES YOU MAY WANT TO WORK SOMEWHERE WHERE THERE ARE NO STEPS AND CLIMBING CABLE, E.G. ABOVE THE ROTOR, ALONGSIDE THE WEATHERSTATION. IN THESE SITUATIONS, USE A LIFE-LINE, WHICH SHOULD BE ATTACHED TO THE CLIMBING HARNESS.

To descend the turbine, proceed as follows:

- 1) Hook the climbing harness onto the hook of the fall protection device. Screw the lock of the fall protection hand tight.
- 2) Start to descend, resting on the resting platforms if necessary.
- 3) When you are down, detach the climbing harness from the fall protection device
- 4) Leave the tower and lock the door.

4.4. Safety features

4.4.1. Pictograms

At several places in the wind turbine, warning pictograms can be found:



4.4.2. Lightning protection

Each blade is fitted with a lightning conductor. This conductor enters the rotor frame, where it is lead to through the brake disc and ends in a sharp point, forming a small spark gap to the nacelle, which is on ground potential. Consequently the current from a lightning stroke is diverted from the main bearing and can do no harm to the control system.

4.4.3. Emergency stop button

On the converter cabinet door, on the nacelle control box 540A001, and on the local hub panel 541C001, emergency stop buttons have been fitted.

4.4.4. Ladder with fall protection

This ladder ensures a safe climb to the nacelle.

4.4.5. Resting platforms

In the tower, there are resting platforms at various places.

4.4.6. Emergency lighting

In the nacelle, emergency lighting is installed. Harakosan standard specifications prescribe emergency lighting in the tower and the switchgear room as well.

4.4.7. Grounding

All wind turbine components that are not fitted together conductively are connected with earth cables to equalise potentials between components.

4.4.8. Fire extinguishers

Harakosan standard specifications prescribe fire extinguishers in the tower base and the switchgear room. Make sure that you know how these work.

4.4.9. First aid box

It is strongly recommended to install a first aid box in the nacelle.

4.4.10. Safety features of the turbine control system

The wind turbine control system is equipped with the following safety features:

- Battery powered emergency trip system shuts down turbine by bringing the blades to vane position in approximately 15 seconds, even when grid power is lost or the pitch drive inverter system is faulty.
- Various trip and alarm conditions shut down the turbine or inform the operator
- PLC watchdog checks if PLC is alive, and issues an emergency trip when the PLC is not responding for > 2.0s.
- Periodical emergency trip system check reports failures in the emergency trip system at an early stage.
- External independent overspeed protection system. Overspeed protection system integrity can be tested when wind speed is low, by temporary lowering the overspeed trip threshold.
- Emergency buttons on the ABB ACS1000 converter front door quickly switches off the auxiliary low voltage supply to turbine control system and the ABB ACS1000 converter resulting in stopping the turbine with the emergency trip system, switching off the converter and disconnecting the converter from the main transformer.
- Emergency buttons on several places inside the turbine (nacelle control box, local hub panel) quickly switch off the main consumers in nacelle and hub, resulting in stopping the turbine with the emergency trip system.
- Maintenance switch for switching off the power to the yaw motors. This switch is used when working on or near the yaw motors / yaw bearing.
- Maintenance switch for switching off the DC power of the battery packs inside the rotor control box. This switch is used when working in the hub, to avoid unwanted emergency pitch actions or high DC voltages on components of the control system.

5. Z72 Wind turbine overview

5.1. Wind turbine definitions and abbreviations

Hub	Central part of the turbine rotor, connecting the rotor blades and pitch bearings to the main bearing and generator rotor and containing the pitch control system.
Nacelle	Wind turbine housing on the top of the tower, connecting wind rotor and the generator to the tower. The nacelle contains the yaw orientation system and auxiliaries such as hydraulic power unit, grease pumps, fans, etc.) and weather station.
Pitching	The process of adjusting the blade angles to control the speed of the rotor.
Pn-curve	Relation between generated electrical energy and rotor speed.
Power converter	Power electronics device used to change electrical power from the variable frequency permanent magnet synchronous generator to the fixed frequency utility grid. Also called AC-DC-AC converter.
SCADA	Supervisory Control and Data Acquisition, remote monitoring and control system for a wind farm, including information gathering and reporting, monitoring for operation.
Synchronous generator	Generator type where the frequency of the generated power is proportional to the rotor speed.
Switch gear room	Room housing the high voltage equipment (power converter, transformer and switch gear) required for connection to the utility grid.
Yawing	The process of keeping the rotor properly aligned with the wind by changing the nacelle direction relative to the tower/foundation by means of the yaw drive.

5.2. Introduction



Figure 1 The Z72 2MW variable speed wind turbine

The Harakosan Z72/2000 wind turbine is a 3-bladed, direct drive, variable speed, pitch controlled wind turbine with a rotor diameter of 70.6 m and a nominal power rating of 2000 kW. The turbine is designed for use in areas for which IEC wind class IIB applies.

The Z72 turbine was designed with demanding offshore and remote application in mind, and has as few moving parts as possible.

Key design considerations were:

- High wind loads (IEC Wind Class I)
- Low maintenance requirements (designed for maintenance interval of 1 year)
- Easy access to key components, also in extreme weather conditions (generator, single main bearing, pitch control mechanism)
- Excellent power quality characteristics
- Erection with a relatively light, inexpensive crane

Major advantages of the permanent magnet design are the lower generator losses and the superior partial load efficiency.

Elimination of the gearbox in the drive train takes away an important potential source of system failure.

The generator is fully integrated into the structural design of the turbine. This has resulted in a very compact nacelle layout. The drive train comprises only one automatically lubricated main bearing. The bearing and generator support structure, together with the absence of a main shaft facilitates easy access to hub internals where the blade pitch mechanism is housed. The nacelle itself is practically airtight and in addition put under permanent overpressure. This measure prevents saline particles from entering the generator internals and blade pitch system

5.3. Principle of operation

A variable speed wind turbine adapts its rotor speed to speed of the wind. By matching the rotor speed to the speed of the wind, the conversion of wind energy to mechanical energy is optimal. This variable rotor speed also causes the frequency of the power generated by the generator to be variable and this power cannot be directly coupled to the grid. For this reason, a power converter, or AC-DC-AC converter, first rectifies the variable frequency power to an intermediate DC power, and then inverts this DC power into AC power with the frequency of the grid. Advantage of this mechanism is that the inverter from DC power to AC power can control the reactive power between certain limits (optional), supporting the grid quality.

Matching the speed of the rotor to the speed of the wind is done by controlling the power coupled to the grid as a function of the actual rotor speed according to a so-called Pn curve. This Pn curve has a maximum at 2 MW: the maximum power of the wind turbine. Should the wind speed increase when the turbine is running at 2 MW, then the excess energy of the wind is diverted from the blades by pitching the blades to vane position.

Each individual blade is controlled by means of an electrically actuated system consisting of a DC servo motor. Emergency operation, essential when the coupling to the grid is lost or the normal pitch control is faulty, is achieved by fast positioning the blades to vane position by batteries. In case the overspeed protection system inside the PLC is not functioning, a backup system consisting of an external electrical overspeed switch will guarantee safe shutdown of the turbine.

It should be noted that the capacity of the emergency batteries is limited. For this reason, too many subsequent emergency pitch actions should be avoided, since this may influence the negatively influence the integrity of the safety system during operation.

5.4. Z72 Main components

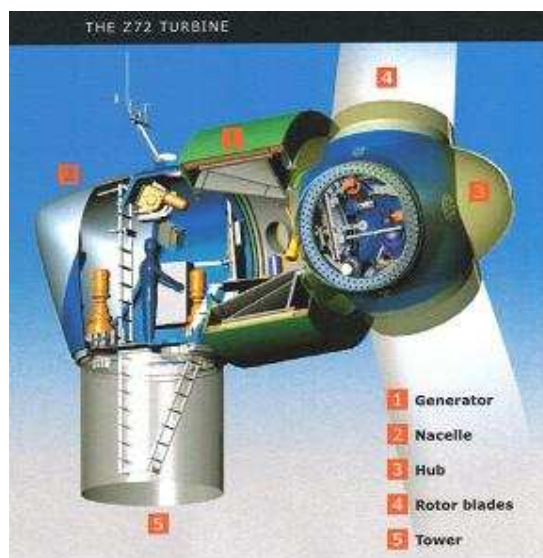


Figure 2 The Z72 Main components

Main components of the Z72 wind turbine are:

- Hub (see section 5.4.1)
- Rotor blades (see section 5.4.2)
- Generator (see section 5.4.3)
- Nacelle (see section 5.4.4)
- Tower and foundation (see section 5.4.5)
- Switchgear room (see section 5.4.6)

Inside these components that are easily recognised, the following systems are accommodated:

- Power converter (see section 5.4.7)
- Grid connection (see section 5.4.8)
- Low voltage switchboard / safety system (see section 5.4.9)
- Lightning and grounding system (see section 5.4.10)
- Turbine control system (see section 5.4.11)
- Interface to the SCADA system (see section 5.5.1 and [Garrad Hassan])

The Switch gear room is outside the scope of Harakosan.

5.4.1. Hub

The rigid hub is composed of nodular cast iron and fitted directly to the main bearing. The blades are connected to the hub using a so-called four point pre-stressed ball bearing. The hub houses the blade pitch control system and a slip-ring unit for the power supply and control system communication.

The hub is sufficiently large to provide a working environment for two service technicians.

The mass of the fully assembled hub measures 19 tons.

Its dimensions are 2.5 x 2.2 x 2.2 m (l x w x h).

Each pitch bearing is driven by a pitch drive system consisting of a DC motor and a gearbox.

One grease pump has been fitted to lubricate the pitch bearings and one grease pump has been fitted to lubricate the pitch gears.

The pitch drive units and the grease pumps are operated by the rotor control box 541A001. This control box also contains the batteries that are required for emergency pitching operations.

5.4.2. Rotor blades

The three rotor blades are constructed from glass fibre reinforced polyester composite (GFRP). The mass of each blade is approximately 5.5 tons. The length of the blades is 34.0 metres with an outer blade root diameter of 1.9 metres and a maximum cord of 3.1 metres. All blades are equipped with a lightning protection system.

5.4.3. Generator



Figure 3 The Z72 Generator

The synchronous generator is a medium voltage 4kV, 60-pole permanent magnet generator manufactured by ABB Finland. The permanent magnet design reduces failure rate and service intervals of the turbine. The use of medium voltage makes efficient transport of power with low currents to the converter possible.

The stator windings are vacuum impregnated with a 7 kV insulation value commonly used in medium voltage generators. Circulation of air assures an equal temperature distribution inside the generator. The single generator bearing is a ring bearing that also functions to carry the wind rotor. The inner stationary ring is fitted to the generator stator. Access to the hub and blades is possible through the hollow stator tube. The outer rotating ring fits between the hub and the generator rotor. The bearing takes axial and radial loads, as well as bending moments and is equipped with an automatic lubrication system, continuously monitored by the turbine control system.

The generator and bearing assembly is directly bolted onto the nacelle. The generator mass is 49 tons, diameter and length are 3.8 x 2.2 m.

Pt100 temperature sensors in each stator phase (three sensors + three spare), two Pt100 sensor for the cooling air temperature (temperature in the air gap) and two Pt100 temperature sensors in the main bearing are wired to a separate junction box in the generator, connected to the nacelle control box 540A001 and monitored by the turbine control system.

5.4.4. Nacelle

The nacelle is a compact structure composed of nodular cast iron. The geometry of the casting assures an almost ideal transfer of loads to the tower. The mass of the nacelle assembly is 12 tons and it measures 3.5 x 2.8 x 2.8 m (l x w x h).

The nacelle contains the yaw bearing. The yaw bearing is driven by a yaw drive consisting of 4 yaw motors. A hydraulic power unit provides the power required for yaw brakes, holding the nacelle in position.

An automatic grease pump lubricates the yaw bearing.

The main bearing and hub are cooled by a cooling system located in the nacelle.

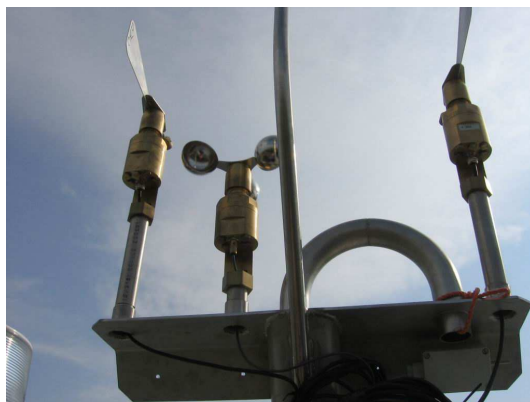


Figure 4 Weather station

On top of the nacelle, a weather station (two wind vanes, one anemometer, one temperature sensor) is mounted.

Another component to be found in the nacelle is the local hub panel. This is an operator panel used for locking the rotor for service. Rotor locking is assisted by a service brake that receives its power also from the hydraulic power unit.

The nacelle control cabinet 540A001 operates most systems in the nacelle: yaw system, grease pumps, cooling fan etc.

Other equipment in the nacelle are a service winch with remote operation tool (UP, DOWN, STOP), to be used for hoisting tools etc. and a transformer, required to supply power to the hub 3x150VAC/60Hz.

5.4.5. Tower and foundation

The nacelle is put on a fully corrosion protected tubular steel tower. The complete tower consists of three sections, bolted together during erection, to give a total length of 65 m. The total tower mass is 100 tons.

A ladder with fall protection and several resting platforms ensure a safe climb up to the top of the tower.

Inside the tower, a cable ladder supports the following cables:

- Generator cables 3 phases Radox for transport of power from generator to converter
- Fibre optic cable HCS (Hard Clad Silica) 200/230 µm interconnection of main turbine control system 542A001 PLC control system and PLC station in the nacelle control box 540A001
- 400 VAC / 60 Hz low voltage supply to nacelle control box 540A001 and rotor control box 541A001
- Grounding cable for grounding / lightning protection
- Local 110 VAC / 60 Hz lighting

Because the top part of the wind turbine can rotate, the cables in the top part of the tower are not mounted on the cable ladder, but are suspended in a lug, that allows 2.5 turns clockwise and counter clockwise rotations of the wind turbine top, without damaging the cables in the tower.

In the bottom of the tower, a generator disconnection switch cabinet is placed. Also, a junction box for low voltage supply to the control systems in nacelle and hub is present on the bottom of the tower.

It is advised to install a switch near the entrance of the tower, for switching on/off the lighting in the tower.

5.4.6. Switch gear room

Standing next to the turbine, the concrete switch gear room is used to accommodate the converter, the grid connection components (transformer and switch gear), the low voltage distribution / switchboard and the SCADA interface.

The switchgear room is outside the scope of Harakosan.

5.4.7. Power converter

The power converter converts the power with variable frequency from the generator to 60Hz power suitable for connection to the grid. The converter is a standard ABB ACS 1000 type medium voltage AC-DC-AC unit, adapted for use in the Z72 turbine. The entire output power is fed through the converter, which is a standard feature of synchronous type generators. The reactive power is fully controllable (option) but set to zero ($\cos\phi = 1$) by default.

The power converter is manufactured by ABB Switzerland.



Figure 5 The ABB ACS1000 Power Converter with Local Operator Panel (left)

The central processing part of the Main turbine control system 542A001 and the Local Operator Panel of the wind turbine are accommodated in the control cabinet (leftmost cabinet) of the ABB converter, as can be seen in fig 5.

Cooling of the converter is accomplished by an external heat transfer agent, connected with a heat exchanger to the internal converter cooling circuit. Both the heat transfer agent and the heat exchanger are outside scope of Harakosan.

The converter is located in the switch gear room. For operation of the ABB ACS1000 converter, see [ABB].

5.4.8. Grid connection

The grid connection comprises the following components:

- Main high voltage transformer
- Transformer cooling
- Main Circuit Breaker between power converter and main transformer
- Main Circuit Breaker on the primary side of the main transformer

The components required for the grid connection of the Z72 wind turbine (Main transformer, MV switchgear and transformer cooling system) are installed in a switchgear room near the turbine tower.

The components for the grid connection are outside the scope of Harakosan.

5.4.9. Low voltage switchboard / safety system

The low voltage switchboard consists of:

- 400 VAC / 60 Hz transformer
- Fused groups 400 VAC / 60 Hz, at least required 50A group for nacelle and rotor control system and 20A group for converter control system
- Emergency circuit consisting of contactor to switch off above mentioned groups of 50A for nacelle and rotor control system and 20A group for converter control system when the emergency mushroom button on the converter cabinet has been pressed
- 400VAC/110VAC transformer
- Fused groups 110VAC for local use (lighting, wall sockets etc.)

Note that the low voltage switchboard plays an essential role as safety system: pressing the emergency mushroom buttons results in disconnection of power to control systems and converter, shutting down the turbine and the converter immediately.

The low voltage power system is TN-S.

For this project, the low voltage switchboard of the Z72 wind turbine is installed in the switchgear room near the turbine tower.

5.4.10. Lightning and grounding system

The complete grounding system of the wind turbine comprises:

- Safety grounding: grounding system to prevent dangerous voltages for humans between different parts of the wind turbine which can be touched simultaneously
- Lightning protection: provision of a predictable way for overvoltage / overcurrent caused by lightning

In the Z72 wind turbine, these grounding systems are combined.

The main lightning route begins with the lightning conductors at the blades, via spark gap and generator stator frame, nacelle and tower to the grounding rod and foundation reinforcement of the wind turbine.

In the nacelle, an earth rail is mounted, connected with nacelle, weather station and with a separate earth cable connected to the grounding rod in the foundation.

5.4.11. Turbine control system

The turbine control system is used to operate the turbine: starting/stopping of the turbine and coordination of yaw and pitch actions (incl. emergency pitching), the power converter on/off control, grease pumps, fans, protection of the turbine in case of failures or harmful operating conditions.

The control system of the wind turbine is distributed over the locations hub, nacelle and converter.

The turbine can be controlled by means of a Local Operator Panel that is mounted on the leftmost cabinet door of the ABB ACS1000 power converter.

Because the turbine control system is essential from an operator's point of view, a separate chapter is devoted to this subject (Chapter 6 Z72 Wind turbine control system description).

5.5. Z72 Options

5.5.1. SCADA system (option)

Harakosan can install a SCADA system to monitor and control a wind farm consisting of one or more Z72 turbines.



Figure 6 SCADA Remote Interface Unit

Each turbine contains a so called RIU (Remote Interface Unit) that is connected to the main PLC station by Ethernet. The RIU exchanges information between the Main PLC and the SCADA server that stores events, real-time samples, as well as 10 minute statistics of various signals. The server runs a SQL database, and is processing and storing the

information from the turbines. Access to the information on this server is Web-based using a standard Internet browser.

The SCADA system is supplied by **Garrad Hassan**.
For operation of the Garrad Hassan SCADA system, see [Garrad Hassan].

5.5.2. Obstruction light (option)

As option, an obstruction light that conforms to local conditions, can be mounted on top of the nacelle.



Figure 7 Obstruction light

In case two obstruction lights per turbine are installed, both obstruction lights are controlled by a separate controller (flash bulb synchronisation and day / night intensity control). This obstruction lights controller operation is described in a manual from the supplier [Orga].

6. Z72 Wind turbine control system description

As the operation of the Z72 wind turbine is done with the turbine control system, this chapter provides a detailed description of the Z72 wind turbine control system and the Local Operator Panel.

Local operation of the turbine using this control system and the Local Operator Panel is described in chapter 8.

Remote operation of the turbine is described in a separate manual [Garrad Hassan].

6.1. Control system definitions and abbreviations

Alarm	This is an event that requires operator attention, but does not shutdown the turbine. A persisting alarm can often be followed by a trip, if the condition causing the alarm is not corrected
CAN	Controller Area Network. Fieldbus originating from automotive industry
CANOpen	Implementation of CAN application layer
ECM	Emergency Controller Module. Custom made electronics board, controlling emergency trip for one blade
EMI	Electromagnetic interference
FastBus	Bachmann specific fast communication bus between main PLC and remote I/O, based on fibre optic HCS cable
HCS	Hard Clad Silica. 200/230 micrometer fibre optic cable
IP	Internet Protocol. Internet addressing scheme
JAVA	Simple, object-oriented, distributed, interpreted, robust, secure, architecture neutral, portable, high performance, multithreaded and dynamic computer language
LAN	Local Area Network
LV	Low Voltage (< 1 kV)
MCB	Main Circuit Breaker
PDO	CAN Process Data Object
PLC	Programmable Logic Controller
RIU	Remote Interface Unit, GH SCADA hardware interface of Z72
SBP	Smart Battery Pack Loader Module. Custom made battery loader, loading and supervising emergency pitch battery for one blade
SDO	CAN Service Data Object
SQL	Structured Query Language, computer language to search in databases
TCP	Transmission Control Protocol. Internet data transport service.
UPS	Uninterruptable Power Supply
VxWorks	Real-time operating system by WindRiver
Trip	This is an event that indicates a severe malfunction and causes the control system to shut down the turbine by means of the normal servo control system.

6.2. Main functions of the turbine control system

The turbine control system is designed to fulfil the following main tasks:

- Automatic and manual starting and stopping of the wind turbine
- Pitch and torque control algorithm to limit rotor speed and optimise energy yield
- Yaw control to turn the turbine into the wind
- Wind speed and wind direction monitoring
- Turbine protection system
- Emergency (backup) protection system and periodical emergency system check
- Various monitoring functions (wind speed, wind direction, generator stator temperatures, bearing temperatures, battery voltages etc.)
- CANOpen communication interface with the converter control system
- Auxiliary functions (grease pumps, service brake, locking pin, cooling fan, winch etc.)
- Operator interface: Local Operator Panel

The battery powered backup emergency protection system (monitoring overspeed and PLC watchdog) in the rotor control box is completely separate from the control system and guarantees safe operation of the turbine in case the normal protection system inside the PLC's does not respond properly (due to for example a power failure).

6.3. Pitch and torque control algorithm

The pitch and torque control algorithm is aimed to regulate and limit rotor speed and to optimise energy yield. The control structure comprises:

- 2 operation modes: partial and full load
- 2 actuation set-points: pitch rate and electric torque
- 3 measurements: pitch (reference) angle, rotor speed, electric power

In the Z72, two different algorithms can be selected for pitch and torque control.

A basic algorithm uses a static Pn curve and a simple pitch control algorithm. The basic algorithm is a robust, field-proven algorithm, that has the disadvantage of a suboptimal energy yield round rated speed.

The extended algorithm has the option to use a dynamic Pn curve (see explanation below) to increase the production round rated speed. In addition to the pitch control structure of the basic control algorithm, the extended algorithm uses a control structure that estimates the wind speed on the turbine, and used that estimate to realise an additional control action, which optimises the turbine power production and improves its behaviour at sudden wind gusts. In addition to the normal pitch angle set-point, an offset based on the wind speed estimate is aimed to maintain stationary rated power production.

Simulation results have shown that the extended algorithm using wind speed estimate based control results in:

- an increase in energy yield of app. 1%
- considerable reduction of rotor speed variations: app. 0.2 rpm standard deviation
- improved turbine behaviour to large wind gusts

Sections 11.7, 11.8 and 11.9 describe how to configure the control system to use either the basic or the extended control algorithm.

6.3.1. Pitch control algorithm

Partial load operation

Partial load operation will take control as soon as the measured pitch angle is close to the pitch angle set-point for partial load operation. This set-point is dependent of the measured rotor speed and is close to zero. The pitch angle during partial load operation aims for maximum aerodynamic efficiency (Maximum power coefficient C_p), but requirements like noise and thrust loading are also decisive. The set-point value is realised by servo control using a maximum pitch speed.

Transition for partial load to full load

The transition to the full load control algorithm interacts with the electric torque control in order to maximise energy yield. A transition to partial load will only take place if the measured pitch angle already has reached its working position and the rotor speed has decreased below rated speed. Reversely, the transition from partial to full load operation is based on a rotor speed threshold level, such that too many transitions are avoided.

Full load operation

During full load control the rotor speed is limited to a maximum value, while rated power is aimed to be maintained. The maximum value of the rotor speed is limited by the control range of the converter and by mechanical limits. The pitch actuators of each blade will then set the pitch angle simultaneously to a suitable position to regulate the rotor speed between its rated and its maximum value.

6.3.2. Torque control algorithm

Electric torque control – static and dynamic Pn curves

The electric torque control is used for power production in a specific speed range. In addition it is an actuation port to optimise the energy yield around rated conditions if small electric torque variations (3%) of its rated value are permitted.

Electric power is generated in accordance with the static (stationary) Pn curve, which can be separated into four ranges: start-up, constant tip-speed ration, transition and constant power.

In order to maximise energy yield round rated conditions, it is possible to activate a control that dynamically implements the Pn curve from partial load to full load. The transition part is in that case valid in case of partial load control. If the wind speed drops when the turbine is in full load control, constant power is extended by keeping the power

set-point constant until the pitch angle is in working position. When the wind speed increases again, the power set-point will be according to the static Pn curve. In addition, production of mean rated power (2MW) is guaranteed by additional small power set-point corrections in full load.

Sections 11.7, 11.8 and 11.9 describe how to configure the control system to use a static (stationary) Pn curve only, or to use a dynamic torque-speed relationship.

6.4. Safety philosophy

Should an event arise that may possibly cause damage to the turbine or people in the vicinity of the turbine, the turbine is stopped as quickly as possible. Two different mechanisms are available to stop the turbine:

Standard protection system: this system stops the turbine by bringing the blades to vane position by means of the pitch driver inverter system (consisting of rotor PLC for all three blades and one inverter per blade, driving one DC pitch motor per blade)

Emergency (backup) protection system: in case reliability of the standard protection system can not be guaranteed, this system brings the three blades to vane position by the emergency protection system consisting of one 168 Vdc, 4Ah battery per blade and a power relay, driving the DC pitch motor.

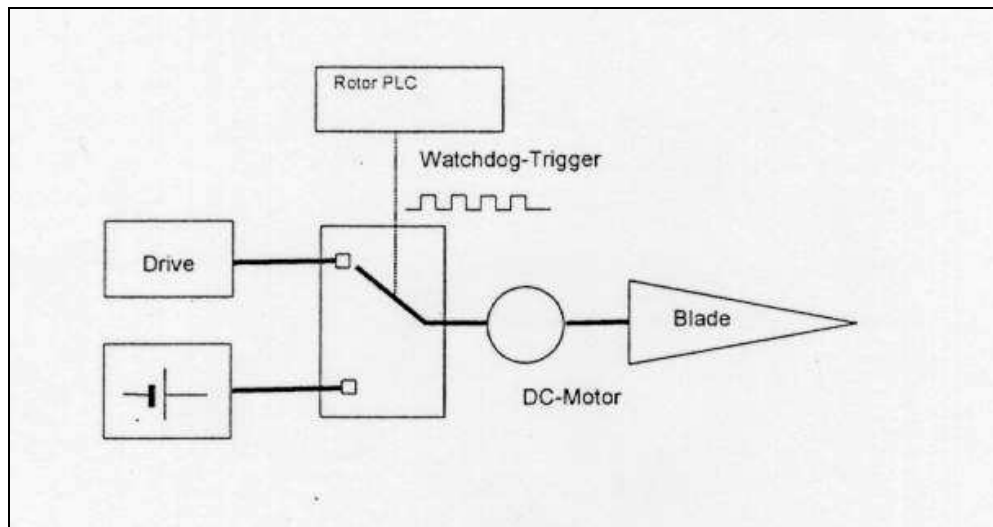


Figure 8 Schematic sketch of standard and emergency protection system

The functionality of the system depicted in fig. 6 is implemented by the ECM Emergency Controller Module. Each blade contains one ECM module.

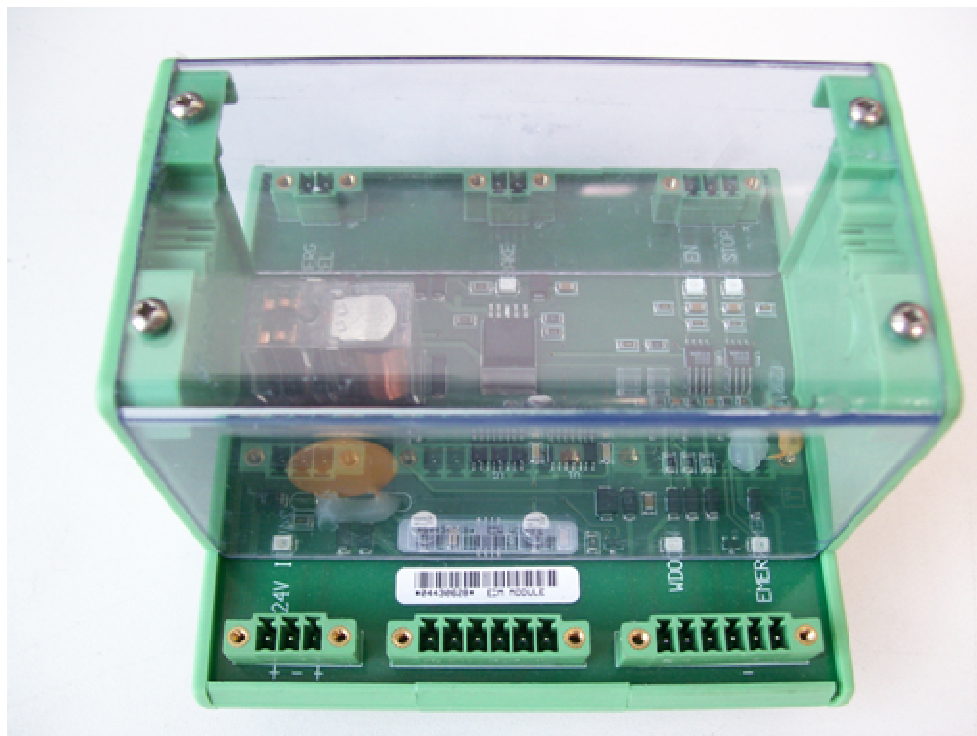


Figure 9 Emergency Control Module

6.5. External power supply and external emergency pushbutton system

The power to the turbine control system is supplied from only 2 distribution groups:

Consumer	Breaker rating	Supply type
Rotor box 542A001 and nacelle control box 540A001	50A	3x400 VAC/60Hz TN-S
Converter, including main turbine control system 542A001	20A	3x400 VAC/60Hz TN-S

Pressing the emergency pushbutton on the converter panel should at least interrupt the supply to these 2 groups, resulting in emergency pitching of the rotor (due to loss of the so-called PLC watchdog) and switching off and disconnection of the converter. Supply to these 2 groups should only be continued after manually pressing a reset button in the switch gear room.

Total power consumption of the control system is < 25 kW (main consumers: 4 yaw motors 4.4 kW, rotor 12 kW). In total 50 kVA power should be available for auxiliaries.

The rotor control cabinet is supplied with 3x150 VAC/60Hz. This (unusual) voltage is required by the pitch drive inverters. A 12kVA transformer in the nacelle converts 3x400 VAC/60Hz to 3x150 VAC/60Hz.

The main turbine control system is supplied with a 24 VDC UPS voltage from the converter cabinet. This allows storing grid outage events on Non-volatile memory of the Main turbine control system PLC.

6.6. Connection of the turbine control system to the SCADA system infrastructure

A park optical ring network provides the infrastructure to monitor and control each turbine of a wind park from a remote location.

Each turbine can therefore be integrated in a wind park fibre optical (ring-) network. Each turbine PLC and each Local Operator Panel touch screen is assigned a unique IP address, based on a uniquely assigned turbine ID (see section 11.1).

The connection with the fibre optic ring LAN network is done with a Moxa Ethernet switch (see section 6.7.2).

The fibre optic cable required for connection to the Moxa should conform to the following specification:

- 50 micrometer or 62.5 micrometer multimode / graded index fibre or 9/125 micrometer single mode fibre, the latter in case the distance between turbines or between turbine and SCADA server exceeds 2 km. For single mode a different type of Moxa switch should be used, suitable for use with single mode fibre.
- Suited for outdoor use, direct burial, with armour
- rodent proof / rodent resistant when applicable
- Number of fibres: minimal 2, 2 spare fibres recommended
- Multi mode: 50/125 μm (preferred), 800MHz.km, 5km @1310nm or 62.5/125 μm , 500MHz.Km, 4km @1310nm,
- Single mode: 9/125 μm
- terminated with SC connectors

The park network is extensively used by the Garrad Hassan SCADA system, but can also be used for example by ABB for troubleshooting the converters (which all have a unique IP address).

6.7. Turbine control system hardware description

6.7.1. Components of the turbine control system

In order to achieve the functions mentioned in 6.2, the turbine control system is divided into the following main hardware components:

- Main turbine control system 542A001. This system is mounted on a plate inside the converter cabinet. The main turbine control system is responsible for all processing and is responsible for starting and stopping the turbine, yaw control, pitch control, control of all auxiliaries, protection, monitoring and coordinating the rotor, nacelle and converter actions. The main turbine control system contains the Local Operator Panel, mounted on the converter.

- Nacelle control box 540A001. This box is mounted in the nacelle, and is mainly used for remote I/O in the nacelle used for yaw control, nacelle auxiliaries control, tower acceleration monitoring and generator protection
- Rotor control box 541A001. This box is mounted in the hub, and is mainly used for remote I/O in the hub used for pitch control, hub auxiliaries control, and emergency pitching

Note that the ABB converter is not part of the turbine control system. The ABB converter is an autonomous system, but is switched on/off by the turbine control system.

The main turbine control system 542A001 and the nacelle control box 540A001 are interconnected by means of a so called Bachmann FastBus proprietary fibre optic link, robust to EMI, based on HCS 200/230 micrometer fibre optic cable. This FastBus provides transmission times in the order of microseconds, which is similar to access times via a conventional backplane.

The connection between the rotor control box 541A001 and the nacelle control box 540A001 is by means of a shielded copper connection over a conventional slip ring. The protocol running over this link is CANOpen. The ABB converter is also connected with a copper link running CANOpen with the Main Turbine Control System 542A001.

The touch screen of the Local Operator Panel is not a vital part of the control and safety system. This means that the touch screen can be switched off without influencing the control and safety system of the turbine, although in normal circumstances the touch screen should be switched on.

6.7.2. Main turbine control system 542A001

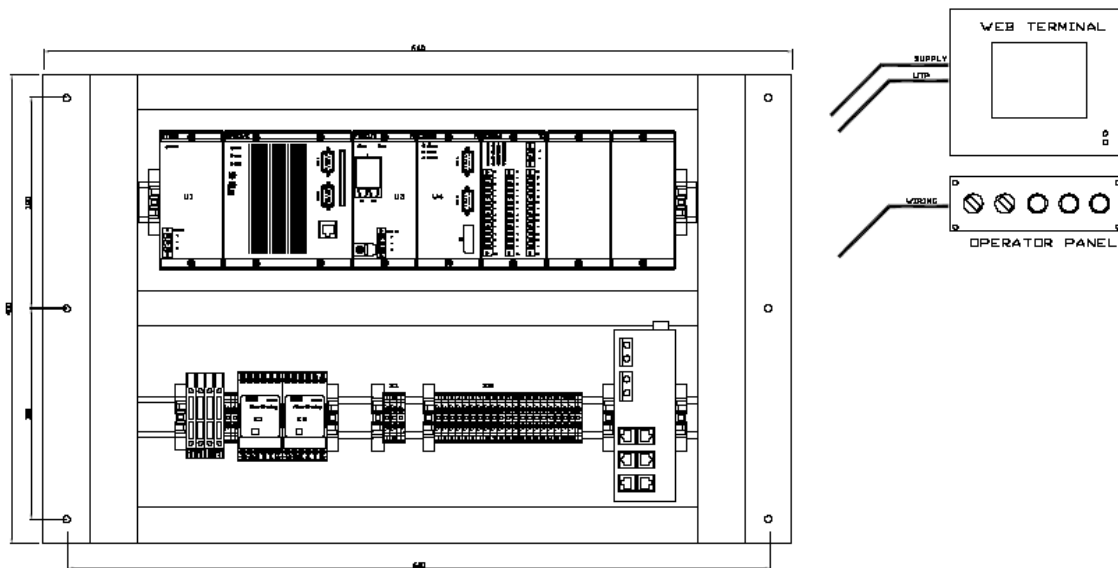


Figure 10 Main turbine control system 542A001 layout

The main turbine control system 542A001 is located on the front door of the leftmost cabinet section of the ABB ACS1000 converter. It is not part of the converter, but because of its small size, and its interface with the converter, is easily accommodated by the converter cabinet.

The tasks of the Main turbine control system 542A001 are:

- Central processing of the turbine control system functions: PLC
- Operator interface: Local Operator Panel, consisting of buttons, switches and a 5.7" LCD touch screen panel running the VxWorks operating system
- Communication: Bachmann FastBus fibre optic HCS interface to nacelle control box 540A001
- Communication: Interface to the ABB converter (hardwired trip contact and CANOpen connection)
- Communication: Ethernet switch, interconnecting PLC, touch screen panel and ABB converter with the Park fibre optic ring LAN network

Main components of the main turbine control system are:

Component	Description
Bachmann PLC	Central processing of all turbine control functionality
Fibre optic Ethernet switch	Moxa Ethernet switch
Bachmann Web-terminal WT305	Touch screen of Local Operator Panel
Various circuit breakers, relays, and terminal blocks	Various functions

The control wiring diagram of the main turbine control system and descriptions of these components can be found in [Z72-2000-2B-TD-18-14-R03].

Software required for the main turbine control system is:

Software	Version	Target hardware
Main turbine PLC application software	V3.0	Bachmann PLC
Local Operator Panel application software	V2.1	Bachmann Web-terminal WT305

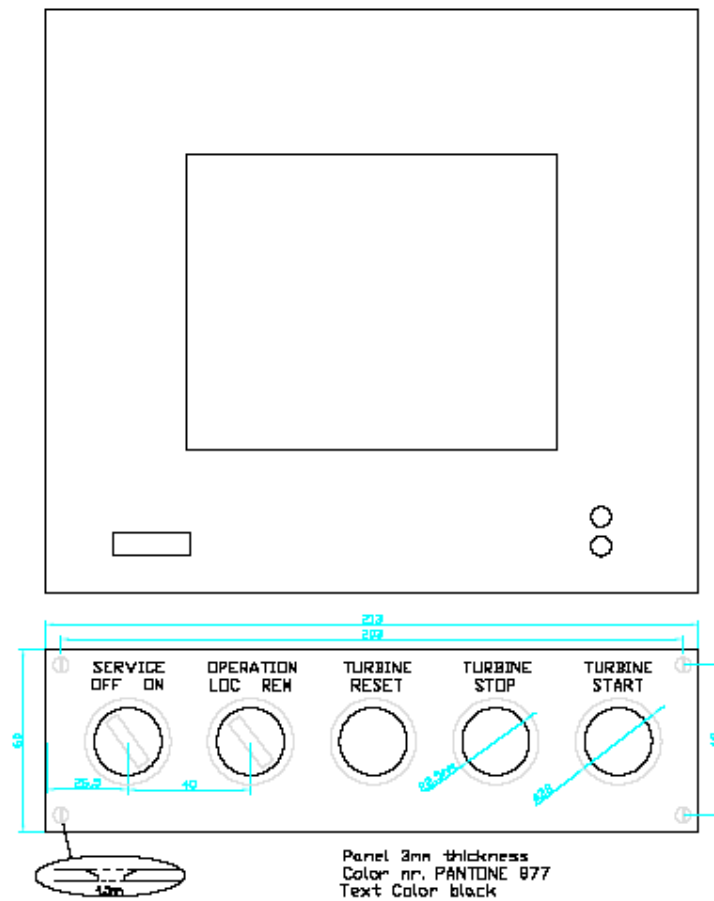


Figure 11 Local Operator Panel

The main turbine control system hardware contains the following hardwired button / switches panel:

Console text	Description
SERVICE	Service switch
OPERATION LOC / REM	Selection operation from Local Operator Panel (LOC) or SCADA (REM)
TURBINE RESET	Manual turbine reset button
TURBINE STOP	Manual turbine stop button
TURBINE START	Manual turbine start button

Use of this panel is described in chapter 8.

There are no connections to instruments or electric consumers connected to the main turbine control system.

6.7.3. Nacelle control box 540A001

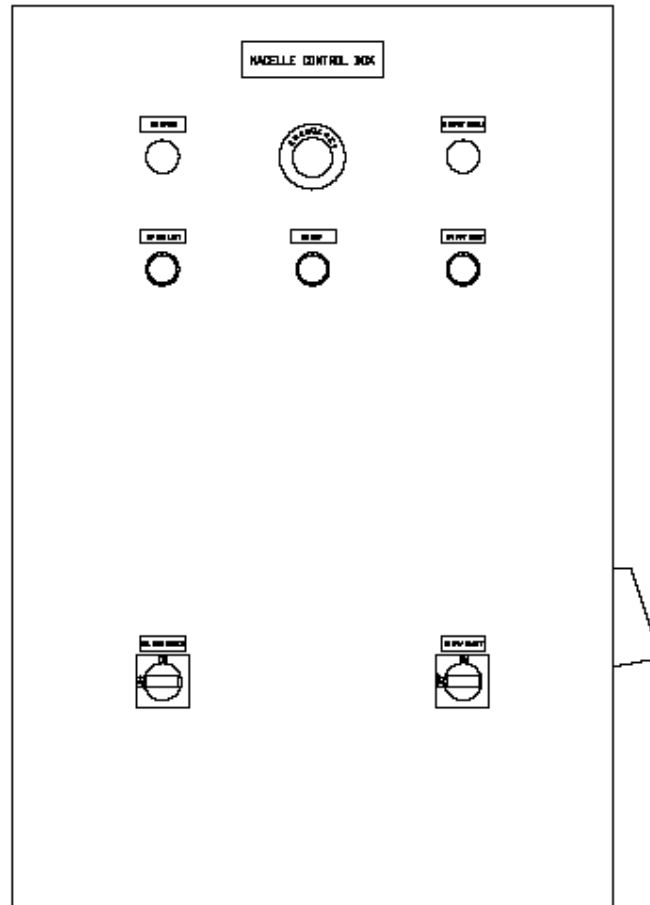


Figure 12 Nacelle Control Box 540A001

The nacelle control box 540A001 is located inside the nacelle.

The tasks of the nacelle control box are:

- Local LV distribution / switching in nacelle / hub
- Nacelle and hub emergency system
- Local manual yaw operation
- Auxiliary function: service winch
- 400 Vac / 60 Hz three phase socket for service equipment
- Connection for laptop PC, to be used as Local Operator Panel for service / maintenance
- Nacelle PLC remote I/O: yaw control, including hydraulic yaw brakes
- Nacelle PLC remote I/O: weather station
- Nacelle PLC remote I/O: main bearing grease system
- Nacelle PLC remote I/O: yaw bearing grease system
- Nacelle PLC remote I/O: generator monitoring
- Nacelle PLC remote I/O: nacelle temperature monitoring
- Nacelle PLC remote I/O: smoke detector

- Nacelle PLC remote I/O: cooling fan
- Nacelle PLC remote I/O: obstruction lights
- Nacelle PLC remote I/O: locking pin and service brake
- Local hub panel 540C001 interface
- Communication: FastBus HCS fibre optic link connection with main turbine control system 542A001
- Communication: CANOpen connection with rotor control box 541A001
- Tower acceleration measurement and protection
- 230 Vac / 60 Hz socket for service equipment (e.g. a laptop PC)

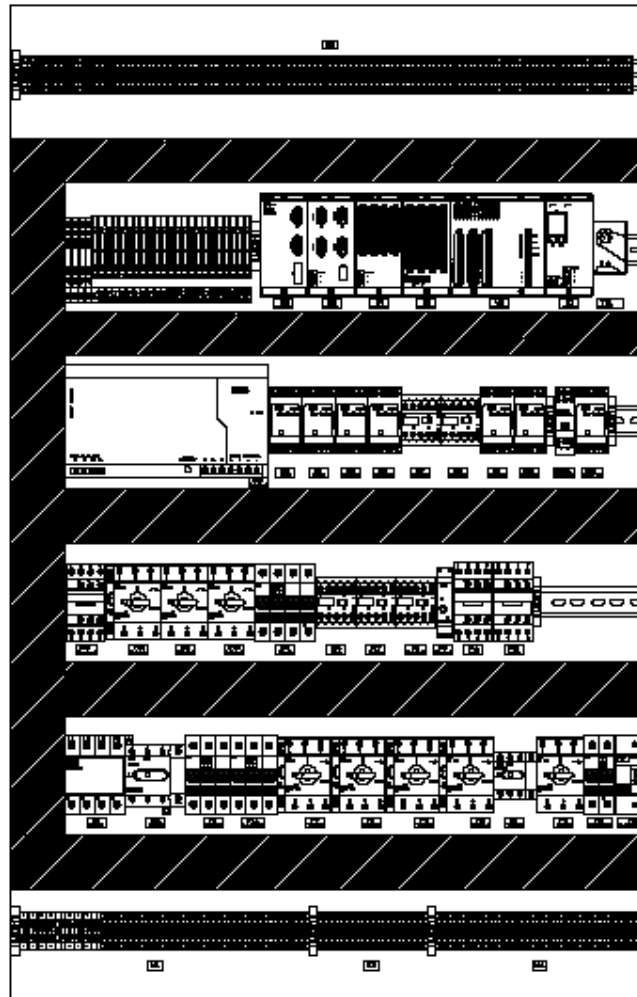


Figure 13 Nacelle Control Box 540A001 components layout

Main components of this cabinet are:

Component	Description
Bachmann PLC station	Remote I/O and communication facilities in nacelle
PCH 1026 Vibration Monitor	Tower X-Y acceleration sensor
Phoenix PSU QUINT 3x400-500AC	24 VDC 30A power supply

Component	Description
24VDC/30A	
KNS1 Telemecanique XPS-AL5110	Emergency trip relay
Various circuit breakers, relays, switches, indicators, and terminal blocks	Various functions safety/control

The control wiring diagram of the nacelle control box and descriptions of these components can be found in [Z72-2000-2B-TD-18-04-R03 Nacelle control box 540A001 electrical diagram].

Software required for the nacelle control box 540A001 is:

Software	Version	Target hardware
Accelerometer configuration	V1.0	PCH 1026

The nacelle cabinet contains the following hardwired operator interface:

Console text	Description
EMERGENCY	Emergency pushbutton
MAIN SWITCH	Main power switch to nacelle and rotor
YAW ON/OFF	Yaw motors maintenance switch
YAW LEFT	Manual yaw left button
STOP	Manual turbine stop and yaw stop button
YAW RIGHT	Manual yaw right button
24 VDC	24 VDC present white indicator lamp
START ENABLE	Turbine start conditions rotor and nacelle OK

6.7.4. Local hub panel 541C001

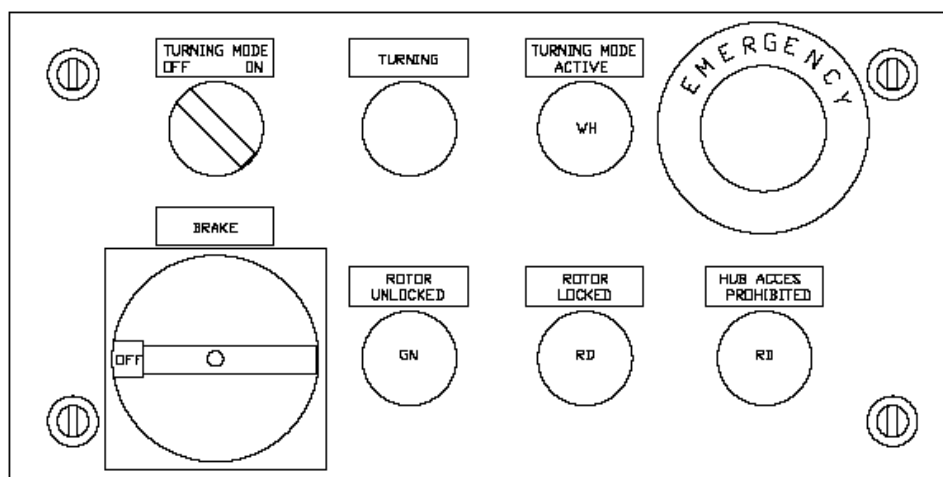


Figure 14 Local Hub Panel 540C001

The local hub panel 541C001 is located on one of the mounting plates in the nacelle, facing the hub.

The local hub panel is used to lock the rotor for service / maintenance of the hub.

The tasks of the local hub panel are:

- Operation of service brake
- Operation of generator turning mechanism
- Monitoring of locking pin
- Emergency pushbutton

Use of this interface is described in chapter 9.

The local hub panel is connected to the nacelle control box 540A001 by one cable.

6.7.5. Rotor control box 541A001

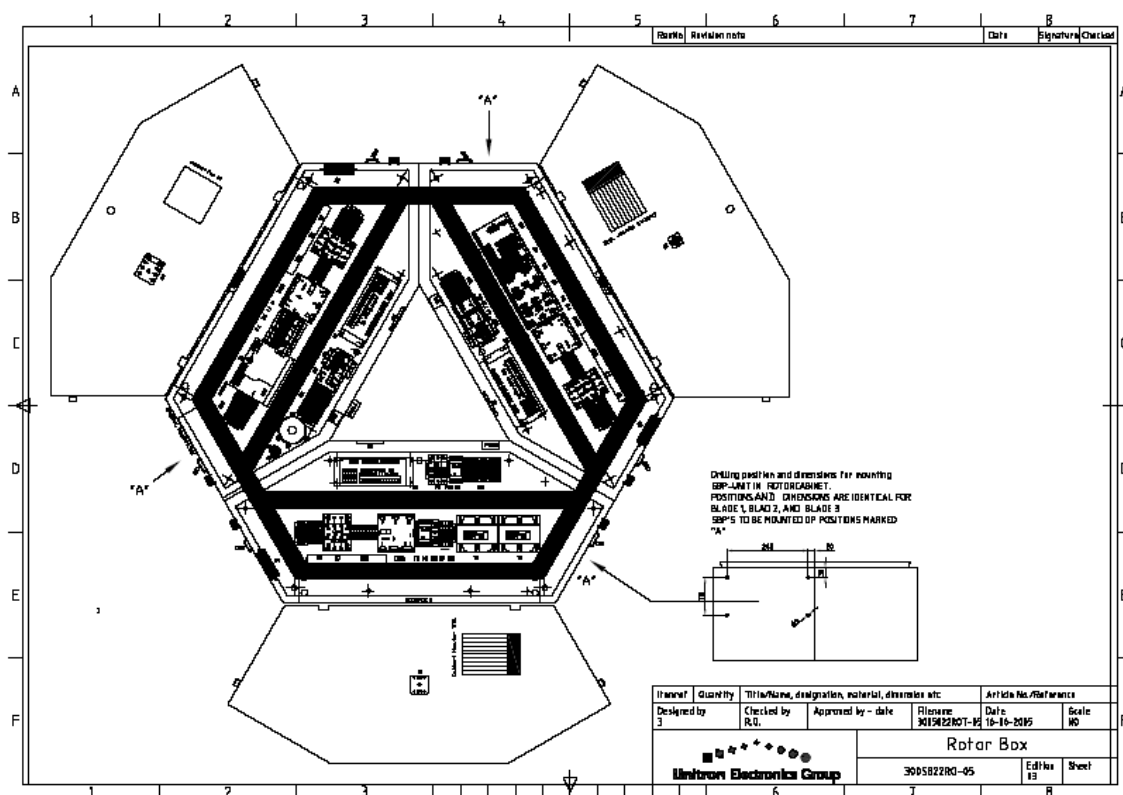


Figure 15 The hexagonal Rotor Control Box 541A001

The rotor control box 541A001 is located in the hub. It consists of a hexagonal shaped cabinet, divided in three sections, one section per blade. Its 3 x 150 VAC power is

supplied by the 12 kVA 400VAC/150VAC 60Hz transformer 004T003 in the nacelle, via slip ring 541X001.

The tasks of the rotor control box are:

- Pitch drive inverters
- Emergency pitch batteries
- Battery loading and monitoring: Smart Battery Pack SBP
- Pitch power source control (see also fig. 8 and fig. 9): Emergency control module ECM
- Independent electrical overspeed protection
- Rotor PLC remote I/O: rotor speed measurement
- Rotor PLC remote I/O: pitch control of the blades
- Rotor PLC remote I/O: pitch bearing grease pump
- Rotor PLC remote I/O: pitch gear grease pump
- Rotor PLC remote I/O: rotor temperature monitoring
- Communication: CANOpen connection with nacelle control box 540A001 via slip ring 541X001

Main components of this cabinet are:

Component	Description
G1 Phoenix PSU Quint PS-100-240AC/24DC 20	Power supply 24 VDC 20A
3 x custom made battery pack 168 Vdc/4Ah blade 1-3	Emergency pitch battery
Control Techniques DCD200-25/50 Maxi Maestro blade 1-3	Pitch drive inverters
Bachmann PLC station	Remote I/O and communication facilities in rotor
SBPC1-SBPC3 Unitron Smart Battery Pack Loaders blade 1-3	Emergency pitch battery loading and monitoring
ECM1-ECM3 Unitron Emergency Control Modules blade 1-3	Pitch source control
Jaquet FTF123 overspeed relay	Independent electrical overspeed protection relay
Various circuit breakers, relays, switches, indicators, and terminal blocks	Various functions safety/control

The control wiring diagram of the nacelle control box and descriptions of these components can be found in [Z72-2000-2B-TD-18-10-R03 Rotor control box 541A001 electrical diagram].

Software required for the rotor control box 541A001 is:

Software	Version	Target hardware
Rotor PLC station application software	V1.0	Bachmann PLC station
Overspeed sensor configuration	V1.0	Jaquet FTF123 overspeed relay

The rotor cabinet contains the following hardwired operator interface:

Console text	Description
MAIN SWITCH	Main power switch to rotor
SERVICE	DC batteries maintenance switch (off=DC batteries connected, on=DC batteries off-line)
24 VDC	24 Vdc present white indicator lamp

6.8. Turbine control system functional description

The following functional groups can be distinguished:

- Main turbine control system functions: main sequence coordination, event processing, production / availability reporting, serving communication of 542A001: power converter communication, Local Operator Panel touch screen, SCADA interface
- Nacelle functions: serving remote I/O in nacelle control box 540A001
- Rotor functions: serving remote I/O in rotor control box 541A001

In fact all processing takes place in the main turbine control system 542A001, so also processing of nacelle and rotor modules is done inside 542A001, although the I/O related to these modules is at a remote location.

6.8.1. Main turbine control system functions

6.8.1.1. Main sequence coordination: starting and stopping of the turbine

Automatic / manual starting and stopping of the turbine is controlled from a sequence containing the following turbine states:

STATE	DESCRIPTION
SHUTDOWN	This state is active in case of a turbine trip, when the maximum number of automatic restarts per hour has been exceeded.
STANDBY	This state is active if all trips are cleared
WAIT RESTART	This state is active when the turbine waits for the conditions for an automatic restart. This can be either due to high wind speed, cable unwinding, periodical safety check or a turbine trip, where the maximum number of automatic restarts per hour has not been exceeded.
PRE-PRODUCTION	This state is active after a valid START command. In this state, the converter is switched to STANDBY and the yaw control is switched to AUTO YAW (if these states were not already active)
START PRODUCTION	When the yaw position is OK, the pitch

STATE	DESCRIPTION
	control system moves the blades to a start-up position (default 30 deg), allowing the rotor to speed-up. When rotor speed > 3.5 rpm, the blades move to work position, increasing rotor speed
PRODUCTION	When the converter production has started (rotor speed exceeds 6.0 rpm) and the production pitch control is active, this state becomes active
→ SHUTDOWN	This is a transition state to SHUTDOWN, when the maximum number of automatic restarts per hour has been exceeded.
→ RESTART	This is a transition state to WAIT RESTART when the maximum number of automatic restarts per hour has not been exceeded, in case of high wind speeds, cable unwind or periodical safety check.
NORMAL STOP	This is a transition state to STANDBY, in case a NORMAL STOP command is issued

The following conditions must be met in order to be able to issue a START command:

- o Trips cleared
- o Pitch system ready
- o Converter ready

The following conditions must be met to enter START PRODUCTION state:

- o Wind speed OK
- o Auto yaw active
- o Yaw angle OK
- o Converter switched on

The following conditions must be met to enter PRODUCTION state:

- o Production pitching active
- o Converter production active

The RESTART state is entered if one of the following conditions is met:

- o Wind speed very high
- o Unwinding cable
- o Periodical emergency system check
- o Over speed test
- o Trip, and maximum number of automatic restarts /hr has not been exceeded

For each trip, a parameter denoting the maximum allowable number of restarts/hr must be configured. When this maximum has been exceeded, the turbine will not be restarted automatically. In this case, an operator needs to investigate the cause of the trip first before restarting the turbine (see Appendix C List of events).

6.8.1.2. Local Operator Panel and SCADA (remote operator interface)

On the converter cabinet, a hardwired operator interface with the following buttons is present, providing the most elementary commands to operate the turbine:

SERVICE: Service mode on / off

OPERATION: Local (touch screen) or Remote (SCADA) operation

TURBINE RESET: Reset trips / alarms command

TURBINE STOP: Stop command,

TURBINE START: Start command, if all start conditions

A LOCAL / REMOTE selection switch transfers the operation of the turbine to the Local Operator Panel or the remote SCADA system.

The Garrad Hassan Remote Interface Unit (RIU) exchanges values with the main turbine control system 542A001. The remote interface is based on Ethernet.

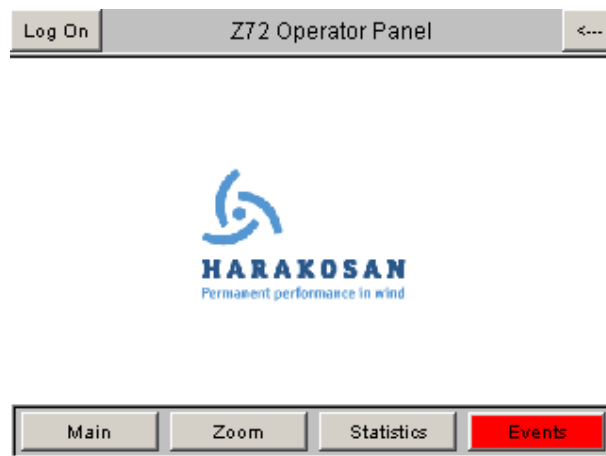


Figure 16 Local Operator Panel Start-up screen

The Local Operator Panel touch screen runs a user interface and provides the user interface functionality complementary to the basic START / STOP / RESET TURBINE push button functions. After starting up the Main turbine control system, a start-up screen with the Harakosan logo is shown.

Navigation is based on 4 items shown on the lower button bar on the screen:

- Main: used to display main operator panel items
- Zoom: used to zoom in into specific turbine components
- Statistics: production and availability data, operational statistics
- Events: active trips and alarms

The description of the actual screen is shown on the top bar.

Navigation to previous pages is accomplished by pressing the "←" button on the top bar of the screen.

The top bar of the screen also shows a button to log on / off. After entering the appropriate password (default "0000"), the user is granted the right to issue commands from the touch screen panel. If the user is logged off, or the LOCAL/REMOTE selection switch is set to REMOTE, the buttons on the panel are greyed out, and operation is blocked.

The touch screen Local Operator Panel has the following menu structure:

- Main Main operator panel items
 - Overview Overview screen of wind speed, rotor speed, etc.
 - Startup conditions Sequence steps and conditions for start-up
 - Parameters List of parameters
 - Auto Restart Settings Maximum allowed number of restarts/hr for each trip
 - Logbook Logbook of most recent 100 events with timestamp
 - System information Software versions information
- Zoom Details of specific components
 - Rotor Rotor details
 - Generator Generator details
 - Nacelle Nacelle details
 - Converter Converter details
 - Grid connection Grid connection details
 - Weather Station Weather station details
- Statistics Operational statistics
 - Production log Daily, monthly and yearly statistics
 - Availability log Daily, monthly and yearly availability
 - Totals Accumulated production, operational statistics
- Events Active trips and alarms

Screens are selected by pressing one of the 4 screen buttons on the bottom, followed by selecting one of the items on the menu shown.

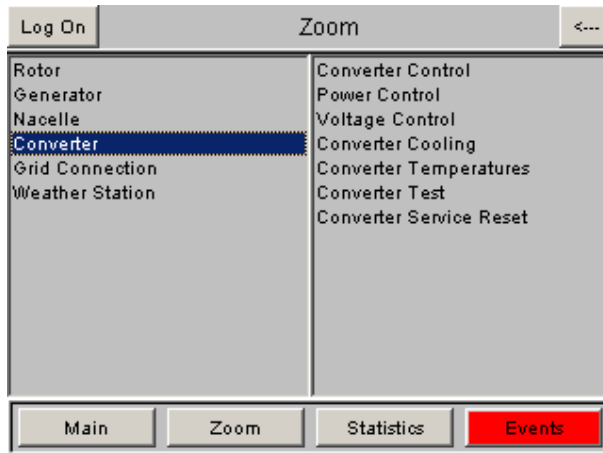


Figure 17 Detail of the Zoom menu

A PC version of the touch screen panel software is required for service. This program is called Service Panel, and provides extra menu items for service activities. By connecting the laptop to an Ethernet port on the EM203 module of the nacelle PLC station a service tool for technicians in the nacelle/hub is available (see Chapter 9).

The touch screen panel is also used for operation of the ACS1000 converter. Use of the interface is described in chapter 8/9.

6.8.1.3. Event processing

An event processing module is used to process the following events:

- Normal trips. A normal trip results in a normal stop (by pitch inverters) of the turbine. Depending on the configured maximum number of restarts/hr, the turbine may or may not restart automatically.
- Emergency trips. An emergency trip results in an emergency stop (by batteries) of the turbine. In general the turbine is not restarted.
- Alarms. An alarm is an unusual situation that requires operator attention.
- Info. Information message, for example results of tests or operation sequence messages

The event processing module contains counters that monitor frequency of occurrence of trips and maximum number of restarts/hr, that are used to determine if a turbine is allowed to restart automatically.

The most recent 100 events are stored in a buffer, together with their associated time stamps. This allows easy troubleshooting of failures.

The possible events are listed in Appendix C List of events.

6.8.1.4. Production / availability reporting

This module accumulates and buffers production pulses generated inside the ABB converter (resolution in steps of 10kWh or 10 kVAh):

- Actual production: daily, monthly and yearly kWh production, kWh consumption, kVAh production, kVAh consumption of the actual day, month and year
- Daily production: 31 buffers (one month history) of daily kWh production, kWh consumption, kVAh production, kVAh consumption
- Monthly production: 13 buffers (one year history) of monthly kWh production, kWh consumption, kVAh production, kVAh consumption
- Yearly production: 40 buffers (lifetime history) of yearly kWh production, kWh consumption, kVAh production, kVAh consumption

And availability data:

- Actual availability: daily, monthly and yearly turbine availability and production availability of the actual day, month and year
- Daily availability: 31 buffers (one month history) of daily turbine availability and production availability
- Monthly availability: 13 buffers (one year history) of monthly turbine availability and production availability
- Yearly availability: 40 buffers (lifetime history) of yearly turbine availability and production availability

The turbine availability is defined as follows:

$$(Time\ with\ trips\ cleared\ /\ Total\ time) \times 100\%$$

The production availability is defines as:

$$(Time\ state\ is\ START\ PRODUCTION\ or\ PRODUCTION\ /\ Total\ time) \times 100\%$$

6.8.1.5. Converter interface

The interface with the converter consists of a hardwired part for control (turbine trip command from converter), and a fieldbus part based on CANOpen, used for converter on/off control, power mode control, grid voltage control and monitoring of converter data.

A separate CAN network (Network number 2, 500 kbits/s) is dedicated for this converter communication. Communication is based both on both so called CAN PDO's for regular data transfer) and SDO's for less time critical information (like temperatures).

6.8.1.6. Periodical safety check

A timer based on production hours periodically issues an emergency trip (once every 96 production hours).

6.8.2. Nacelle functions

6.8.2.1. Automatic and manual yaw control, including cable unwind

Yaw control is used to turn the turbine into the wind.

Automatic / manual yaw control or cable unwind is based on the following yaw states:

STATE	DESCRIPTION
YAW INHIBIT	Using the yaw system may cause damage to the yaw mechanism or the generator cable.
YAW STANDBY	Yaw inhibit trips are cleared. Manual yaw left and manual yaw right commands are enabled.
YAW AUTO	The nacelle is aligned upwind automatically. Yaw brakes in automatic mode.
CABLE UNWIND	The yaw system is unwinding the generator cable. Yaw brakes in automatic mode.

Yaw inhibit

Yawing is inhibited under the following conditions:

- Event 223. Yaw timeout. When yaw alignment within 5 deg within 5 minutes (this is 15 times the nominal time for yawing 10 deg with an average yaw speed of 0.5 deg/s)
- Event 224. Yaw motor protection switch tripped
- Event 225. Work switch yaw motors active
- Event 226. Cable twist limit switch active
- Event 227. Cable twist sensor fault
- Event 211. Emergency system nacelle tripped

Yaw standby

Manual yawing can be useful when servicing the turbine. Manual yawing is possible when the YAW STANDBY state is active.

To start manual yawing, do the following:

- Select manual yaw left or right from the local touch screen panel or the YAW LEFT / YAW RIGHT buttons on the nacelle control box 540A001

To stop yawing in manual mode, do the following:

- Select manual stop right from the local touch screen panel or by pressing the STOP button on the nacelle control box 540A001

The YAW LEFT / YAW RIGHT commands issued from the nacelle cabinet override the PLC and are interrupted by cable twist end switches only.

Yaw right command (issued either automatic or manually) are blocked by relays in 540A001 when the yaw motors turn left or vice versa.

Yaw auto

To start yawing in automatic mode the following conditions should be fulfilled:

- Automatic yawing active (default after START is issued)
- Vwind, 5s ≥ 3.5 m/s
- Vwind, 60s ≥ 3.0 m/s
- Yaw angle (measured with respect to delayed wind vane) not within (-15deg,+15deg)

To stop yawing in automatic mode the following conditions should be valid:

- When yaw angle based on 5 sec. average within [-2deg, +2 deg]

Cable unwind

Due to frequent yawing in one direction, the generator- and other cables that connect the fixed tower to the rotating nacelle can twist too much, resulting in broken cables/damaged junction boxes/short circuits/machine damage etc.

A (potentiometer based) cable twist sensor is positioned at the yawing bearing. The number of revolutions - in whole numbers of 360 deg left or right - is read out by the control system.

When unwinding the cable, the turbine is shut down, but will restart automatically when finished.

To unwind the cable, the following conditions should be fulfilled:

- |Twist| > 1 & Vwind,10 min < 5.0 m/s: |unwind target| : < 0.5 turn
- |Twist| $> 1,5$ & 10 min < 10.0 m/s: |unwind target| : < 1.0 turn
- |Twist| > 2 : |unwind target| : < 0.5 turn

To stop unwinding, the following condition should be fulfilled:

- Unwind target reached & yaw angle based on 5 sec. average within [-2 deg, +2deg]

Hydraulic Yaw brakes

In the states YAW AUTO and CABLE UNWIND, the hydraulic yaw brakes are automatically lifted during movement and automatically dropped when the movement is stopped. The brake system operates with 3 pressures: 0 bar no friction (brakes lifted), 20 bar for low friction brake (currently not used) and 210 bar for high friction brakes (brakes dropped).

In the states YAW INHIBIT and YAW STANDBY the yaw brakes can be manually dropped or lifted by commands on the Local Operator Panel.



Figure 18 Yaw brakes

6.8.2.2. Weather station

On top of the nacelle, a cup anemometer, two wind vanes and a temperature sensor are mounted

- The anemometer is used to shut down the turbine in case the wind speed becomes very high, and for determination of the number of turns to unwind the cable (see above). The anemometer can also be used for Pv performance analysis. The signal coming from the anemometer is filtered with moving average filters to achieve 5 second average values, 60 second average values and 10 minute average values.

- The wind vane is used to align the nacelle to the wind (see above), and for determination of the absolute wind direction for monitoring purposes. For yaw alignment, the signal is filtered with a first order digital filter with time constant 25s for yaw misalignment detection and with a filter with time constant 5s for yaw alignment detection.
- The temperature sensor is used to determine if the turbine is operating inside its normal operating range. It can assist in Pv performance analysis corrections and in the interpretation of temperature monitoring results of for example generator stator temperature measurements or bearing temperatures measurements.

6.8.2.3. Tower acceleration monitoring

Excessive tower/nacelle movements can be dangerous for the turbine and therefore are monitored.

Tower vibration is measured with a device from PCH (see [PCH]) in two perpendicular directions (X: left-right; Y: forward-afterward) and sent to the remote I/O PLC station in the nacelle.



Figure 19 Vibration monitor

A trip is issued if the vector sum of the accelerations in these directions exceeds a pre-set level (1.75 m/s²).

6.8.2.4. Nacelle emergency system

The emergency pushbutton system in the nacelle can be monitored and reset with the remote I/O PLC station in the nacelle.

6.8.2.5. Yaw bearing grease system

The yaw bearing is lubricated by a electric grease pump supplied by Vogel. The grease to the yaw bearing is distributed with a flow divider based on sequentially activated pistons to different greasing points on the bearing. One of the outgoing lines is equipped with a proximity switch that detects piston movement in that line. The grease tank is equipped with a level switch.

The grease pump can be operated in automatic and in manual mode.

In manual mode, the pump can be switched on and off manually.

In automatic mode, the pump is switched on after a preset interval time has expired. The pump is active for a preset number of piston cycles, corresponding to a preset fixed dose of grease per interval. The time required for these cycles is measured and compared with a maximum time. In case this time is exceeded, grease flow is assumed to be stuck, and an alarm is generated. Interval time and number of cycles can be set with a parameter (see also Appendix B)

A level sensor detects if the grease tank level is low. An alarm will inform the operator that the grease tank needs to be refilled.

6.8.2.6. Nacelle and nacelle cabinet temperature monitor

Both the temperature inside the nacelle control box 540A001 and the temperature outside the nacelle control box 540A001 are monitored.

6.8.2.7. Cooling fan

A cooling fan in the nacelle, providing cooling air to the bearing and hub environment is controlled either automatically (based on whichever temperature is higher: bearing temperature or hub temperature) or manually (for service and test purposes). The air produced by the fan is guided to the hub by means of a duct. The threshold for automatic operation can be set with a parameter (see section 11.2).

6.8.2.8. Obstruction lights monitoring

The intensity (day / night) and the proper functioning (strobe fail or power fail) of the obstruction lights controller is monitored.

6.8.2.9. Generator temperatures monitoring

The following generator stator temperatures are monitored:

- generator stator temperature U phase: 12 O'clock and 6 O'clock
- generator stator temperature V phase: 2 O'clock and 8 O'clock
- generator stator temperature W phase: 4 O'clock and 10 O'clock
- generator air gap temperature 1
- generator air gap temperature 2

The protection system uses these values to initiate a turbine trip or an alarm in case the temperatures exceed trip- or alarm thresholds.

Also, air temperatures of the air gap are monitored:

- air gap temperature 1
- air gap temperature 2

The protection system uses these values to initiate an alarm in case the temperatures exceed alarm thresholds.

6.8.2.10. Main bearing temperatures monitoring

The main bearing temperatures are monitored by two temperature sensors in the bearing. The protection system uses the temperature measurements to initiate a turbine trip or an alarm in case the temperatures exceed trip- or alarm thresholds.

6.8.2.11. Main bearing grease system

The main (generator) bearing is lubricated by a electric grease pump supplied by Vogel. The associated grease tank is equipped with a level switch.

The grease pump can be operated in automatic and in manual mode.

In manual mode, the pump can be switched on and off manually from the Local Operator Panel.

In automatic mode, the pump is switched on after a preset interval time has expired. The pump is active for a preset time. Interval time and "on"-time can be set with a parameter (see also Appendix B).

A level sensor detects if the grease tank level is low. An alarm will inform the operator that the grease tank needs to be refilled.

6.8.2.12. Service PC operator interface

On the nacelle remote I/O PLC station, an Ethernet module (Bachmann EM203) is used for connection of a Service laptop PC. A socket in the nacelle control box can provide 230 VAC power for a laptop PC with a wide input range switched power supply.

6.8.2.13. Service brake, turning mode and locking pin

A rotating permanent magnet rotor produces high voltages on converter if the rotor is not properly locked. For this reason, and to firmly stop the rotor for service / maintenance in the hub or the converter it is required to lock the rotor. Locking of the rotor is done with the locking pin. This can be controlled on the local hub panel.

During a rotor locking procedure, a hydraulic service brake, fed by the hydraulic pressure unit (also used by the yaw brakes), fixes the rotor, enabling a locking pin to be applied manually. The position of the locking pin is read in by a sensor. Based on this sensor, a ROTOR LOCKED or ROTOR UNLOCKED indicator on the local hub panel 540C001 will illuminate.

Alternatively, it is possible in the so called generator turning mode, to slowly turn the generator, until the correct rotor position for locking is achieved.

The PLC system reads in the state of the brake (applied or not applied) and used this feedback as a turbine start condition.

See the chapter 10.6 for complete locking instructions.

6.8.3. Rotor functions

6.8.3.1. Pitch control

The pitch control of the blades is used to start/stop the turbine, to limit the rotor speed or to move (individual) blades for service / maintenance or storm park.

Blade pitch control is based on the following states:

STATE	DESCRIPTION
EMERGENCY	The emergency batteries are connected to the DC motors. This will cause an emergency pitch of the 3 blades to vane position. When arriving at the vane position, the batteries are disconnected from the motors. When arrived at the vane switches, the brakes of the motors are applied.
MAN OPEN LOOP	Service mode; Blade 1, 2 or 3 can be pitched towards vane position or towards work position. Only one blade can be <> vane position at any time. The encoders need not to be calibrated for this mode.
MAN CLOSED LOOP	Service mode; Blade 1, 2 or 3 or all blades simultaneously can be pitched to an adjustable target. The encoders need to be calibrated for this mode.
MAN PITCH RAMP	Service mode; Blade 1, 2 or 3 can be positioned sequentially between an upper and a lower limit to run-in blade bearings
INIT	Watchdog is activated. Pitch motor brakes are lifted. The pitch drive inverters are enabled or reset in case of a failure.
STOP	The blades are pitched from the limit switches to the vane position of the servo system.
STANDBY	The rotor is ready to start.
STARTUP	The blades are pitched to the start-up position.
PRODUCTION	When rotor speed exceeds 3.5 rpm, the blades are pitched to work position.
STORM PARK	The blades are pitched to the vane switches. When arrived at the vane switches, the brakes of the motors are applied.

The pitch system is used to stop the rotor in case a trip condition is detected.

The state STOP is used when a trip is issued by the standard protection system.
The state EMERGENCY is used when an emergency pitch is issued.

The state STORM PARK is used when the wind speed is very high. The turbine is stopped and the blades are locked at the vane switch position by the pitch motor brakes. Event T_204 If event The 10m wind speed has to be higher than SDVwind_Storm_10m (27 m/s) or the 3s wind speed has to be more than SDVwind_Storm_3s (35 m/s) for activating STORM PARK.

Start-up of the turbine is accomplished by pitching the blades to a start-up position. This is a position where aerodynamic torque is high when the rotor speed is low. When the turbine speed exceeds 3.5 rpm, the blade angle is ramped to -1 degrees (work position), that enable optimal energy extraction when the turbine is in partial load control (between 9 rpm and nominal speed setpoint).

When wind speed is high, the production pitch control algorithm will become active by continually adjusting the blades, controlling rotor speed at a high constant value to maximize energy yield, but low enough to prevent overspeed. The nominal speed setpoint (see parameters Appendix B) determines when production pitching starts.

Manual pitch control modes are used by service technicians.

6.8.3.2. Emergency pitch system

Normal pitch control drive train consists of a Control Techniques Maxi Maestro DC servo drive, and a Control Techniques DC pitch motor. This pitch control mechanism is active only when a PLC watchdog signal is received and the external Jaquet overspeed protection (see section 6.8.3.8) is not tripped.

In case the normal pitch control mechanism is faulty or unavailable (for example due to grid loss), or the external overspeed protection is tripped, the DC batteries are connected to the DC motors, causing an EMERGENCY PITCH action.

6.8.3.3. Battery Monitoring

The batteries of the emergency protection system are built up using 14x12Vdc cells in series. This total voltage is used for emergency pitching. The first two sections of 12 Vdc are used to power the emergency system electronics. These sections consequently have a higher power consumption than other sections.



Figure 20 Smart Battery Pack Charger

The SBP loaders check the condition of the 14 cells, and start charging cells in order of increasing condition. The SBP loaders are integrated in the CAN network in the rotor (Network number 1, 500 kbit/s). Node numbers are related to the blade, so the SBP loader for blade 1 has node number 1, the SBP loader for blade 2 has node number 2, and the SBP loader for blade 3 has node number 3. The SBP loaders give indications of the loading status of each battery pack:

- Initial charging (red LED)
- Top-up charging (yellow LED)
- Maintenance charging (green LED)

6.8.3.4. Periodical safety check

A timer based on production hours periodically (once every 96 production hours) issues an emergency trip by temporarily withdrawing the rotor watchdog signal.

During this trip, the time it takes to shutdown the turbine is measured and pitch ramp rates are compared with minimum values. Should the control system detect a malfunction in the emergency system, then a subsequent restart is blocked. In that case an operator needs to investigate the cause of the malfunction first and needs to reset locally before the turbine can be restarted.

If the result of the safety check is satisfactory, the turbine is restarted automatically.

6.8.3.5. Rotor and rotor cabinet temperature monitor

Both the temperature inside the rotor control box 541A001 and the temperature outside the rotor control box 541A001 are monitored.

Both under temperatures and over temperatures initiate alarm and trips.

6.8.3.6. Pitch bearing grease pump control

The pitch bearings are lubricated by a electric grease pump supplied by Vogel. The grease to the pitch bearings is distributed by flow dividers. One of the outgoing lines of the divider is equipped with a proximity switch that detects piston movement in that line. The grease tank is equipped with a level switch.

The grease pump can be operated in automatic and in manual mode.

In manual mode, the pump can be switched on and off manually.

In automatic mode, the pump is switched on after a preset interval time has expired. The pump is active for a preset number of piston cycles, corresponding to a preset fixed dose of grease per interval. The time required for these cycles is measured and compared with a maximum time. In case this time is exceeded, grease flow is assumed to be stuck, and an alarm is generated.

Interval time and number of cycles can be set with a parameter (see also Appendix B)

A level sensor detects if the grease tank level is low. An alarm will inform the operator that the grease tank needs to be refilled.

6.8.3.7. Pitch gear grease pump control

The pitch gears are lubricated by a electric grease pump supplied by Vogel. The grease tank is equipped with a level switch.

The grease pump can be operated in automatic and in manual mode.

In manual mode, the pump can be switched on and off manually.

In automatic mode, the pump is switched on after a periodical emergency pitch test. The pump is active for a preset time, corresponding to a preset fixed dose of grease per interval. The time required for these cycles is measured and compared with a maximum time. In case this time is exceeded, grease flow is assumed to be stuck, and an alarm is generated.

Interval time and number of cycles can be set with a parameter (see also Appendix B)

A level sensor detects if the grease tank level is low. An alarm will inform the operator that the grease tank needs to be refilled.

6.8.3.8. Electrical overspeed protection

An electrical overspeed protection shuts down the turbine in case the software overspeed protection has failed. The electrical overspeed protection is based on a Jaquet ferostat speed sensor, sensing the teeth of a pole wheel connected to the slip ring. The time between successive teeth is measured, the corresponding speed is calculated and compared with a preset threshold (the threshold can be set with a Jaquet software tool and a Jaquet serial configuration cable). An emergency system trip is initiated if the overspeed trip threshold set at 28.5 rpm is exceeded.



Figure 21 Jaquet T401 Over speed relay

It is possible to temporarily lower the trip set-point to 10 rpm. By doing so, the integrity of the electrical over speed protection mechanism can be verified during periods with low wind speeds.

7. Powering on the turbine control system

To power on the turbine control system:

- 1) Switch on the nacelle control box 540A001 HS1 Main Switch
- 2) The indicator H2 24 VDC on the nacelle control box 540A001 should be lit
- 3) Switch on the rotor control box 541A001 HS1 Main Switch
- 4) The indicator H1 24 VDC on the rotor control box 541A001 should be lit
- 5) Switch on the 50A supply to nacelle control box 540A001 and rotor control box 541A001 in the switch gear room.
- 6) Switch on the 20A supply to the converter in the switch gear room

The Local Operator Panel touch screen should now boot-up.



After power up of the control system, the turbine is in a safe state (shutdown).

If the turbine control system is not yet configured, the touch screen will show the "Off-line" screen. In that case, the control system first needs to be configured as described in Chapter 11 Configuration of the turbine control system.

8. Operating & Monitoring the turbine from the Local Operator Panel

When power is applied to the control system and the converter, the turbine can be started.

WARNING:

Before starting the turbine, be sure to understand the procedures to issue a normal stop and an emergency stop.

8.1. Emergency stop

When situations occur that are dangerous for the service personnel working at the turbine, or the turbine itself, yellow/red emergency "mushroom" buttons can be activated at several places in the turbine.



The locations of the yellow/red coloured emergency mushroom buttons are:

- 1) On the converter control panel.
- 2) In the nacelle, on the nacelle control box.
- 3) In the entry to the hub, on the local hub panel.

The result of pressing the mushroom button is depending on the location:

Pressing the emergency button on the converter will switch off the converter, disconnect the converter from the main transformer, and disconnect power to the nacelle and hub, resulting in emergency shutdown of the turbine.

Pressing the emergency button in nacelle or on local hub panel will only disconnect power to power consumers in the nacelle and hub, resulting in emergency shutdown of the turbine.

WARNING:

BE VERY CAREFUL TO ACTIVATE THE EMERGENCY BUTTON WHEN PEOPLE ARE WORKING INSIDE THE HUB WHILE THE SERVICE SWITCH IS OFF. ACTIVATING THE EMERGENCY BUTTON WILL IMMEDIATELY ACTIVATE THE EMERGENCY PITCH SYSTEM, WITH POSSIBLE INJURIES TO PEOPLE WORKING IN THE VICINITY OF THE PITCH GEAR !

HINT:

IF ONLY A TURBINE STOP IS REQUIRED, PRESS THE "STOP" BUTTON ON THE NACELLE CONTROL BOX. THIS WILL STOP THE TURBINE, BUT WILL NOT TRIP THE MCB.

An emergency trip issued from the nacelle or local hub panel can be reset with the blue RESET button on the right hand side in the nacelle control box 540A001 or with the RESET button on the Local Operator Panel of 542A001 on the converter.

When an emergency trip from the converter is issued, follow the guidelines in ABB manual [ABB] to reset the emergency trip circuit.

8.2. Manual starting and stopping the turbine from the Local Operator Panel

STARTING THE TURBINE



NOTE:

Before starting the turbine, make sure that trips are reset according to the procedures outlined in Appendix C List of events. An active trip is shown as a red event on the Events screen of the touch screen Local Operator Panel.

The turbine can be reset with the TURBINE RESET button on the Local Operator Panel of 542A001. This will reset inactive trips and/or reset the emergency circuit in the nacelle.

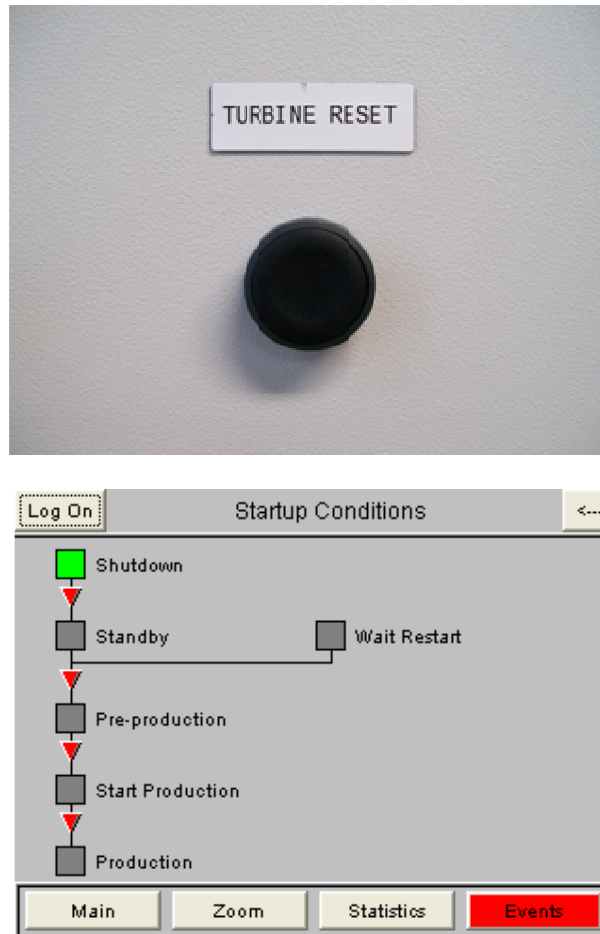


Figure 22 Start-up sequence screen

- Go to screen Main/Start up conditions.
- Click on the transition triangle below the STANDBY state to pop-up the conditions.
- Verify that all pre-production conditions are fulfilled.

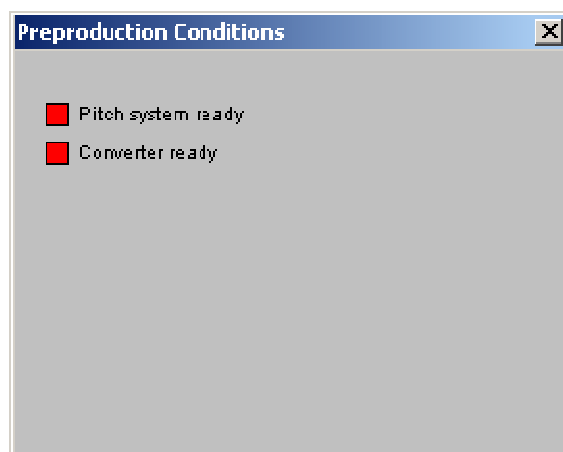


Figure 23 Preproduction pop-up



- Switch the LOC / REM switch to LOC for local operation.
- The turbine is started by pushing the TURBINE START button on the Local Operator Panel of 542A001.
- the converter is switched on (if not already switched on)
- the yaw control is switched to automatic yawing (if not already active). The turbine will turn left or right to the wind direction, as indicated by either the "yaw left" (turbine turning counter clockwise) or "yaw right" (turbine turning clockwise) indication lamps.

When the rotor is aligned upwind, the Start Production state is entered. In this state:

- the blades will pitch from the vane position (default 86 deg.) to the start-up position (30 deg.). In this position, the rotor starts-up easily when rotor speed is low. Depending on the wind speed, the rotor will now start to turn. If the rotor speed exceeds 3.5 rpm (this will be the case when wind speed > 4 m/s), the

blades will pitch to the working position. When 3.5 rpm cannot be reached, because wind speed is too low ($< 4\text{ m/s}$), the blades will stay at the start-up position, and the turbine will wait until the wind speed becomes high enough to start.

- When the rotor speed exceeds 9 rpm, the converter enters the Production state, and the start-up is completed. When the wind speed $> 12\text{ m/s}$, the blades will keep the rotor speed constant by pitching the blades towards vane position when the speed becomes too high, and by pitching back to working position when the speed is lower.

The start-up sequence can be followed on the Main/Startup conditions screen or on the Main/Overview screen.

After a successful start-up, the corresponding counter on Statistics/Totals/Operation will be increased by 1.

When the start-up sequence is finished, the turbine will now run until a manual stop command is issued, or a trip occurs. The auto restart frequency determines if the turbine will restart automatically after recovery of the trip, or stays in a shutdown state.

STOPPING THE TURBINE



- The turbine can be stopped by pushing the TURBINE STOP button on the Local Operator Panel 542A001. Also the turbine can be stopped by pressing the STOP button on the nacelle cabinet, in the undesirable situation where the turbine has been started while people are present inside the turbine.

The stop sequence can be followed on the Main/Overview screen. The manual stop sequence is completed, when the turbine state is STANDBY (or SHUTDOWN, in the event that a trip occurred during the manual stop), and rotor speed $< 2.5\text{ rpm}$.

After a normal stop, the corresponding counter on Statistics/Totals/Operation will be increased by 1.

8.3. Automatic shutdown of the turbine after a trip and notification of alarms

If during running a problem is detected by the Main turbine control system 542A001, the turbine will be either shut down or it stays running and activates a notification alarm, depending on the severity of the problem.

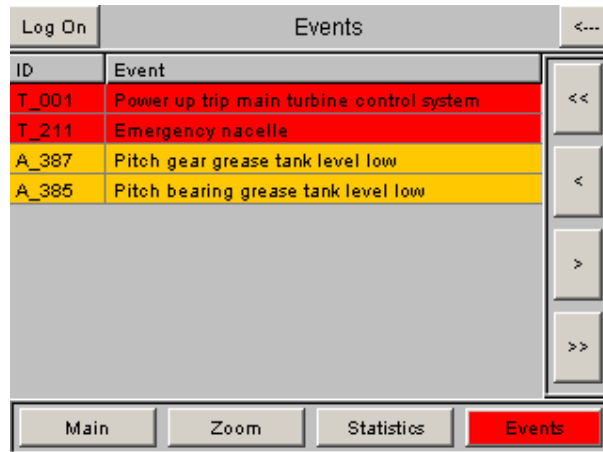


Figure 24 Events screen

If a trip is active, the colour of the Event button on the Local Operator Panel touch screen will turn red. If an alarm is active, the colour of the Event button will turn orange.

On the Events screen, the event, including its associated event ID (see Appendix C and the corresponding colour is shown.

For each trip, a so called auto restart frequency is configured. This auto restart frequency determines the maximum allowed number of identical trips within one hour.

In general, the main PLC will try to restart the turbine automatically after a trip. Should the number of restarts within the last hours exceed the maximum allowed auto-restart frequency, then the turbine will not restart, but instead will go to a SHUTDOWN state.

Only by pressing the TURBINE RESET button on the Local Operator Panel, the turbine can be reset. This will assure that somebody is at the turbine for investigation of the problem.

This investigation of the problem is done using the sequence of events in the Main/Logbook of the Local Operator Panel, and the guidelines of Chapter 11 Troubleshooting. When the problem is solved, the turbine can be started.

If an alarm is active, the operator has some time to anticipate necessary actions. In general, if the alarm is not followed up on, it will result in a turbine trip.

8.4. Manual emergency trip test

After every 96 production hours, an emergency trip test is issued automatically. This test can also be issued manually from the Local Operator Panel screen Zoom/Rotor/Test Emergency Trip.

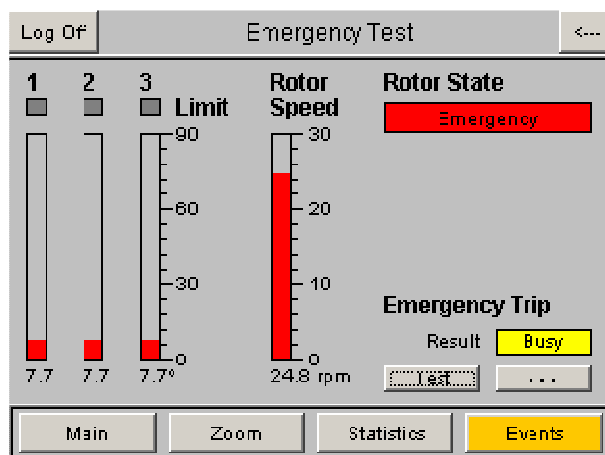


Figure 25 Test emergency trip screen

To manually issue the emergency trip test, follow the following procedure:

- Go to the emergency trip test screen by selecting Zoom/Rotor/Test Emergency Trip.
- Verify that the 10 min avg. wind speed < 10.0 m/s, the turbine is running and the blade angles are -1 deg.
- Start the test by pressing "Test".
- The emergency pitch batteries are now connected with the pitch motors; the blades will pitch to vane position.
- When the blades are at vane position, or otherwise after a timeout, the test result will be shown in the Result box.
- More detailed information regarding the test can be found by pressing the "... button.

WARNING:

DO NOT RESTART THE TURBINE WHEN THE EMERGENCY TRIP TEST HAS FAILED !

8.5. Manual overspeed trip test

NOTE:

The periodical over speed test must be performed "on-line", that means that it is not possible to do while the Service switch on the Local Operator Panel is switched ON.

With the control system it is possible to check the integrity of the overspeed protection system also while wind speed is low, by temporarily lowering the trip setpoint from 28.5 rpm to 10 rpm.

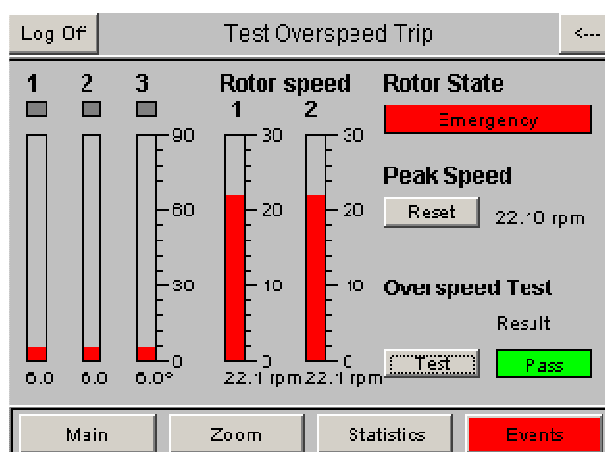


Figure 26 Test emergency trip screen

The procedure for a manual test of the overspeed protection system is as follows:

- wind speed $V_{wind, 10 \text{ min. avg.}}$ should be $> 6 \text{ m/s}$ and $< 20 \text{ m/s}$.
- Reset peak speed with the Reset button on the screen /Zoom/Rotor/Test Overspeed Trip.
- When rotor speed $> 10.0 \text{ rpm}$, press the Test button.
- The emergency pitch system should now stop the turbine.
- The test result will be shown in the Result box.

WARNING:

DO NOT RESTART THE TURBINE WHEN THE PERIODICAL OVERSPEED TEST HAS FAILED !

8.6. Grid voltage control

Reactive power up to a maximum value (2MVar) can be produced or consumed, even if the turbine is not running. Reactive power control is achieved based on grid voltage droop control.

Grid voltage setpoints can be adjusted from screen Zoom/Converter/Voltage control. Both the desired nominal grid voltage [kV], as the desired voltage droop [%] (static) can be configured.

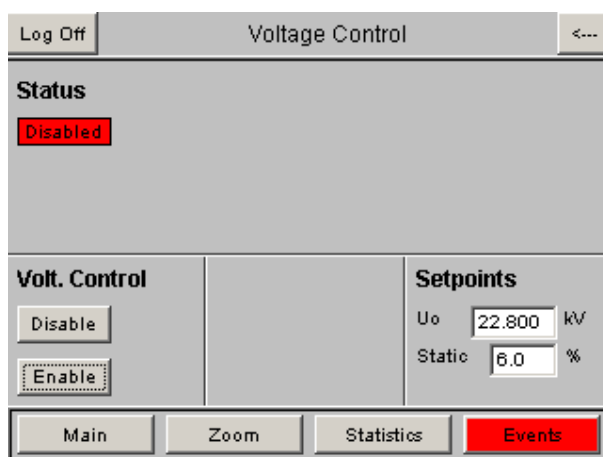


Figure 27 Grid voltage control screen

See [ABB] for a thorough treatment of this issue.

The setpoints for voltage control can be either set at screen Zoom/Converter/Voltage control, or at the Main/Parameters screen.

Switching voltage control on or off can be done while the turbine is connected to the grid and running.

To switch on grid voltage control, using the selected values of nominal grid voltage and static press the Volt. Control "On" button on the Zoom/Converter/Voltage control.

An indicator indicating that voltage control is active will lit when the command is accepted.

The Volt. Control "Off" button is used to switch voltage control off.

8.7. Limiting converter grid power

Grid power can be limited by a limit set on the Local Operator Panel. This feature can be used for peak shaving of excess power.

Mode	Status	Pref,max
☐ Pconst	Pact 4 kW	PLC 2200 kW
☒ Pn curve	Ngen 0.0 rpm	Conv. 2237 kW

P constant Pref,plc 0 kW	Pn curve Request 0 Actual 0	Alarms ☐ Power reduced
------------------------------------	--	---

Main
Zoom
Statistics
Events

Figure 28 Converter power control screen

A grid power limit can be adjusted when the turbine is online.

Setting a grid power limit is accomplished as follows:

- Go to screen Main/Overview.
- Check actual wind speed and actual grid power.
- If the turbine is on-line, the grid power limit needs to be decreased in small steps to avoid overspeeding the turbine.
- Go to screen Zoom/Converter/Power control.
- In the entry box $P_{refmax, PLC}$ enter a value for the new grid power limit, that is high enough to avoid overspeed. Depending on actual wind speed and grid power, lower the grid power limit for example with 100 kW.
- Go to screen Main/Overview.
- Wait until the new situation has settled and go back to Zoom/Converter/Converter .control to lower the limit if desired.
- Repeat this until the new setpoint is reached.

When the grid power limit is no longer desired, remove the limit as follows:

- Go to screen Zoom/Converter/Converter control.
- In the entry box $P_{refmax, PLC}$ enter default value 2200 kW.
- The grid power limit is now removed.

WARNING:

The power limiter can be adjusted while the turbine is running, but be careful to change the limit in small steps as it can be very dangerous to make large jumps in the limit set-point.

8.8. Monitoring the turbine from the Local Operator Panel

There are several reasons to monitor the turbine during running:

- 1) Check the events screen for unnoticed alarms.
- 2) Check the logbook for recent events (frequent trips etc.).
- 3) Check the emergency pitch report.
- 4) Collecting production information / operational statistics.

The Local Operator Panel provides the primary interface to monitor the turbine.

An overview of the turbine is shown on Main/Overview. This is used to get an overall impression of the turbine status. It shows turbine state, actual wind speed, corresponding rotor speed, blade angles for blade angles 1-3, and actual grid power.

From there on, details of the turbine can be viewed with the Zoom menu:

- Rotor.
- Generator.
- Nacelle.
- Converter.
- Grid connection.
- Weather station

8.9. Check the events screen for unnoticed alarms

If the turbine is running and the Events button on the button bar on the bottom of the touch screen is orange, there is an alarm active.

Check which alarm is active and look in Appendix C List of events how to solve the associated problem.

8.10. Check the logbook for recent events

The logbook can be found at screen Main/Logbook.

Log Off		Logbook		<---
Date	Time	ID	Logbook Entry	
05-08-26	09:13:28	236	Acceleration very high	<<
05-08-26	09:13:28	225	Work switch yaw motor	
05-08-26	09:13:28	224	Yaw motor MCB tripped	
05-08-26	09:13:28	211	Emergency nacelle	
05-08-26	09:13:28	202	Beacon day mode	
05-08-23	08:47:17	301	Emergency pitch test OK	>
05-08-23	08:47:15	5	Service switch on	
05-08-23	08:47:15	31	LV distribution MCB trip	
05-08-23	08:47:15	30	Ethernet switch flt	
05-08-23	08:47:15	25	LV distribution MCB flt.	>>
Main		Zoom		Statistics
				Events

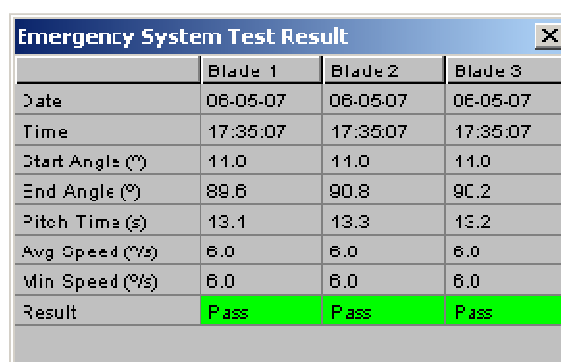
Figure 29 Logbook screen

The sequence of events shown in the logbook shows the most recent 100 events (trips, alarms and information messages) that have occurred, together with a timestamp (yy-mm-dd).

The frequency of occurrence or severance of trips and alarms is important information derived from the logbook, because it provides an indication of which components in the turbine need service/replacement.

8.11. Check the emergency pitch report

The results of the periodical emergency system check can be viewed at screen Zoom/Rotor/Test Emergency Trip.



	Blade 1	Blade 2	Blade 3
Date	06-05-07	06-05-07	06-05-07
Time	17:35:07	17:35:07	17:35:07
Start Angle (°)	11.0	11.0	11.0
End Angle (°)	89.6	90.8	90.2
Pitch Time (s)	13.1	13.3	13.2
Avg Speed (°/s)	6.0	6.0	6.0
Min Speed (°/s)	6.0	6.0	6.0
Result	Pass	Pass	Pass

Figure 30 Emergency system test report

When the "... " - button next to the TEST button is pressed, the most recent emergency pitch test report pops-up.

The actual blade speed during an emergency pitch test must exceed 5 deg/s. This condition must be met to pass the test successfully.

The result contains the date and time of the test, and for each blade the start angle, end angle, emergency pitch duration, average and minimum emergency pitch speed, and the result Pass/Fail of the test.

8.12. Collecting production information, availability information and operational statistics

Production data is shown on screens of the Statistics menu.

The following submenu's can be selected:

- Production log: kWh production (+kWh) and kWh consumption (-kWh), kVARh production (+kVARh) and kVARh consumption (-kVARh)
 - o Actual : today, this month, this year
 - o Daily : 31 days: production of 1 month buffered on daily basis
 - o Monthly : 13 months: production of 1 year buffered on monthly basis

8.13. Transferring operating & monitoring to a remote location



Operation of the turbine can be transferred to a remote location by switching the LOC / REM switch to REM. This will enable remote commands from the Garrad Hassan SCADA system and disable commands from the Local Operator Panel (with the exception of the hardwired pushbuttons TURBINE STOP, SERVICE).

Operation from the SCADA system is described in a separate manual [Garrad Hassan].

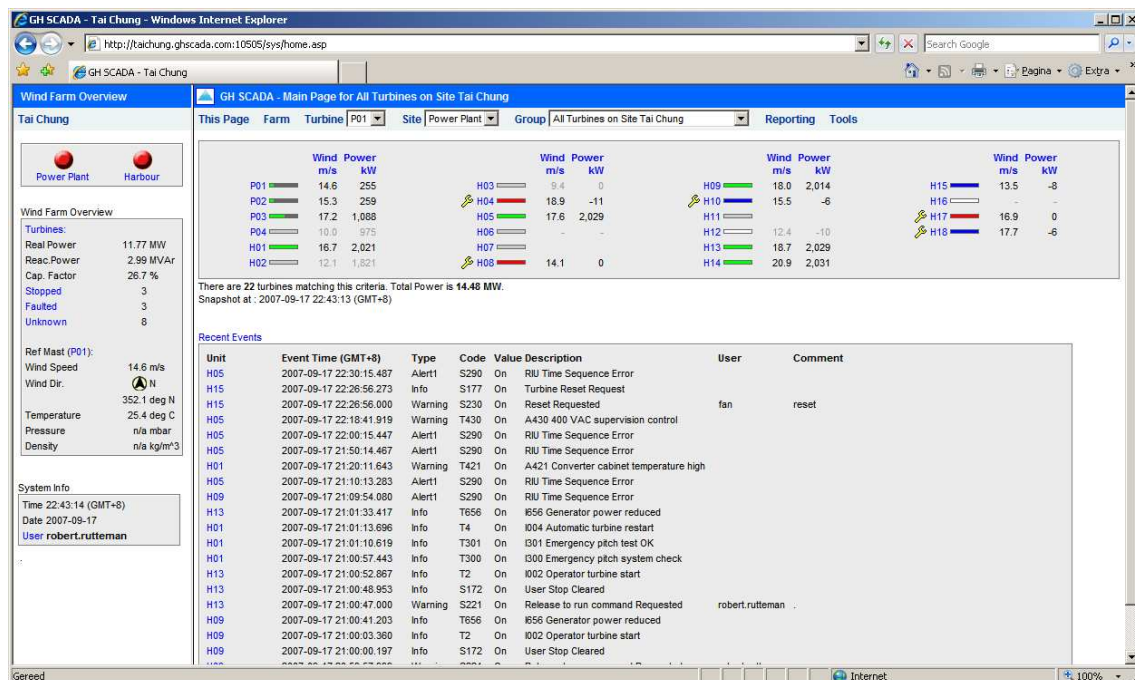


Figure 32 Garrad Hassan summary SCADA screen

Operation can be transferred back to the Local Operator Panel by switching the LOC/REM button back to LOC. This will disable commands from the SCADA system.

9. Service

This chapter describes how to use the turbine control system service features. Service procedures are not described here, but can be found in document [Z72-2000-2B-TD-24-03].

9.1. Shutting down the turbine and activating service mode

Service mode needs to be activated when working on the turbine. Activating service shuts down the turbine (by emergency pitching), and when the shutdown sequence is completed, switches off the ABB converter. All motors/actuators of the control system are switched off and all control modes are set to manual operation.

Service mode is also a prerequisite for certain service activities, like manual blade pitch angle adjustment. It is not possible to start the turbine if the service mode is set to ON.



To enter service mode:

- Set the LOCAL / REMOTE switch to LOC
- Manually stop the turbine with TURBINE STOP on the Local Operator Panel 542A001.
- The turbine shuts down, and when shutdown, switch Service switch on Local Operator Panel 542A001 to the "ON" position.
- The converter is switched off, blades go to end switches, yaw control is switched off, grease pumps are switched off.

NOTE:

Switching off the converter causes the converter MCB's to open.

NOTE:

During service mode, the manufacturer availability calculation (see previous section) assumes the turbine is available.

To leave service mode:

- Be sure that manual blade adjustment modes have been terminated properly (see section 10.11).
- Be sure that all people have left the hub
- Switch Service switch on the Local Operator Panel of 542A001 to the "OFF" position.

The blades should now go from the end switches to the standby position 86.0 deg.



9.2. Maintenance switches for yaw and emergency pitch batteries

Before doing any work on the yaw system, follow the following procedure:

- switch Service switch on Local Operator Panel 542A001 to the "ON" position
- switch off automatic yaw control by pressing STOP on the nacelle control box 540A001.
- switch maintenance switch YAW ON/OFF to "OFF" to switch off the supply to the yaw motors.



Before doing any work in the hub, switch off the emergency batteries:

- switch SERVICE switch on Local Operator Panel 542A001 to the "ON" position.
- switch SERVICE switch "ON" .

9.3. Power sockets for connection of service equipment

At several places 230 Vac/60 Hz sockets for connection of small hand tools like laptop PC's, portable lights etc. are present. Note that the usual European equipment used for service is based on 50Hz, although especially laptops normally have a switched power supply with a wide input range, that enable running laptops on this power.

- The converter container contains one sockets (in the leftmost cabinet).
- The nacelle cabinet: WCD1 three phase 400 VAC/60Hz socket on right hand side of cabinet) and the cabinet lighting contains a 230VAC/60Hz socket for power to for example a service laptop.

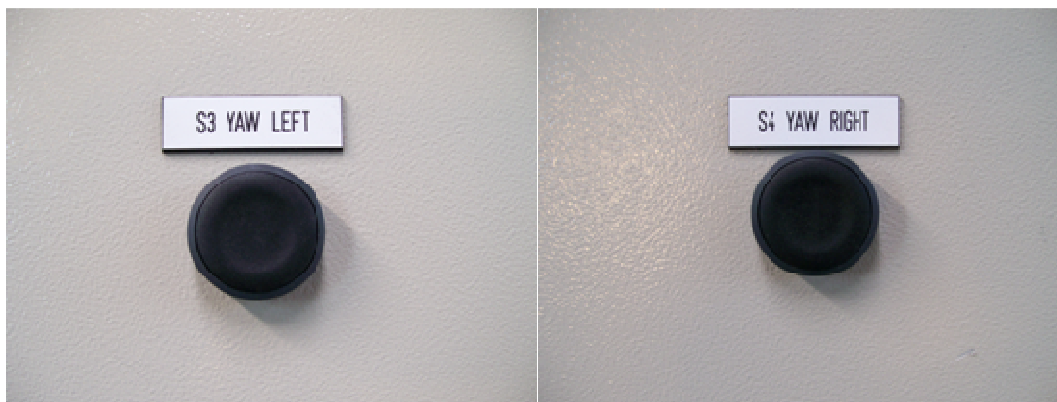
At several places in the switch gear room sockets might be installed. These sockets are outside the scope of Harakosan .

9.4. Local reset of the nacelle emergency system after disconnection of power

When power to the nacelle control system 540A001 is interrupted, the emergency system is tripped. This will cause all major power consumers (yaw system, hub, grease pumps etc.) in the nacelle and the hub to be disconnected from power.

When power has been reconnected, the nacelle emergency system can be reset inside the nacelle control box, at the right hand side of the cabinet, with the blue RESET button. This will re-apply power to the major power consumers in nacelle and hub.

9.5. Local manual yaw control from nacelle control box 540A001



The nacelle control box 540A001 contains a simple manual yaw control operator interface:

STOP Use STOP to stop automatic yawing.
YAW LEFT use YAW LEFT to yaw the nacelle counter clockwise.
YAW RIGHT Use YAW RIGHT to yaw the nacelle clockwise.

Note that the nacelle emergency system needs to be reset (see section 8.4) before manual yaw control is available !

9.6. Locking the rotor with the Local Hub Panel 541C001 and the locking device

For several service activities, it is required to work inside the hub. The hub is easily accessed through the main bearing.

WARNING:

Do not enter the hub when the indicator HUB ACCESS PROHIBITED on the Local Hub Panel is illuminated !

WARNING:

Before the hub is entered, the rotor should always be locked with the locking pin !

Locking of the rotor is done with the local hub panel 541C001 and a manually operated hydraulic locking pin.

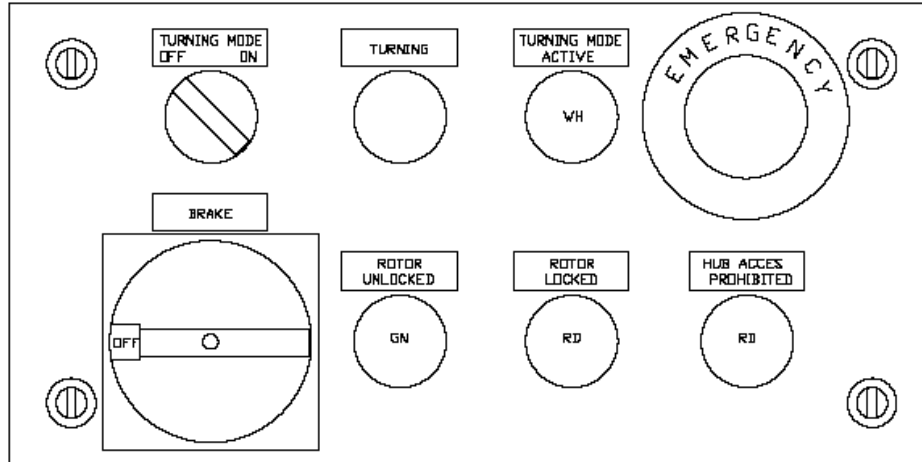


Figure 33 Local Hub Panel 540C001

The procedure to apply the locking pin is as follows:

- Check if the red indicator HUB ACCESS PROHIBITED is off. If the indicator is lit, the wind speed is too high, or the Service switch on the Tower Control Box is still OFF. In this case, entering the hub is not safe.
- Now turn TURNING MODE to "ON". In this mode, the converter will drive the generator as a motor. If the converter is accepting the turning mode, the indicator TURNING MODE ACTIVE will be lit.
- If the TURNING MODE ACTIVE indicator is lit, the rotor can be turned slowly by pressing the TURNING pushbutton.
- Turn slowly until the "^" sign at the rotor is opposite to the mark on the static ring.
- When the "^" sign is opposed to the mark on the static ring, pump in the hydraulic pin with the hand crank until the indicator ROTOR LOCKED is lit.
- Switch TURNING MODE "Off".

Alternatively, instead of using the Turning mode, it is possible to switch the service brake on and off to position the idling rotor until the locking pin can be inserted. This can only be used if the idling speed is suitable (not too low and not too high):

- Check if the red indicator HUB ACCESS PROHIBITED is off. If the indicator is lit, the wind speed is too high, or the Service switch on the Tower Control Box is still OFF. In this case, entering the hub is not safe.
- Switch on/off the rotor service brake with BRAKE until the rotor position is OK; when the brake is applied the rotor idling should stop and when the brake is released the idling continues.
- Release the brake, turn slowly until the "^" sign at the rotor is opposite to the mark on the static ring.
- Just before the "^" sign is opposed to the mark on the static ring, apply the brake. When the position is OK pump in the hydraulic pin with the hand crank until the indicator ROTOR LOCKED is lit, otherwise wait until another locking hole appears.
- When the rotor is locked, switch off the Service brake.

To retract the locked pin:

- Be sure that the blades are at the limit switches
- Leave the hub.
- Set the direction of the manual locking pin to "Unlock".
- Pump out the locking pin until the indicator ROTOR UNLOCKED is lit.

9.7. Connecting a laptop with Local Operator Panel in the nacelle

The Local Operator Panel touch screen program is available in a laptop version for use by service technicians. This Service Panel program runs on a laptop PC connected to the nacelle PLC.

To connect the laptop to the nacelle remote I/O PLC station:

- 1) Disconnect the Local Operator Panel touch screen in the Tower Control Box by disconnecting the Ethernet cable, to avoid IP conflicts.
- 2) Connect a normal Ethernet cable to the laptop PC's Ethernet connector and a free Ethernet connector on the EM203 module in the nacelle control box
- 3) Adjust the IP address of the laptop: Choose "Start", then "Settings", then "Control Panel", then "Network connections". Choose "Local Area Connection". Right mouse click on "Properties". At "This connection uses the following items", scroll down to "Internet Protocol (TCP/IP)". Click "Properties", choose "Adjust the following IP address".
Fill in at IP address: "192.0.0.*", where "*" is turbine ID
At subnet mask fill in "255.255.255.0"
At gateway fill in "192.0.0.224"

Choose "OK".

Choose "Close".

- 4) Run the Z72 HMI Service Panel program

Verify that the Z72 HMI Service Panel now shows actual instrumentation values on the screen.



Figure 34 Connecting a laptop in the nacelle

NOTE

FOR CONNECTION TO THE INTERNET, DO NOT FORGET TO SET YOUR LAPTOP IP SETTINGS BACK TO DHCP IP ASSIGNMENT AFTER USE OF THE SERVICE PANEL !

NOTE

DO NOT FORGET TO RECONNECT THE LOCAL OPERATOR PANEL IN THE TOWER BASE CONTROL BOX AFTER USE OF THE SERVICE PANEL !

9.8. Manual converter control

When the converter is in “manual” mode, it is possible to manually switch the converter on or off.

WARNING:

DO NOT SWITCH OFF THE CONVERTER MANUALLY WHILE THE TURBINE IS IN OPERATION. DUE TO THE PERMANENT EXCITATION OF THE ROTOR (PERMANENT MAGNETS), THE IGCT'S ARE EXPOSED TO HIGH VOLTAGES WHILE SWITCHED OFF. THIS CAN DESTROY IGCT'S !

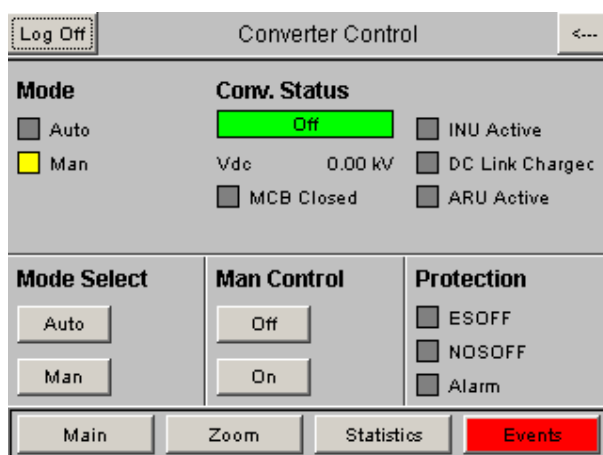


Figure 35 Converter control screen

To manually switch on the converter, follow the following procedure:

- Go to screen Zoom/Converter/Converter control.
- Select MAN mode.
- Press Man Control "On".
- The converter status on the screen should now show "Standby".
- The VM1 MCB of the converter should close with a loud noise.
- The following indicators should now lit: MCB closed, INU active, DC link charged, ARU active.

To manually switch off the converter, follow the following procedure:

- Stop the turbine.
- Go to screen Zoom/Converter/Converter control.
- Select MAN mode.
- The corresponding indication on the screen should be lit now.
- VERIFY THAT THE TURBINE'S SPEED IS BELOW < 2.5 RPM !
- Press Man Control "Off".
- The VM1 MCB of the converter should open with a loud noise.
- The converter state indicator should show the transition between STANDBY and OFF, and MCB closed, INU active should not be lit and ARU active should not be lit.

WARNING:

DO NOT SWITCH OFF THE CONVERTER MANUALLY WHILE THE TURBINE IS IN OPERATION. THIS CAN RESULT IN OVERSPEED OF THE ROTOR AND CONSEQUENTLY HIGH VOLTAGES ON THE CONVERTER COMPONENTS !

9.9. Converter Test

From the screen Zoom/Converter/Converter Test, different Tests on the converter can be selected:

- First switch on Service mode as explained in section 9.1.
- Switch the Test mode on with the ON button. The indicator TEST MODE ACTIVE should now be lit
- Choose the appropriate test by entering a test number in the entry box (see [ABB] for a table with different tests, currently 16 tests have been defined).

Figure 36 Converter test screen

9.10. Converter Service Reset

Some converter trips result in a start blocking of the converter. If this is the case, the indicator on screen Zoom/Converter/Converter Service Reset is lit. If the event causing the start blocking has disappeared, it is possible to reset this start blocking condition by pressing the RESET button.

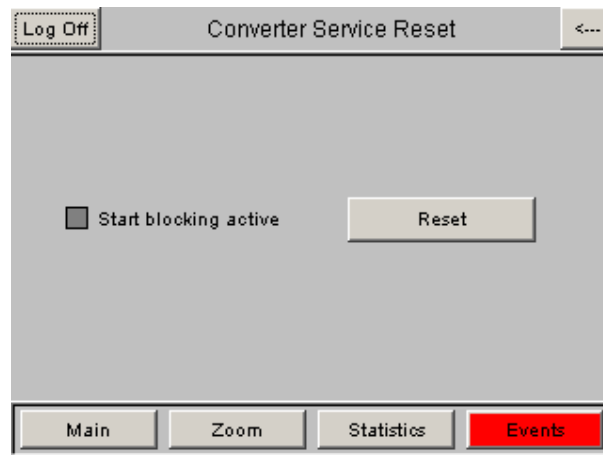


Figure 37 Converter Service Reset

9.11. Manual yaw control

Manual yaw control is done with the Local Operator Panel from screen Zoom/Nacelle/Yaw control. Manual yawing can be done when yaw control is not in state "YAW INHIBIT".

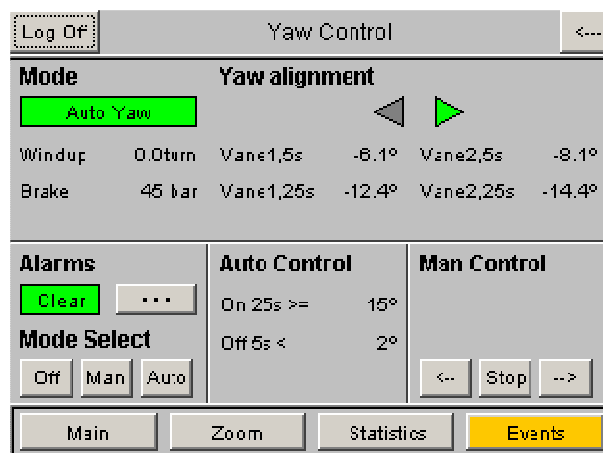


Figure 38 Yaw control screen

To manually yaw the nacelle, follow the following procedure:

- If control from the nacelle is required, connect a laptop PC running Service Panel as described in section 9.7 Connecting a laptop with Service Panel in the nacelle.
- Go to the yaw control screen by pressing Zoom/Nacelle/Yaw control.
- If auto yawing is active, select mode MAN. The yaw state should now become "Manual".
- To yaw counter clockwise, select Man Control "←".
- To stop yawing counter clockwise, select Man Control "Stop".

- To yaw clockwise, select Man Control "→".
- To stop yawing clockwise, select Man Control "Stop".
- Different indicators "yaw left" and "yaw right" on the nacelle screen indicate if the turbine is yawing counter clockwise or clockwise.
- When finished, restart auto yawing, by selecting mode AUTO.
- The yaw state should become "YAW AUTO".

With mode control "Off" it is possible to switch off yawing for service. The turbine should not be running when yaw control is switched off !

9.12. Manual hydraulic yaw brake control

Manual control of hydraulic yaw brakes is done with the Local Operator Panel from screen Zoom/Nacelle/Yaw brakes.

Manual control of yaw brakes is possible when yaw control is in state "YAW INHIBIT" or "YAW STANDBY".

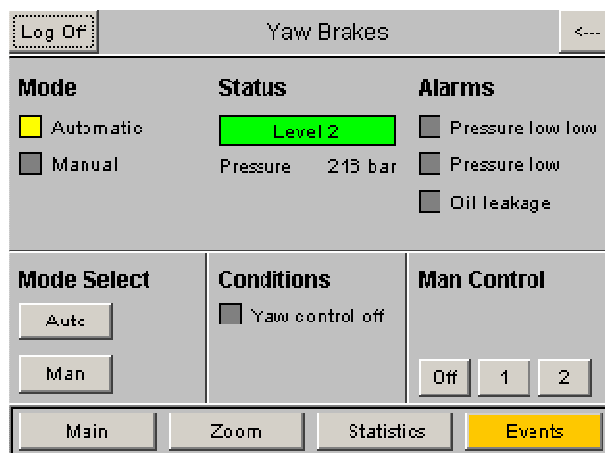


Figure 39 Yaw brakes screen

To manually yaw the nacelle, follow the following procedure:

- If control from the nacelle is required, connect a laptop PC running Service Panel as described in section 9.7. Connecting a laptop with Service Panel in the nacelle.
- Go to the yaw brakes screen by pressing Zoom/Nacelle/Yaw brakes.
- Verify that yaw control is off (yellow indicator is lit), otherwise go to screen Zoom/Nacelle/Yaw control and switch off yaw control.
- With Man Control Off/1/2, select 0 bar, 20 bar or 210 bar yaw brake pressure

9.13. Manual pitch control

WARNINGS:
DO NOT ENTER THE HUB BEFORE THE LOCKING PIN IS INSERTED.
BE AWARE THAT THE BLADES MOVE TO THEIR END BUFFERS WHEN THE SERVICE SWITCH IS ACTIVATED.
WHEN WORKING INSIDE THE HUB, ALWAYS BE AWARE OF WHICH BLADE IS MOVING.
DO NEVER STAND ON LOCATIONS NEAR THE BLADE PINIONS.
BE SURE THAT CLOTHES, FEET ETC. DO NOT GET STUCK IN THE PINIONS.
BE AWARE THAT AT ANY TIME DURING MANUAL MOVEMENT, THE EMERGENCY TRIP MECHANISM MAY BECOME ACTIVE, CAUSING SUDDEN MOVEMENT OF THE PITCH MECHANISM.

Blades can be pitched manually when the Service switch is switched On.

The modes can be selected from the Zoom/Rotor menu of the Service Panel only.

The blades can be pitched in one of three possible manual modes:

- Manual Pitch Control Open Loop – with this mode, the blades can be moved manually towards work position or towards vane position with a slow speed while the encoders do not need to be operational or calibrated.
- Manual Pitch Control Closed Loop – with this mode, the blades can be pitched manually to a configurable target blade angle. Encoders should be operational and calibrated in this control mode.
- Manual Pitch Ramping – with this mode a blade can be continuously ramped back and forth between two positions to run-in the blade bearing. Encoders should be operational and calibrated in this control mode.

9.13.1. Manual Pitch Control – Open Loop

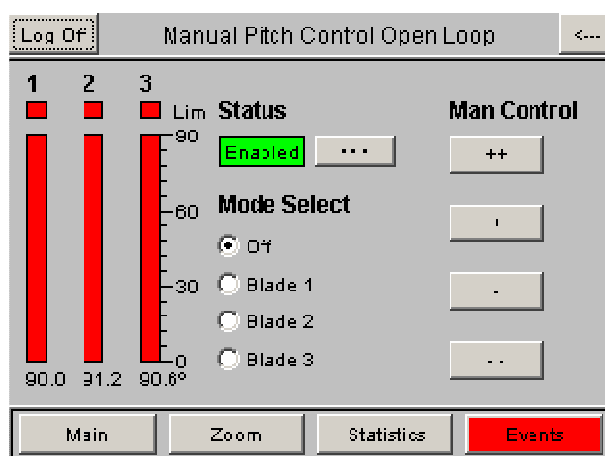


Figure 40 Manual Pitch Control – Open Loop screen

To manually control the pitch of a blade in open loop mode, follow this procedure:

- Switch on the Service switch on the Local Operator Panel.
- Connect a laptop PC running the Service Panel as described in section 9.7 Connecting a laptop with Service Panel in the nacelle.
- Verify that the Status box now shows "Enabled", otherwise click on "...".
- Select one of the buttons "Blade 1", "Blade 2", or "Blade 3" and check that the status shows "active".
- The rotor state changes to MAN OPEN LOOP.
- The active blade can now be moved towards working position ("-" : very slowly; "- -" : slowly) or towards vane position ("+" : very slowly; "+ +" : slowly).
- When the vane limit switch is active, any movement towards vane position is blocked. Between the servo vane position (default 86.0 deg) and the limit switch the "+ +" button is blocked; in this region, it is only possible to move very slowly with the "+" button.

To quit Manual Pitch Control – Open Loop, or to select another blade for pitching, first bring back all blades to the vane position switch. When the blades are at the vane position switch, the small rectangle above each blade with text "Limit" will become red coloured. Only then, this mode can be quitted, or another blade can be selected for pitching.

9.13.2. Manual Pitch Control – Closed Loop

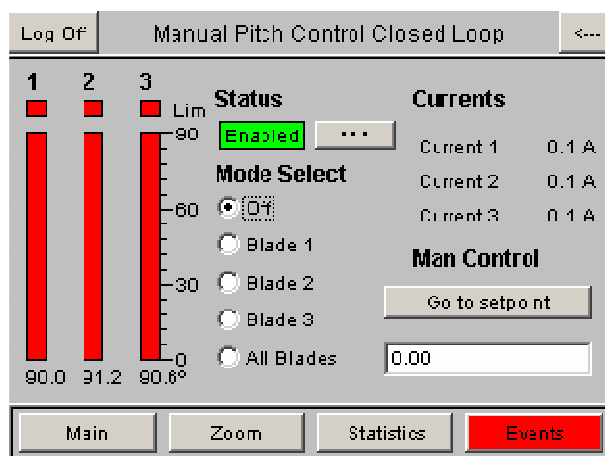


Figure 41 Manual Pitch Control – Closed Loop screen

This mode is used to manually pitch an individual blade, or all blades simultaneously, to an adjustable target blade angle. An additional condition for this mode is that the locking pin of the rotor must be applied.

To manually control the pitch of a blade in closed loop mode, follow the following procedure:

- Switch on the Service switch on the Local Operator Panel.

- Connect a laptop PC running the Service Panel as described in section 9.7 Connecting a laptop with Service Panel in the nacelle.
- Verify that the Status box now shows "Enabled", otherwise click on "...".
- The rotor should be locked.
- Select one of the buttons "Blade 1", "Blade 2", "Blade 3" or "All blades" and check that the status shows "active".
- The rotor state changes to MAN CLOSED LOOP.
- The selected blade or blades can now be moved by entering a target set-point. This is done by touching the entry window, and by entering a suitable value.
- By pressing "Go to set-point", the blade(s) will move to the set-point with a slow rate.
- When the blade(s) is(are) at the target angle, a new target can be selected.

By selecting "Off", the blade(s) will move towards vane servo position, and from there on, will move with a very slow rate until the vane switch(es) is (are) reached.

It is possible to select another blade from any position of the blades (in contrast to the open loop control).

9.13.3. Manual Pitch Control – Pitch ramping

This mode is used to automatically ramp one blade between an upper and a lower position. This mode is used to run in pitch bearings, or to determine if pitch bearings run smoothly. A condition for this mode is that the locking pin of the rotor must be applied.

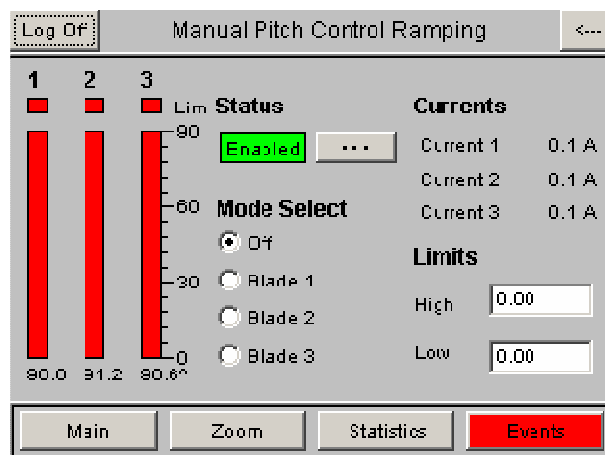


Figure 42 Manual Pitch Control Ramping screen

To use this pitch ramping mode, follow the following procedure:

- Switch on the Service switch on the Local Operator Panel.

- Connect a laptop PC running the Service Panel as described in section 9.7 Connecting a laptop with Service Panel in the nacelle.
- Verify that the Status box now shows "Enabled", otherwise click on "... " and check the conditions.
- The rotor should be locked.
- Enter the desired upper blade position of the ramp in the entry box. Values from lower limit up to 86 deg can be entered.
- Enter the desired lower blade position of the ramp in the entry box. Values from upper limit down to -2 deg can be entered.
- Select one of the buttons "Blade 1", "Blade 2", "Blade 3" and check that the status shows "active".
- The rotor state changes to MAN PITCH RAMP.
- The blade will now ramp up and down between the upper and the lower position with a slow rate.
- Select "Off" to stop the ramping
- Select another blade, if desired.

By selecting "Off", the blade will move towards vane servo position, and from there on, will move with a very slow rate until the vane switch is reached.

9.14. Manual grease pump operation

In total the turbine contains four grease pumps:

- Main bearing grease pump, screen Zoom/Generator/Main Bearing Grease System.
- Yaw bearing grease pump, screen Zoom/Nacelle/Yaw Bearing Grease System.
- Pitch bearing grease pump, screen Zoom/Rotor/Pitch Bearing Grease System.
- Pitch gear grease pump, screen Zoom/Rotor/Pitch Gear Grease System.

The grease pumps all have a similar screen on the Local Operator Panel.

Main Bearing Grease System		
Log Off	<---	
Mode	Status	Alarms
☒ Automatic ☐ Manual	☐ Pump On	☐ Tank Level Low Reset
Mode Select	Auto Control	Man Control
Auto Man	Interval 332 min Dutytime ' min	Off On
Main	Zoom	Statistics Events

Figure 43 Main bearing grease pump screen

Each of these grease pumps can be operated in automatic and in manual mode. By default, the pumps work in automatic mode.

In manual mode, the pump can be switched on and off manually.

In automatic mode, the pump is switched on after a preset interval time has expired (and depending on the pump also a condition to determine if greasing is required: is the turbine running; or, is auto yaw active). The pump is active for a preset number of piston cycles, or a preset time, corresponding to a preset fixed dose of grease per interval time. The values of these parameters that are used in the automatic mode can be entered from the Parameters screen (see section 11.2), and are shown on the "Auto control" section of the grease system screens.

To manually operate the grease pump, follow the following procedure:

- Go to the appropriate screen.
- If automatic mode is active, select mode MAN. The mode should now become "Manual".
- To manually switch the pump on, press Man Select "On".
- The Status indicator "Pump On" should now be lit.
- If the pump is equipped with a piston detector, the corresponding indicator should be lit with a frequency of about once per minute.
- The pump can be switched off by pressing Man Select "Off".
- After manual operation, select automatic mode with Mode Select "Auto".

Should the pump have an alarm, as indicated by the level low or flow low alarm indicator, then the alarm can be reset with the Reset button on the grease pump screen. A pump that has an alarm will be stopped. Resuming grease pump operation is only possible after the operator has reset the alarm.

9.15. Manual cooling fan operation

Operation of the cooling fan is done from screen Zoom/Nacelle/Cooling fan.

Nacelle Cooling Fan		
Log Off	<---	
Mode	Status	Alarms
☒ Automatic	☒ Fan On	☐ Sensor Fault
☐ Manual	Temperatur82.9 °C	
Mode Select	Auto Control	Man Control
Auto	Fan On >= 35.0 °C	Off
Man	Fan Off < 30.0 °C	On
Main	Zoom	Statistics
Events		

Figure 44 Nacelle cooling fan screen

The cooling fan can be operated in automatic and in manual mode. By default, the fan works in automatic mode.

In manual mode, the fan can be switched on and off manually.

In automatic mode, the fan is switched on after a preset generator bearing temperature threshold has been exceeded. The pump is active until the temperature has decreased 5 °C below the setpoint. The values of these parameters that are used in the automatic mode can be entered from the Parameters screen (see section 11.2), and are shown on the "Auto control" section of the cooling screen.

To manually operate the cooling fan, follow the following procedure:

- Go to the screen Zoom/Nacelle/Cooling Fan.
- If automatic mode is active, select mode MAN. The mode should now become "Manual".
- To manually switch the fan on, press Man Control "On".
- The Status indicator "Fan On" should now be lit.
- The fan can be switched off by pressing Man Control "Off".
- After manual operation, select automatic mode with Mode Select "Auto".

Should the fan have an alarm, as indicated by the alarm indicator, then the alarm is reset automatically when the alarm condition disappears.

A fan that has an alarm will be stopped. Resuming fan operation is only possible after the alarm condition has disappeared and the operator has reset the alarm.

9.16. Restarting the turbine after Service

After service, restore all settings in the nacelle and hub

Before descending, be sure that

- The rotor control box MAIN SWITCH is switched on.
- The rotor control box SERVICE SWITCH is switched off.
- The turning mode is switched off.
- The TURNING MODE ACTIVE indicator is not lit.
- The ROTOR UNLOCKED indicator is lit.
- Service brake is switched off.
- The nacelle control box MAIN SWITCH is switched on.
- The YAW ON/OFF maintenance switch is switched on.
- The nacelle emergency system is reset.

If these conditions are fulfilled, the START ENABLED indicator on the nacelle control box 540A001 is lit, indicating that all turbine start conditions related to nacelle and rotor are met.

NOTE:

Before descending the tower, be sure that the START ENABLED indicator on the nacelle control box is illuminated. This indicates that all turbine start conditions related to nacelle and hub are fulfilled.

When the turbine has been started, switch the LOCAL/REMOTE operation selector back to REM before leaving the turbine. This will enable control from SCADA when crew has left.

10. Troubleshooting

This chapter provides instructions on how to proceed when the control system is not available, the turbine has tripped, or the turbine has an active alarm.

10.1. Safety instructions

Note: before starting work on the turbine control system, the general safety instructions in chapter 4 must be read and understood.

10.2. Standard procedure for troubleshooting

- If the Local Operator Panel is out of operation (screen is off-line, the values on the screen are not refreshed or the touch screen is black) see section 10.3.
- If the Local Operator Panel touch screen is on-line, refer to Appendix C for explanation of the fault messages on the Local Operator Panel and for instructions on fault finding and troubleshooting.
- Do not reset the fault, unless the reason has been located !
- Identify the fault.
- Try to rectify the fault.

10.3. What to do when the Local Operator Panel touch screen is out of operation

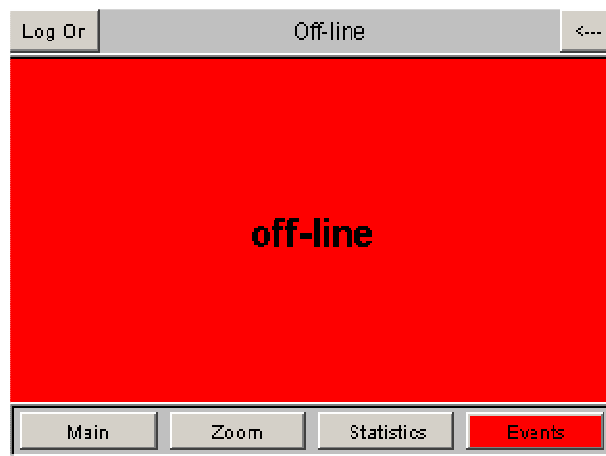


Figure 45 Local Operator Panel touch screen off-line

When the Local Operator Panel touch screen stays black, please check if the 128 MByte FlashCard with the appropriate software is inserted in the back of the Local Operator Panel touch screen, or refer to [Bachmann Web-terminal].

When the Local Operator Panel touch screen shows the Off-line screen, please check the following:

- The turbine ID of the Web-terminal and the turbine ID of the Bachmann PLC of the Main turbine control system 542A001 (as can be adjusted with the rotary switches on the M1 module of the Bachmann PLC and the back of the Web-terminal), as described in section 11.1 are identical.
- The Ethernet cable between Fibre optic Ethernet switch and Bachmann Web-terminal WT305 of 542A001 is connected.
- Check if the green RUN LED of the MP213/E module of the Main turbine control system PLC is running.
- Connect a laptop PC to the Moxa switch, configure it's IP address to 10.128.99.Turbine ID, subnet mask 255.0.0.0, try to ping the Webterminal at 10.128.2.Turbine ID (at some sites use 10.129.99.turbineID and 10.129.2.TurbineID)

NOTE	
Sites consisting of more than two sub-parks use 10.129.DeviceID.TurbineID for the second sub-park !	

MP213/E -LED's	
Green LED RUN	Booting finished successfully and ready
Yellow LED INIT	Booting (about 60s)
Red LED ERR	Booting terminated because of failure (see console messages)

If both the INIT and de ERR LED are active: debug mode, waiting for instruction.

When the values on the Local Operator Panel touch screen are not refreshed, please cycle power to the touch screen by temporarily removing the 3A fuse labeled XF4 of the Main turbine control system 542A001.

11. Configuration of the turbine control system

During the commissioning of the turbine, the turbine control system is configured by Harakosan. There are circumstances where changes to this configuration are required or desired. This chapter describes how to configure the turbine control system in such cases. This might be necessary for example after exchange of a PLC, after changing blade encoders on the pitch motors, or on request of the grid operator.

11.1. Configuring the turbine ID

Each turbine has a unique turbine ID, to be able to distinguish between different wind turbines in a wind park.



Figure 46 MX213 Rotary switches, here set at Turbine ID=1

The turbine ID is set with the H (high order) and L (low order) hexadecimal rotary switches on the MX213 module of the main PLC and also on the rear side of the Local Operator Panel touch screen. A maximum of 64 = 40H addresses can be configured.

MX213 and WT305 – rotary switches	
Rotary switch – L	Low order byte setting of turbine ID address (L = 0 ..F)
Rotary switch – H	High order byte setting of turbine's ID (H = 0 = RUN, 1 = D, E = TEST, F = PROG)

The IP address of the main PLC will be:

$$IP \text{ address PLC} = 10.128.1.(Turbine \text{ ID})$$

The IP address of the Local Operator Panel touch screen will be:

$$IP \text{ address touch screen} = 10.128.2.(Turbine \text{ ID})$$

The factory settings of the rotary switches are H=0, L=1, so the factory default Turbine ID setting of PLC and Local Operator Panel touch screen is 1 (default IP address PLC = 10.128.1.1, default IP address touch screen = 10.128.2.1).



Figure 47 Local Operator Panel touch screen rotary switches

NOTE

Sites consisting of more than two sub-parks use 10.129.DeviceID.TurbineID for the second sub-park !

The turbine ID adjustment on the back of the Local Operator Panel touch screen should of course correspond to the turbine ID adjustment on the MX213 module.

WARNING:

WHEN THE TURBINE ID ON THE LOCAL OPERATOR PANEL DOES NOT CORRESPOND WITH THE ID ON THE MX213, THEN THE LOCAL OPERATOR PANEL IS CONNECTED TO A DIFFERENT TURBINE AT ANOTHER LOCATION IN THE PARK ! THERE WILL BE COMMUNICATION CONFLICTS WITH A LOCAL OPERATOR PANEL WITH IDENTICAL SETTINGS !

11.2. Adjusting parameters

A list of parameters that can be configured is shown in Appendix B, together with their default values.

Parameters can be viewed and modified at Local Operator Panel screen Main/Parameters.

Log Off	Parameters		<---
Parameter	Value	Unit	
Rotor nominal speed	22.50	rpm	<<
Rotor overspeed setpoint	26.50	rpm	
Pitch standby position	80.00	deg	<
Pitch startup position	30.00	deg	
Blade limit switch 1 position	89.80	deg	>
Blade limit switch 2 position	91.10	deg	
Blade limit switch 3 position	90.30	deg	>>
Blade work position	-1.00	deg	
Wind direction offset	0.00	deg	
Cooling fan setpoint	35.00	°C	
PLC status	OK		
Main Zoom Statistics Events			

Figure 48 Parameters screen

To change a parameter, it is required to log on (Default Password = "0 0 0 0"). When logged on, pointing at a parameter will pop-up an entry window, where the parameter value can be entered. The entry window will only accept the value if it is inside the safe range shown on the pop-up.

Rotor nominal speed (rpm)

Old: 22.50

Max: 30.00

23.50

Min: 10.00

7

8

9

<--

-->

4

5

6

Backspace

1

2

3

Cancel

0

.

+/-

OK

Figure 49 Parameter entry pop-up screen

11.3. Auto Restart Settings

In general, the turbine is stopped after a severe fault (trip). The operator needs to restart the turbine after each trip.

Some trips can occur more often and are known to be not harmful (nuisance trips). Examples are a loose wire in a generator Pt100, the locking pin position sensor, that takes some time to fix or yaw time-outs at locations where the wind direction can change rapidly in a short time.

Log Off	Auto Restart Settings		<---
ID	Description	#/hr	
210	PLC DI0248 card 22 fault	0	<<
217	card 17 CAN kernel fit	0	
219	Turning mode switched on	0	<
220	Service brake active	0	
221	Rotor not unlocked	0	>
222	Locking pin fit	0	
223	Yaw timeout	2	>>
224	Yaw motor MCB tripped	0	
225	Work switch yaw motor	0	
226	Cable twist limit switch	0	
Main Zoom Statistics Events			

Figure 50 Auto restart settings screen, "Yaw time" restart setting adjusted

To avoid loss of availability, the control system can be configured to restart automatically after trips occurred that are known not to be harmful:

- Go to screen Zoom/Auto Restart Settings
- Enter the maximum allowed number of restarts / hour for the nuisance trip

The standard auto restart setting of each trip is 0

NOTE:
DO NOT FORGET TO RESET THE AUTO RESTART SETTING TO 0 AFTER THE PROBLEM HAS BEEN FIXED !

11.4. Setting the actual date / time

The control system contains a real time clock, used for event time stamping and daily, monthly and yearly availability logging. The date and time must be adjusted, for example after downloading software, or to account for daylight saving time, with the Bachmann M-Manager program.

11.5. Setting the values of the blade limit switch positions

The positions of the blade limit switches differ per turbine. When the blades are in the end buffers (after entering Service mode), read in the positions of the blades and enter the values in the Parameters screen.

11.6. Setting the wind direction offset

Determining the value of the wind direction offset is accomplished as follows:

- Read the average absolute wind direction on Zoom/Weather Station (be sure that the Parameter "Wind direction offset" is set to 0.0).

- Estimate the actual wind direction using a compass.
- Adjust the difference between the actual wind direction and the measured wind direction – measured in deg. - on the parameter "Wind direction offset", on screen Main/Parameters (For example: if measured wind direction is "NE", and value on screen is "W", then adjust an offset of 135 deg.).
- Verify now on the local operator panel screen Zoom/Nacelle/Weather station that the wind direction shown corresponds to the actual absolute wind direction.

NOTE:
THE VALUE OF THE WIND DIRECTION ON THE SCREEN NACELLE/WEATHER-STATION IS NOT CORRECT WHEN THE WIND DIRECTION OFFSET HAS NOT BEEN SET.

11.7. Selecting the active P-n curve

The Pn table defines the relationship between actual rotor speed and generator power, and is an important parameter that influences the power production of the turbine.

With the P-n curve, the partial load behaviour and the maximum power can be configured. In total one of 22 different Pn curves with a resolution of 20 points can be selected:

Curve Nr.	Curve Type	Tip-speed ratio (TSR)	Const."slip" [%]	Source	Pn / Pconst	Extended pitch
0 – standard curve	Static	8.5	25	ABB	Pn	Possible
1 – low slip	Static	8.5	5	ABB	Pn	Possible
2 – low slip, lower TSR	Static	8.25	5	ABB	Pn	Possible
3 – low slip, lower TSR	Static	8.0	5	ABB	Pn	Possible
4 – low slip, lower TSR	Static	7.75	5	ABB	Pn	Possible
5 – higher slip	Static	8.5	10	ABB	Pn	Possible
6 – higher slip, lower TSR	Static	8.25	10	ABB	Pn	Possible
7 – higher slip, lower TSR	Static	8.0	10	ABB	Pn	Possible
8 – higher slip, lower TSR	Static	7.75	10	ABB	Pn	Possible
9 – low slip	Static	8.5	5	ABB	Pn	Possible
10 – very low slip	Static	8.5	2	ABB	Pn	Possible
11 – as 1 from PLC	Static	8.5	5	PLC	Pconst	Possible
12 – as 2 from PLC	Static	8.25	5	PLC	Pconst	Possible
13 – as 3 from PLC	Static	8.0	5	PLC	Pconst	Possible
14 – as 4 from PLC	Static	7.75	5	PLC	Pconst	Possible
15 – as 5 from PLC	Static	8.5	10	PLC	Pconst	Possible
16 – as 6 from PLC	Static	8.25	10	PLC	Pconst	Possible
17 – as 7 from PLC	Static	8.0	10	PLC	Pconst	Possible

18 – as 8 from PLC	Static	7.75	10	PLC	Pconst	Possible
19 – as 9 from PLC	Static	8.5	5	PLC	Pconst	Possible
20 – as 10 from PLC	Static	8.5	2	PLC	Pconst	Possible
21 – as 0 from PLC	Static	8.5	25	PLC	Pconst	Possible
22 – ECN STX II	Dynamic	ECN	Dynamic	PLC	Pconst	Required

Tables 0-10 are predefined static Pn tables, and are stored inside the ABB ACS1000. Selecting a curve 0-10 will automatically set the converter to so called Pn mode. In this mode, the setpoint for the power is generated inside the converter, based on the Pn Curves 0-10 that are stored inside the converter.

Tables 11-21 are the predefined static Pn tables as 0-10, but now stored inside the Main Turbine Control System. Selecting a curve 11-21 will automatically set the converter to so called Pconst mode. In this mode, the setpoint for the power is generated inside the Main Turbine Control System, based on the Pn Curves 11-21 that are stored inside the PLC.

The tables 0-21 work in combination with either the extended or the basic control algorithm.

Power table 22 is the default 2MW power table (see Appendix B). This table uses a dynamic Pn table in combination with the so-called extended pitch algorithm.

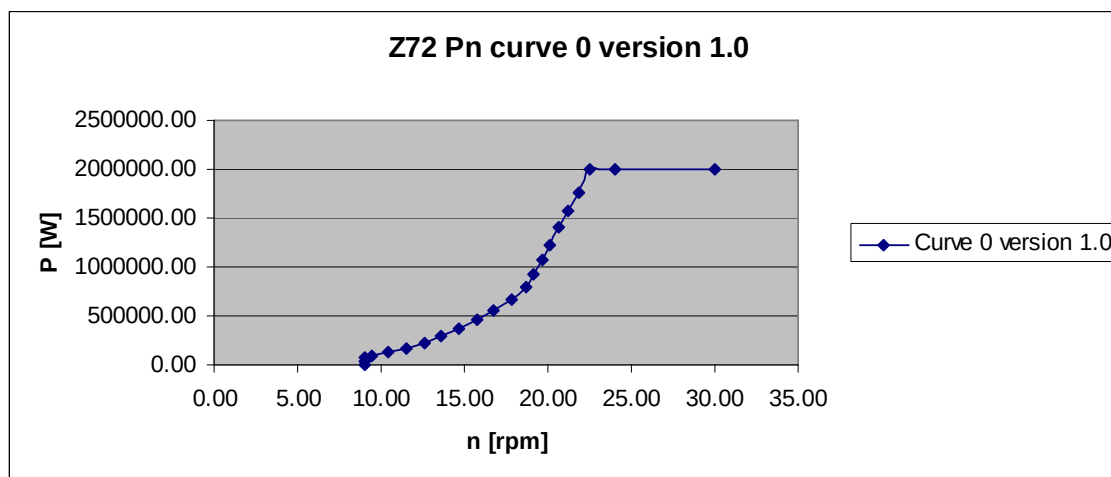


Figure 51 Example P-n curve 2000 kW

Selecting a different P-n curve is accomplished as follows:

- Manually stop the turbine.
- Switch the service switch to the "On" position. This will cause the converter to go to the "OFF" state (converter MCB will be opened).
- Go to screen Main/Parameters
- Set Parameter "Pn power-speed table" to the number of the desired static Pn table

- Check on screen Zoom/Converter/Power Control that the actual table corresponds to the selected table

NOTE:

WHEN TABLE 22 IS SELECTED, EXTENDED MODE IS NOT CHANGED AUTOMATICALLY TO 1. DO NOT FORGET TO SET EXTENDED PITCH MODE TO 1 !

11.8. Using the Basic control algorithm (Parameter Extended Mode = 0): setting static Pn tables (0-21), nominal rotor speed and a separate power limit set-point

The basic algorithm is based on 3 separate settings: the nominal rotor speed, the Pn table number (0-21) and the PLC power limiter set-point.

Rotor nominal speed or PLC power limiter set-point can be adjusted as described in section 11.2 and section 8.7.

The Pn table can be adjusted as described in section 11.7.

When a curve 0-21 is selected the rotor speed and the power limit value remain uncoupled.

With the power limiter, an absolute generator power limit can be adjusted. This feature can be used for peak shaving of excess power (for example to limit output power to 1500 kW).

11.9. Using the Extended control algorithm (Parameter Extended Mode = 1): setting the nominal rotor speed and the power limit simultaneously

In the basic algorithm, rotor speed is controlled by means of the blade pitch angle in the full load range, and by using so called static Pn curves in the partial load range.

By default, the control algorithm is set to the extended algorithm.

In this algorithm, the pitch control uses a more advanced algorithm. In addition to this, a dynamic Pn curve provides the set-point to the converter.

The extended pitch algorithm keeps rotor speed more constant and provides more energy capture; the dynamic Pn curve provides more output power as well and uses converter output power as an additional control variable for rotor speed control in the full load range.

The dynamic Pn curve is selected with Pn curve 22. It is also possible to run the extended pitch algorithm with static Pn curves by simultaneously selecting extended mode and a Pn curve 1-20. In this case, only the benefits of extended pitch control are used; the dynamic Pn curve are not used in this case.

When the extended algorithm is active and curve 22 is selected the nominal rotor speed value and the power limit value are coupled: selecting one of the two will influence also the other. By doing so, when adjusting a power limit of for example 1 MW, the nominal

rotor speed will automatically be adjusted to a lower value. This will result in a decrease of the noise produced from the turbine.

When a curve 0-21 is selected the rotor speed and the power limit value remain uncoupled.

Switching on the extended control algorithm is accomplished as follows:

- Manually stop the turbine.
- Switch the service switch to the "On" position. This will cause the converter to go to the "OFF" state (converter MCB will be opened).
- Go to screen Main/Parameters
- Make sure the converter is switched off.
- Set Parameter "Extended pitch control" to "1".
- Switch the Service switch on the Local Operator Panel to the "Off" position and press RESET.

Switching to the basic control algorithm is accomplished as follows:

- Manually stop the turbine.
- Switch the service switch to the "On" position. This will cause the converter to go to the "OFF" state (converter MCB will be opened).
- Go to screen Main/Parameters
- Make sure the converter is switched off.
- Set Parameter "Extended pitch control" to "0".
- Set Parameter "Pn power-speed table" to the number of the desired static Pn table (0-21)
- Switch the Service switch on the Local Operator Panel to the "Off" position and press RESET.

Rotor nominal speed or PLC power limiter set-point can be adjusted as described in section 11.2 and section 8.7. The Pn table can be adjusted as described in section 11.7.

12. Reference list

[ABB]

– User's manual of ABB ACS1000 Power Converter:
Luxe User's Manual 3BHS209619 ZAB E01, Rev. A, ABB, Turgi, Switzerland.

[Bachmann PLC]

– Bachmann Operating Manual, Revision 9, May 2006, D-M1-OP-MANUAL/09E

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Web-Terminal WT305e Technical Description (General), Revision 02, Dezember 2004, D-WT-TB-305/02E.

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User Guide Midi-Maestro Maxi-Maestro drives Variable Speed drives for permanent-magnet DC servo motors 1kW, 2kW and 5 kW. Part Number: 0437-0006-04, issue nr. 4.

[Garrad Hassan]

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Generic wind farm supervisory control and data acquisition system User's Manual Software Release v 2.1 Oct 2006, Garrad Hassan and Partners Limited, Bristol, UK.

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Frequenzmess- und Schaltgeraete Typenreihe FT100. Betriebsanweisung 830D-36299 / 5419, 2 Februar 1999, Jaquet AG, Basel, Switzerland.

[Orga]

– User's manual of the obstruction lights controller:
Medium Intensity Strobe Light System Controller CIP100 Installation and Operation Manual. Revision 0. ORGA B.V., Schiedam, Holland.

[PCH]

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Digital Wind Turbine Structural Vibration Monitor Type PCH1026, A member of the FlexMON family. Jan, 2001. User Manual Version CHF2026_UK12.

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– User's manual of the hydraulic power unit:
Installation and Maintenance Manual Hydraulic Power Unit for operating Rotor- and Yaw Brakes Harakosan Windturbine Type Z72 – Rev. 2, edition WU-02/01-02-2007.

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KFG, KFGS Pump Unit – Operating Instructions, Willy Vogel AG.

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[Z72-2000-2B-TD-18-01-R03]

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[Z72-2000-2B-TD-18-04-R03]

Nacelle control box 540A001 electrical diagram, Harakosan Europe B.V.

[Z72-2000-2B-TD-18-10-R03]

Rotor control box 541A001 electrical diagram, Harakosan Europe B.V.

[Z72-2000-2B-TD-18-14-R03]

Main turbine control system 542A001 electrical diagram, Harakosan Europe B.V.

Appendix A. Technical data Z72 2MW wind turbine

General	
Rotor diameter	70.65 m
Rotor speed	Variable, nominal 22.5 rpm
Nominal power	2.0 MW*
Transmission	Direct drive generator, single main bearing
Rated wind speed	13 m/s
Cut-in / cut-out wind speed	3-25 m/s
Survival wind speed	70 m/s
Rotor speed control	Blade pitch
Wind class	2 and S according to IEC61400-1
Generator mass	49 tons
Rotor mass	36 tons
Nacelle mass	12 tons

* For noise critical sites nominal power can be reduced to an output level of 1.5 MW. This corresponds to a nominal rotational speed of 18 rpm

Power conversion	
Generator	ABB multi-pole, 4kV, permanent magnet, synchronous machine
Protection	IP54
Insulation class	F
Converter	ABB AC-DC-AC full size converter
Power factor control	Fully adjustable, with active grid support (capacitive and inductive)
Power ride-through	Converter can remains grid connected during voltage dips of up to 200 ms
Tower	
Type	Tubular steel
Hub heights	65.0 m (100 tons) or 80.0 m (150 tons)
Safety	
Power/speed regulation	Blade pitch control (3 safety systems)
Safety aspects	3 independent pitch drive units

Appendix B. Default parameter settings

Default parameter settings. Parameters are accessible via Local Operator Panel Main menu.

Parameter	Value	Unit
Rotor nominal speed	22.5	rpm
Rotor overspeed setpoint	26.5	rpm
Pitch standby position	86.0	deg
Pitch start-up position	30.0	deg
Blade limit switch 1 position	Adjust	deg
Blade limit switch 2 position	Adjust	deg
Blade limit switch 3 position	Adjust	deg
Blade work position	-1.0	deg
Wind direction offset	Adjust	deg
Cooling fan setpoint	35.0	Deg C
PLC power limiter setpoint	2200	kW
Local nominal grid voltage	22.8	kV
Local nominal grid static	6	%
Main bearing grease pump int.	332	Min.
Main bearing grease pump on	1	Min.
Yaw bearing grease pump int.	1367	Min.
Yaw bearing grease pump cycl.	3	cycles
Pitch bearing grease pump int.	240	Min.
Pitch bearing grease pump cycl.	3	cycles
Pitch gear grease pump on.	3	sec
Pn power-speed table	22	
Extended pitch control	1	
Enable cos(phi) control	1	

Default hardware over speed protection and accelerometer protection settings: these parameters are programmed and not accessible through the Local Operator Panel.

Parameter	Value	Unit
Hardware over speed trip set-point	28.5	rpm
Tower acceleration trip set-point	1.75	m/s ²

Appendix C. List of events

The following list shows all possible events, their meaning, and possible hints for recovery.

The following abbreviations are used:

- T1: Turbine trip by normal servo control system. This an event that indicates a severe malfunction and causes the control system to shut down the turbine by means of the normal servo control system. If the event disappears without operator intervention, the turbine can restart automatically, unless a preset maximum number of identical trips / hour has been reached.
- T2: Turbine trip by emergency pitch system. This an event that indicates a severe malfunction (probably inside the normal servo control system), or a failing software overspeed protection and causes the control system to shut down the turbine by means of the emergency pitch system. If the event disappears without operator intervention, the turbine can restart automatically, unless a preset maximum number of identical trips / hour has been reached.
- A: Turbine alarm. This is an event that requires operator attention, but does not shutdown the turbine. A persisting alarm can often lead to a trip, if the condition causing the alarm is not corrected.
- I: Information. In general information about turbine state transitions (start / stop / cable unwind etc.).

For a detailed description on displaying and resetting trips and alarms, see chapter 8.

Event ID	Message	Meaning of the message – possible causes	Test and rectification	Event	Max #/hr
001	Power up trip main turbine control system	The power to the main turbine control system PLC has been interrupted.	Apply power to main turbine control system PLC.	T1	2
002	Operator start	A command to start the turbine has been issued		I	N.A.
003	Operator stop	A command to stop the turbine has been issued		I	N.A.
004	Automatic turbine restart	An automatic start command to restart the turbine was issued by the control system		I	N.A.
005	Service switch on	Service switch on Local Operator Panel is switched to "on" position	Temporary situation – service	T2	1
006	Service switch off	Service switch on Local Operator Panel is switched to "off" position		I	N.A.
007	SMC initialisation failed	The SMC software module could not be initialised	Reboot PLC	T1	1

		during boot-up of the PLC			
008	Operator reset	A reset command was issued by the control system		I	N.A.
010	Main turbine PLC FM211 card 4 fault	Main turbine PLC FM211 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
011	Main turbine PLC CM202 card 5 fault	Main turbine PLC CM202 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
012	Main turbine PLC DIO216 card 6 fault	Main turbine PLC DIO216 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
013	Main turbine PLC CM202 card 6 CAN kernel fault	Kernel fault CAN network 2 (Main turbine PLC to ABB converter) communication	Check if connection is interrupted	T1	1
014	Converter CAN node not operational	The link check bit has timed out	Check converter CAN AnyBus module	T1	2
015	Local operation selected	LOC/REM switch on Local Operator Panel is switched to "LOC" position		I	N.A.
016	Remote (SCADA) operation selected	LOC/REM switch on Local Operator Panel is switched to "REM" position		I	N.A.
019	Parameter file R/W fault	Reading from or writing to flash card of parameters was not successful	Check if flash card is not write protected. Reboot and try again. If problems persist, replace flash card.	T2	1
020	Parameter Configuration fault	A parameter has been set to a value that disables protective functions	Check parameter ranges, and re-configure	T2	1
021	Turbine trip request from converter	A turbine trip request is issued by the converter control system	This trip comes in combination with a trip in the range 400-700. See [ABB] for troubleshooting	T1	1
022	Converter could not be switched on	A command from the PLC to switch on the converter was not successful	Check CAN communication. Check converter	T1	1

			alarms/trips		
023	Generator disconnect open	The generator disconnect in the tower base is open	Close the generator disconnect before a turbine start	T1	1
024	GCB open	The grid breaker of the turbine is open	Close the grid circuit breaker before a turbine start	T1	1
025	Critical mini MCB trip LV distribution	The MCB of the LV distribution has tripped	Check if the emergency push button has tripped, reset the emergency trip.	T1	1
026	Converter is not switched on	A command from the PLC to switch on the converter was not successful	Check CAN communication. Check converter alarms/trips	T1	1
027	Generator overspeed	The speed measured with the generator frequency exceeds 28.0 rpm	Sudden wind gust. If occurring too frequently, please contact Harakosan for investigation	T1	1
028	Maximum grid power exceeded	$P_{elec, 5s \text{ avg.}} > 110\% \text{ of } P_{nom}$ or $P_{elec, 10 \text{ min. avg.}} > 105\% \text{ of } P_{nom}$	Notify Harakosan for investigation	T1	1
029	Rotor speed sensor fault	The rotor speed measured by the control system deviates $> 10\%$ with the generator speed measured by the converter	Check connections of rotor speed encoder in the hub	T1	1
030	Ethernet switch fault	The Ethernet switch has detected a fault.	Check presence of both power supplies PWR1 and PWR2 to the Ethernet switch.	A	N.A.
032	Converter could not be switched off	A command from the PLC to switch off the converter was not successful	Check CAN communication. Check converter alarms/trips	A	N.A.
037	Communication with Remote Interface Unit (RIU) lost	The RIU has failed to poll the PLC for at least 60s	Check RIU and RIU communication.	A	N.A
039	Auto restart blocked	Automatic restart of the turbine is not possible because the number of Max #/hr is exceeded.	See Chapter 8.3	A	N.A.
200	Unwinding cable	A cable unwind procedure is initiated	Wait until unwind completed	I	N.A.

201	Wind speed too high	$V_{\text{wind},10 \text{ min. avg.}} > 25.0 \text{ m/s}$ or $V_{\text{wind},5 \text{ sec. avg.}} > 35.0 \text{ m/s}$	Wait until $V_{\text{wind},10 \text{ min. avg.}} < 20.0 \text{ m/s}$	I	N.A.
202	Obstruction lights day mode	The light sensors on the obstruction lights have sensed ambient day light, obstruction light intensity is increased		I	N.A.
203	Obstruction lights night mode	The light sensors on the obstruction lights have sensed absence of ambient day light, light intensity is decreased		I	N.A.
204	Storm park	Due to high wind speeds, the rotor control state STORM PARK is active, see also section 6.8.3.1		T1	1
210	PLC Fastbus fibre optic link fault	The PLC Fastbus master of the Main Turbine Control System 542A001 could not establish connection with the Fastbus slave in the Nacelle Control Box 540A001.	Check fibre optic cable connection between main turbine control system and nacelle control box. Measure damping of the fibre optic link.	T1	2
211	Emergency system nacelle tripped	The emergency mushroom button on the nacelle control box or on the local hub panel has been pressed, or the nacelle control box is powered-up	Reset the emergency mushroom buttons on nacelle control box and/or local hub panel and/or issue RESET command from the Local Operator Panel	T2	2
212	Nacelle PLC CM202 card 17 fault	Nacelle PLC remote I/O CM202 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
213	Nacelle PLC EM203 card 18 fault	Nacelle PLC remote I/O EM203 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
214	Nacelle PLC PTAI216 card 19 fault	Nacelle PLC remote I/O PTAI216 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
215	Nacelle PLC AIO288 card 20 fault	Nacelle PLC remote I/O AIO288 module not found	Check if module RDY LED is on, replace module when faulty	T1	1
216	Nacelle PLC DIO248	Nacelle PLC remote I/O	Check if module	T1	1

	card 22 fault	DIO248 module not found	RDY LED is on, replace module when faulty		
217	Nacelle PLC CM202 card 17 CAN kernel fault	Kernel fault CAN network 1 (nacelle to rotor) communication	Check if connection is interrupted	T1	1
219	Turning mode switched on	The turning mode has been switched on		I	N.A.
220	Service brake active	The service brake is applied while the turbine has been started	Release service brake or check hydraulic brake circuit	T1	1
221	Rotor locking pin not retracted	The locking pin retracted signal is not received while the turbine is started, or the locking pin inserted signal is received while the turbine has been started.	Unlock rotor. Check wiring of the locking pin sensor	T2	1
222	Rotor locking pin sensor fault	The I/O module has sensed a broken wire	Check wiring of the locking pin sensor	T1	1
223	Yaw timeout	Yawing the turbine into the wind took too long	Is there a problem with (one of) the yaw motors ? Alternatively, it is possible that the wind direction changed too much while yawing had started	T1	10
224	Yaw motor protection switch tripped	One of the protection switches F7-F10 of the yaw motors has tripped	Visually inspect yaw mechanism for obstruction / damage. Switch on F7-F10 in nacelle control box 540A001.	T1	1
225	Work switch yaw motors active	The maintenance switch S1 of the yaw motors on nacelle control box 540A001 is "off" while the turbine has been started	Switch on the yaw motors maintenance switch S1 on nacelle control box 540A001	T1	1
226	Cable twist limit switch active	The cable has been wind up too much, without an adequate cable unwind command from the control system.	Visually inspect the cable. Unwind the cable with the YAW LEFT or RIGHT button on the nacelle control box.	T1	1
227	Cable twist sensor fault	The I/O module has sensed a broken wire	Check wiring from twist sensor.	T1	2

			Check potentiometer value (4.7 kOhm)		
228	Yaw brake pressure very low	The pressure of the hydraulic power unit has dropped 25% below nominal value (< 150 bar)	Check hydraulic power unit. Check yaw brakes for leakage. Fix leakage.	T1	1
229	Yaw brake pressure sensor fault	The I/O module has sensed a broken wire	Check wiring to the hydraulic pressure sensor	T1	1
230	Wind meter fault	The wind meter signal is lost, or does not match with the current rotor speed	Check wiring to wind meter. Visually inspect wind meter for icing / damage (are all cups still present ?)	T1	10
231	Wind vane 1 fault	Wind vane 1 signal frozen for more than 1 minute while $V_{wind,1 \text{ min. avg}} > 4.0 \text{ m/s}$	Check wiring to wind vane 1. Check wind vane 1 for icing / damage / obstruction.	T1	10
232	Wind vane 2 fault	Wind vane 2 signal frozen for more than 1 minute while $V_{wind,1 \text{ min. avg}} > 4.0 \text{ m/s}$	Check wiring to wind vane 2. Check wind vane 2 for icing / damage / obstruction.	T1	10
233	Wind vane 1-2 discrepancy	Filtered values (T=25s) of wind vane 1 and wind vane 2 differ by more than 15° while 10 min. avg. wind speed $> 4.5 \text{ m/s}$	Check wind vanes. Is one of the vanes bent ?	T1	1
234	Turbine outside operating temperature	The outside temperature is so low, that a proper functioning of the emergency pitch system cannot be guaranteed. Or the temperature is too high for the equipment (transformers etc.)	Outside temperature too low or too high. Wait for right operating temperature.	T1	1
235	Outside temperature signal fault	The I/O module has sensed a broken wire	Check wiring to the Pt100 sensor	T1	1
236	Tower acceleration very high	The acceleration of the tower is very high	Blade discrepancies and/or very gusty wind	T1	2
237	Acceleration sensor fault	The I/O module has sensed a broken wire	Check accelerometer wiring from PCH sensor to remote	T1	1

			I/O PLC station		
240	Bearing temperature 1 very high	Bearing temperature 1 is very high	Check lubrication main bearing. Check cooling fan.	T1	1
241	Bearing temperature 2 very high	Bearing temperature 2 is very high	Check lubrication main bearing. Check cooling fan.	T1	1
242	Bearing temperature 1 fault	The I/O module has sensed a broken wire	Check wiring to the Pt100 sensor	T1	1
243	Bearing temperature 2 fault	The I/O module has sensed a broken wire	Check wiring to the Pt100 sensor	T1	1
244	Generator stator temperature 1 very high	Stator temperature 1 is very high	Is outside temperature very high in combination with a very high production ? Contact Harakosan for investigation.	T1	1
245	Generator stator temperature 2 very high	Stator temperature 2 is very high	Is outside temperature very high in combination with a very high production ? Contact Harakosan for investigation.	T1	1
246	Generator stator temperature 3 very high	Stator temperature 3 is very high	Is outside temperature very high in combination with a very high production ? Contact Harakosan for investigation.	T1	1
247	Generator stator temperature 4 very high	Stator temperature 4 is very high	Is outside temperature very high in combination with a very high production ? Contact Harakosan for investigation.	T1	1
248	Generator stator temperature 5 very high	Stator temperature 5 is very high	Is outside temperature very high in combination with a very high production ? Contact Harakosan for investigation.	T1	1
249	Generator stator temperature 6 very	Stator temperature 6 is very high	Is outside temperature very	T1	1

	high		high in combination with a very high production ? Contact Harakosan for investigation.		
250	Generator stator temperature 1 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
251	Generator stator temperature 2 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
252	Generator stator temperature 3 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
253	Generator stator temperature 4 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
254	Generator stator temperature 5 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
255	Generator stator temperature 6 fault	The I/O module has sensed a broken wire	Check wiring to Pt100.	T1	1
256	Nacelle cabinet temperature outside range	The nacelle cabinet temperature is so low , that the system cannot be guaranteed. Or the temperature is too high for the equipment (transformers etc.)	Nacelle cabinet temperature too low or too high. Wait for right operating temperature.	T1	1
257	Nacelle cabinet temperature fault	The I/O module has sensed a broken wire	Check wiring to the Pt100 sensor	T1	1
258	Nacelle smoke detected	The smoke detector has detected smoke.	Investigate potential damage. Check for loose wiring. Reset is only possible by disconnecting power supply to nacelle control box 540A001.	T1	1
260	Yaw brake pressure low	The pressure of the hydraulic power unit has dropped 10% below nominal value (< 180 bar)	Check hydraulic power unit. Check yaw brakes for leakage. Fix leakage.	A	N.A.
261	Yaw brake oil leakage detected	The leakage detector has detected a pressure > 3 bar	Check yaw brakes for leakage. Fix leakage.	A	N.A.
262	Main bearing temperature 1 high	Main bearing temperature 1 is high and might culminate into a trip	Check lubrication main bearing. Check cooling fan.	A	N.A.
263	Main bearing temperature 2 high	Main bearing temperature 2 is high and might culminate into a trip	Check lubrication main bearing. Check cooling fan.	A	N.A.
264	Generator stator temperature 1 high	Stator temperature 1 is high and might culminate	Is outside temperature very	A	N.A.

		into a trip	high in combination with a very high production ?		
265	Generator stator temperature 2 high	Stator temperature 2 is high and might culminate into a trip	Is outside temperature very high in combination with a very high production ?	A	N.A.
266	Generator stator temperature 3 high	Stator temperature 3 is high and might culminate into a trip	Is outside temperature very high in combination with a very high production ?	A	N.A.
267	Generator stator temperature 4 high	Stator temperature 4 is high and might culminate into a trip	Is outside temperature very high in combination with a very high production ?	A	N.A.
268	Generator stator temperature 5 high	Stator temperature 5 is high and might culminate into a trip	Is outside temperature very high in combination with a very high production ?	A	N.A.
269	Generator stator temperature 6 high	Stator temperature 6 is high and might culminate into a trip	Is outside temperature very high in combination with a very high production ?	A	N.A.
270	Generator air temperature 1 fault	The I/O module has sensed a broken wire	Check Pt100 wiring	A	N.A.
271	Generator air temperature 2 fault	The I/O module has sensed a broken wire	Check Pt100 wiring	A	N.A.
272	Spare				
273	Nacelle temperature range exceeded	The nacelle temperature is out of range and can culminate into a trip	Nacelle cabinet temperature too low or too high. Wait for right operating temperature.	A	N.A.
274	Nacelle temperature signal fault	The I/O module has sensed a broken wire	Check wiring to Pt100	A	N.A.
275	Main bearing grease tank level low	The level of the grease tank is low.	Refill the grease tank as described in [Vogel]. Check sensor if level is not low.	A	N.A.

276	Main bearing grease pump manual	After maintenance, the pump is not switched to automatic mode.	Switch the pump to automatic mode as described in section 9.14.	A	N.A.
277	Yaw bearing grease flow low	The time required for the grease system to supply the required amount of grease takes longer than expected	Check grease system for air entered in the system / flow obstruction.	A	N.A.
278	Yaw bearing grease tank level low	The level of the grease tank is low.	Refill the grease tank as described in [Vogel]. Check sensor if level is not low.	A	N.A.
279	Yaw bearing grease pump manual	After maintenance, the pump is not switched to automatic mode.	Switch the pump to automatic mode as described in section 9.14.	A	N.A.
280	Cooling fan 1 manual	After maintenance, the fan is not switched to automatic mode.	Switch the fan to automatic mode as described in section 9.15.	A	N.A.
281	Cooling fan 2 manual (option)	After maintenance, the fan is not switched to automatic mode.	Switch the fan to automatic mode as described in section 9.15.	A	N.A.
282	Obstruction lights fault	Loss of power to the obstruction lights controller or the controller has detected a strobe fail condition: 1) Flashhead has missed 4 or more consecutive flashes (=flash fail) 2) System flashing at wrong intensity step (=all photocell fail) 3) Missing flashhead station (=missing station)	Check power to the obstruction lights controller. Check wiring to flashheads. Check that flashheads have different station number after replacement. See also [Orga]	A	N.A.
283	Cable unwind inhibited: Vwind high	During very high wind speeds, cable unwinding is inhibited.		I	N.A.
300	Emergency pitch system check	An emergency system check is conducted.		I	N.A.
301	Emergency pitch test OK	The result of the most recent emergency system check is successful		I	N.A.
302	Overspeed test	An overspeed test is		I	N.A.

		conducted			
303	Overspeed test OK	The result of the most recent overspeed test is successful		I	N.A.
310	Rotor PLC CAN node not operational	The communication link between rotor remote I/O PLC station and the nacelle remote I/O PLC station is not operational.	Check availability of power to the rotor: is F14 in nacelle control box 540A001 tripped ? Check communication cable over slip ring. Check polarity of communication cable. Check presence of CAN termination resistors in nacelle control box 540A001 and rotor control box 541A001. If link is operational, the green RUN LED on the CS200/N module of the rotor remote I/O PLC is blinking. If the green RUN LED is off or is lighting constantly the link is not operational !	T1	2
311	SBP1 CAN node not operational	The communication link between SBP loader of blade 1 and the nacelle remote I/O PLC station is not operational.	Check availability of power to the SBP loader 1. Check communication cable over slip ring. Check polarity of communication cable. Check presence of CAN termination resistors in nacelle control box 540A001 and rotor control box 541A001. Check SBP loader station number on	T1	2

			cover after replacement.		
312	SBP2 CAN node not operational	The communication link between SBP loader of blade 2 and the nacelle remote I/O PLC station is not operational.	Check availability of power to the SBP loader 2. Check communication cable over slip ring. Check polarity of communication cable. Check presence of CAN termination resistors in nacelle control box 540A001 and rotor control box 541A001. Check SBP loader station number on cover after replacement.	T1	2
313	SBP3 CAN node not operational	The communication link between SBP loader of blade 3 and the nacelle remote I/O PLC station is not operational.	Check availability of power to the SBP loader 3. Check communication cable over slip ring. Check polarity of communication cable. Check presence of CAN termination resistors in nacelle control box 540A001 and rotor control box 541A001. Check SBP loader station number on cover after replacement.	T1	2
320	Hardware overspeed or watchdog timeout	The emergency pitch system is activated, but not by the PLC system. This can be caused by a trip of the external Jaquet overspeed relay or a watchdog timeout.	Read peak speed measurement from the Zoom/Rotor/Test Overspeed Trip screen on Local Operator Panel. Reset peak speed. Visually inspect	T2	1

			the hub for damage, and if possible try to download a high resolution logging of the event to send to Harakosan.		
323	Pitch drive blade 1 fault	The DRIVE HEALTHY contact of the Control Techniques Maxi Maestro drive of blade 1 was lost. This indicates either: -DC link voltage not OK -Drive temperature not OK	Verify resistance of braking resistor (8 Ohm). See also [Control techniques]. If problems persist or drive fault frequency is high, replace drive and return faulty for investigation.	T2	3
324	Pitch drive blade 2 fault	The DRIVE HEALTHY contact of the Control Techniques Maxi Maestro drive of blade 2 was lost. This indicates either: -DC link voltage not OK -Drive temperature not OK	Verify resistance of braking resistor (8 Ohm). See also [Control techniques]. If problems persist or drive fault frequency is high, replace drive and return faulty for investigation.	T2	3
325	Pitch drive blade 3 fault	The DRIVE HEALTHY contact of the Control Techniques Maxi Maestro drive of blade 3 was lost. This indicates either: -DC link voltage not OK -Drive temperature not OK	Verify resistance of braking resistor (8 Ohm). See also [Control techniques]. If problems persist or drive fault frequency is high, replace drive and return faulty for investigation.	T2	3
326	Encoder blade 1 fault	The blade angle measured with the encoder of blade 1 is outside range	Check wiring to the encoder of blade 1. Check encoder calibration.	T2	1
327	Encoder blade 2 fault	The blade angle measured with the encoder of blade 2 is outside range	Check wiring to the encoder of blade 2. Check encoder calibration.	T2	1
328	Encoder blade 3 fault	The blade angle measured with the encoder of blade	Check wiring to the encoder of	T2	1

		3 is outside range	blade 3. Check encoder calibration.		
329	Servo system blade 1 timeout	The movement of blade 1 to the target position takes longer than expected.	Check pitch bearing blade 1 for smooth movement. Is greasing OK ?	T2	3
330	Servo system blade 2 timeout	The movement of blade 2 to the target position takes longer than expected.	Check pitch bearing blade 2 for smooth movement. Is greasing OK ?	T2	3
331	Servo system blade 3 timeout	The movement of blade 3 to the target position takes longer than expected.	Check pitch bearing blade 3 for smooth movement. Is greasing OK ?	T2	3
332	Watchdog timeout blade 1	The ECM module feedback indicates that the emergency system of blade 1 has been activated.	Automatic reset when emergency trip condition is cleared	T2	1
333	Watchdog timeout blade 2	The ECM module feedback indicates that the emergency system of blade 2 has been activated.	Automatic reset when emergency trip condition is cleared	T2	1
334	Watchdog timeout blade 3	The ECM module feedback indicates that the emergency system of blade 3 has been activated.	Automatic reset when emergency trip condition is cleared	T2	1
335	Discrepancy blade 1-2	The synchronisation between blades 1 and 2 is lost, resulting in unbalance of the rotor (instantly > 8 deg, during 3 sec. > 3 deg or during 5 sec. > 1 deg)	Are pitch drives 1 and 2 OK ? Check pitch bearing blades 1 and 2 for smooth adjustment. Is greasing OK ? Are there significant differences in current of the blades on screen Zoom/Rotor/Pitch drive inverters ? Is the I2t protection of blade 1 or blade 2 activated frequently ?	T2	3
336	Discrepancy blade	The synchronisation	Are pitch drives 1	T2	3

	1-3	between blades 1 and 3 is lost, resulting in unbalance of the rotor. (instantly > 8 deg, during 3 sec. > 3 deg or during 5 sec. > 1 deg)	and 3 OK ? Is I ² t protection blade 1 active ? Check pitch bearing blades 1 and 3 for smooth adjustment. Is greasing OK ? Are there significant differences in current of the blades on screen Zoom/Rotor/Pitch drive inverters ? Is the I ² t protection of blade 1 or blade 3 activated frequently ?		
337	Discrepancy blade 2-3	The synchronisation between blades 2 and 3 is lost, resulting in unbalance of the rotor. (instantly > 8 deg, during 3 sec. > 3 deg or during 5 sec. > 1 deg)	Are pitch drives 2 and 3 OK ? Check pitch bearing blades 2 and 3 for smooth adjustment. Is greasing OK ? Are there significant differences in current of the blades on screen Zoom/Rotor/Pitch drive inverters ? Is the I ² t protection of blade 2 or blade 3 activated frequently ?	T2	3
350	DC battery maintenance switch on	The DC battery maintenance switch on the rotor cabinet has not been set to the normal "Off" position. The turbine start-up is blocked, because the emergency pitch system is not armed.	Put DC maintenance switch on rotor cabinet to "Off" position. Note: On = Service on / Batteries disconnected Off = Service off / Batteries connected	T1	1
351	Software overspeed	The pitch control system	Read peak speed	T1	5

		was unable to keep the rotor speed inside it's normal operating range	measurement from Local Operator Panel screen Zoom/Rotor/Test Overspeed Trip. Reset peak speed. If this trip occurs too often, the parameters of the pitch control system may have to be fine-tuned. Contact Harakosan for investigation.		
352	Hardware overspeed test failed	The result of an overspeed test was not successful	Check peak speed at Local Operator Panel screen Zoom/Rotor/Test overspeed Trip. Was rotor peak speed below 10.0 rpm ? Otherwise, do not restart turbine !	T1	1
353	Blade 1 battery voltage very low	The voltage of the control voltage of the emergency pitch system battery of blade 1 has decreased below the level where the emergency relay is assumed to work properly. The battery is worn out, the SBP loader of blade 1 is not functioning properly or a connection is lost.	Check connector connection XB1 of Rotor Control Box 541A001. Check connector of battery pack to SBP loader of blade 1. Check if power is available on SBP 1 loader. Check voltage of 24 Vdc section of battery. Replace battery if battery is worn out.	T1	1
354	Blade 2 battery voltage very low	The voltage of the control voltage of the emergency pitch system battery of blade 2 has decreased below the level where the emergency relay is assumed to work properly. The battery is worn out, the SBP loader of blade 1 is not functioning properly or a connection is lost.	Check connector connection XB1 of Rotor Control Box 541A001. Check connector of battery pack to SBP loader of blade 2. Check if power is available on SBP 2 loader. Check voltage of 24 Vdc section of	T1	1

			battery. Replace battery if battery is worn out.		
355	Blade 3 battery voltage very low	The voltage of the control voltage of the emergency pitch system battery of blade 3 has decreased below the level where the emergency relay is assumed to work properly. The battery is worn out, the SBP loader of blade 1 is not functioning properly or a connection is lost.	Check connector connection XB1 of Rotor Control Box 541A001. Check connector of battery pack to SBP loader of blade 3. Check if power is available on SBP 3 loader. Check voltage of 24 Vdc section of battery. Replace battery if battery is worn out.	T1	1
356	Blade 1 emergency pitch speed very low	The emergency pitch speed of blade 1 measured during the last emergency system check is below the set level of 4 deg/s. The battery is worn out, the SBP loader of blade 1 is not functioning properly or a connection is lost	Check connector of battery pack to SBP loader of blade 1. Check if power is available on SBP 1 loader. Check voltage of 168 Vdc section of battery. Replace battery if battery voltage < 140 Vdc. Is greasing blade 1 OK ?	T1	1
357	Blade 2 emergency pitch speed very low	The emergency pitch speed of blade 2 measured during the last emergency system check is below the set level of 4 deg/s. The battery is worn out, the SBP loader of blade 2 is not functioning properly or a connection is lost	Check connector of battery pack to SBP loader of blade 2. Check if power is available on SBP 2 loader. Check voltage of 168 Vdc section of battery. Replace battery if battery voltage < 140 Vdc. Is greasing blade 2 OK ?	T1	1
358	Blade 3 emergency pitch speed very low	The emergency pitch speed of blade 3 measured during the last emergency system check is below the set level of 4 deg/s. The battery is	Check connector of battery pack to SBP loader of blade 3. Check if power is available on SBP 3	T1	1

		worn out, the SBP loader of blade 3 is not functioning properly or a connection is lost	loader. Check voltage of 168 Vdc section of battery. Replace battery if battery voltage < 140 Vdc. Is greasing blade 3 OK ?		
359	Emergency pitch system timeout	The time to reach vane position for all blades exceeded 2 x expected pitch time	This trip should be accompanied with one of the events 353-358. See description events 353-358.	T1	1
360	Pitch motor blade 1 temperature high	The clixon in pitch motors 1 indicates a very high temperature	Check pitch motor 1 for damage. Notify Harakosan. Check if connection of clixon 1 is loose.	T1	1
361	Pitch motor blade 2 temperature high	The clixon in pitch motors 2 indicates a very high temperature	Check pitch motor 2 for damage. Notify Harakosan. Check if connection of clixon 2 is loose	T1	1
362	Pitch motor blade 3 temperature high	The clixon in pitch motors 3 indicates a very high temperature	Check pitch motor 3 for damage. Notify Harakosan. Check if connection of clixon 3 is loose	T1	1
363	Rotor cabinet temperature outside range	The cabinet temperature is outside the normal operating range.	Is outside temperature very high or very low ? Check fan inside rotor control box.	A	N.A.
364	Rotor cabinet temperature fault	The I/O module has sensed a broken wire	Check wiring to Pt100 sensor	A	N.A.
365	Rotor temperature high	The rotor temperature is very high	Is outside temperature very high or very low ? Check cooling fan operation.	A	N.A.
366	Rotor temperature fault	The I/O module has sensed a broken wire	Check wiring to Pt100 sensor.	A	N.A.
367	Pitch motor blade 1 overload	The pitch motor current has exceeded the limits	Check pitch motor and pitch gear. Notify Harakosan	T1	1
368	Pitch motor blade 2 overload	The pitch motor current has exceeded the limits	Check pitch motor and pitch gear. Notify Harakosan	T1	1
369	Pitch motor blade 3	The pitch motor current	Check pitch motor	T1	1

	overload	has exceeded the limits	and pitch gear. Notify Harakosan		
381	I ² t protection servo drive 1 active	The current limiter of the drive of blade 1 is active, possibly causing loss of synchronisation between blade 1 and the other blades	The message should disappear automatically when the limiter becomes inactive. Too frequent occurrence of this alarm is an indication of a mechanical problem of blade 1. See also [Control Techniques].	A	N.A.
382	I ² t protection servo drive 2 active	The current limiter of the drive of blade 2 is active, possibly causing loss of synchronisation between blade 2 and the other blades	The message should disappear automatically when the limiter becomes inactive. Too frequent occurrence of this alarm is an indication of a mechanical problem of blade 2. See also [Control Techniques].	A	N.A.
383	I ² t protection servo drive 3 active	The current limiter of the drive of blade 3 is active, possibly causing loss of synchronisation between blade 3 and the other blades	The message should disappear automatically when the limiter becomes inactive. Too frequent occurrence of this alarm is an indication of a mechanical problem of blade 3. See also [Control Techniques].	A	N.A.
384	Pitch bearing grease flow low	The time required for the grease system to supply the required amount of grease takes longer than expected	Check grease system for air entered in the system / flow obstruction.	A	N.A.
385	Pitch bearing grease tank level low	The level of the grease tank is low.	Refill the grease tank as described in [Vogel]. Check sensor if level is	A	N.A.

			not low.		
386	Pitch bearing grease pump manual	After maintenance, the pump is not switched to automatic mode.	Switch the pump to automatic mode as described in section 9.14.	A	N.A.
387	Pitch gear grease tank level low	The level of the grease tank is low.	Refill the grease tank as described in [Vogel]. Check sensor if level is not low.	A	N.A.
388	Pitch gear grease pump manual	After maintenance, the pump is not switched to automatic mode.	Switch the pump to automatic mode as described in section 9.14.	A	N.A.
389	Rotor cabinet temperature outside range	The cabinet temperature is outside the normal operating range.	Is outside temperature high or low ? Check fan inside rotor control box.	A	N.A.
390	SBP loader blade 1 fault	The SBP has one of the following faults: - unable to disconnect battery - undefined error - general charging error - top-up charging error - CAN error - charger state change timeout - no battery connected	Automatic reset of SBP can node. This implies also automatic reset of SBP software.	A	N.A.
391	SBP loader blade 2 fault	See above	See above	A	N.A.
392	SBP loader blade 3 fault	See above	See above	A	N.A.
401	VLU temperature high positive half	See [ABB]	See [ABB]	A	N.A.
402	VLU temperature high negative half	See [ABB]	See [ABB]	A	N.A.
403	MCB open fail	See [ABB]	See [ABB]	A	N.A.
404	Processor temperature high	See [ABB]	See [ABB]	A	N.A.
417	Transformer temperature high	See [ABB]	See [ABB]	A	N.A.

418	Main circuit breaker in test position	See [ABB]	See [ABB]	A	N.A.
419	Ground isolator feedback fail	See [ABB]	See [ABB]	A	N.A.
420	Cooling water temperature high	See [ABB]	See [ABB]	A	N.A.
421	Converter cabinet temperature high	See [ABB]	See [ABB]	A	N.A.
422	Cooling water flow low	See [ABB]	See [ABB]	A	N.A.
423	Cooling water conductivity high	See [ABB]	See [ABB]	A	N.A.
424	UPS G11 battery discharging	See [ABB]	See [ABB]	A	N.A.
425	UPS G11 battery charging	See [ABB]	See [ABB]	A	N.A.
426	UPS G21 battery discharging	See [ABB]	See [ABB]	A	N.A.
427	UPS G21 battery charging	See [ABB]	See [ABB]	A	N.A.
428	UPS G31 battery discharging	See [ABB]	See [ABB]	A	N.A.
429	UPS G31 battery charging	See [ABB]	See [ABB]	A	N.A.
430	400 VAC supervision control	See [ABB]	See [ABB]	A	N.A.
431	Chiller trip	See [ABB]	See [ABB]	A	N.A.
432	Static water pressure L	See [ABB]	See [ABB]	A	N.A.
433	Cooling water pressure sensor PT1 failure	See [ABB]	See [ABB]	T1	1
434	Cooling water pressure sensor PT2 failure	See [ABB]	See [ABB]	T1	1
435	Cooling water conductivity sensor QIT1 failure	See [ABB]	See [ABB]	T1	1
436	Cooling water temperature sensor TT1 failure	See [ABB]	See [ABB]	T1	1
437	Cooling water temperature very high	See [ABB]	See [ABB]	T1	1
438	Converter cabinet temperature very high	See [ABB]	See [ABB]	T1	1
439	Cooling water flow very low	See [ABB]	See [ABB]	T1	1
440	Cooling water conductivity very high	See [ABB]	See [ABB]	T1	1

441	Cooling water pump1 failure	See [ABB]	See [ABB]	T1	1
442	Cooling water pump2 failure	See [ABB]	See [ABB]	T1	1
443	Cooling water internal fan failure	See [ABB]	See [ABB]	T1	1
444	Transformer temperature very high	See [ABB]	See [ABB]	T1	1
445	Control cabinet temperature high	See [ABB]	See [ABB]	T1	1
446	Static water pressure LL	See [ABB]	See [ABB]	T1	1
449	Overcurrent INU phase R (Hardware)	See [ABB]	See [ABB]	T1	1
450	Overcurrent INU phase R high	See [ABB]	See [ABB]	T1	1
451	Overcurrent INU phase S (Hardware)	See [ABB]	See [ABB]	T1	1
452	Overcurrent INU phase S high	See [ABB]	See [ABB]	T1	1
453	Overcurrent INU phase T (Hardware)	See [ABB]	See [ABB]	T1	1
454	Overcurrent INU phase T high	See [ABB]	See [ABB]	T1	1
455	Short circuit INU positive half	See [ABB]	See [ABB]	T1	1
456	Short circuit INU negative half	See [ABB]	See [ABB]	T1	1
457	The sum of INU currents is non-zero	See [ABB]	See [ABB]	T1	1
465	Overcurrent ARU	See [ABB]	See [ABB]	T1	1
466	Overcurrent ARU phase R (Hardware)	See [ABB]	See [ABB]	T1	1
467	Overcurrent ARU phase R high	See [ABB]	See [ABB]	T1	1
468	Overcurrent ARU phase R very high	See [ABB]	See [ABB]	T1	1
469	Overcurrent ARU phase S (Hardware)	See [ABB]	See [ABB]	T1	1
470	Overcurrent ARU phase S high	See [ABB]	See [ABB]	T1	1
471	Overcurrent ARU phase S very high	See [ABB]	See [ABB]	T1	1
472	Overcurrent ARU phase T (Hardware)	See [ABB]	See [ABB]	T1	1
473	Overcurrent ARU phase T high	See [ABB]	See [ABB]	T1	1
474	Overcurrent ARU phase T very high	See [ABB]	See [ABB]	T1	1
475	Overvoltage ARU phase R-S high	See [ABB]	See [ABB]	T1	1

476	Overvoltage phase R-S high	ARU very	See [ABB]	See [ABB]	T1	1
477	Overvoltage phase S-T high	ARU high	See [ABB]	See [ABB]	T1	1
478	Overvoltage phase S-T high	ARU very	See [ABB]	See [ABB]	T1	1
479	Overvoltage phase T-R high	ARU high	See [ABB]	See [ABB]	T1	1
480	Overvoltage phase T-R high	ARU very	See [ABB]	See [ABB]	T1	1
481	Undervoltage phase R-S low	ARU	See [ABB]	See [ABB]	T1	1
482	Undervoltage phase R-S very low	ARU	See [ABB]	See [ABB]	T1	1
483	Undervoltage phase S-T low	ARU	See [ABB]	See [ABB]	T1	1
484	Undervoltage phase S-T very low	ARU	See [ABB]	See [ABB]	T1	1
485	Undervoltage phase T-R low	ARU	See [ABB]	See [ABB]	T1	1
486	Undervoltage phase T-R very low	ARU	See [ABB]	See [ABB]	T1	1
487	Short circuit positive half	ARU	See [ABB]	See [ABB]	T1	1
488	Short circuit negative half	ARU	See [ABB]	See [ABB]	T1	1
489	Grid frequency out of bound		See [ABB]	See [ABB]	T1	1
497	Overvoltage positive DC link		See [ABB]	See [ABB]	T1	1
498	Overvoltage negative DC link		See [ABB]	See [ABB]	T1	1
499	Short circuit undervoltage positive DC link		See [ABB]	See [ABB]	T1	1
500	Short circuit undervoltage negative DC link		See [ABB]	See [ABB]	T1	1
501	Asymmetrical voltage DC link		See [ABB]	See [ABB]	T1	1
513	Reverse power		See [ABB]	See [ABB]	T1	1
514	MCB operation fail		See [ABB]	See [ABB]	T1	1
515	Trip1 command from Control Builder		See [ABB]	See [ABB]	T1	1
516	Trip2 command from Control Builder		See [ABB]	See [ABB]	T1	1
517	Off command from		See [ABB]	See [ABB]	T1	1

	Control Builder				
518	Block restart command Control Builder	See [ABB]	See [ABB]	T1	1
519	Filter earth fault	See [ABB]	See [ABB]	T1	1
520	Grid lost in Turning mode	See [ABB]	See [ABB]	T1	1
529	Short circuit VLU positive half	See [ABB]	See [ABB]	T1	1
530	Short circuit VLU negative half	See [ABB]	See [ABB]	T1	1
531	VLU temperature high high positive half	See [ABB]	See [ABB]	T1	1
532	VLU temperature high high negative half	See [ABB]	See [ABB]	T1	1
545	IOEC 1 link frame error (IOEC -> PEC)	See [ABB]	See [ABB]	T1	1
546	IOEC 1 link watchdog (IOEC -> PEC)	See [ABB]	See [ABB]	T1	1
547	IOEC 2 link frame error (IOEC -> PEC)	See [ABB]	See [ABB]	T1	1
548	IOEC 2 link watchdog (IOEC -> PEC)	See [ABB]	See [ABB]	T1	1
549	PECFMI LFU ADC fault	See [ABB]	See [ABB]	T1	1
550	PECFMI LFU power supply fault	See [ABB]	See [ABB]	T1	1
551	PECFMI LFU LEM supply fault	See [ABB]	See [ABB]	T1	1
552	PECFMI ARU ADC fault	See [ABB]	See [ABB]	T1	1
553	PECFMI ARU power supply fault	See [ABB]	See [ABB]	T1	1
554	PECFMI ARU LEM supply fault	See [ABB]	See [ABB]	T1	1
555	PECFMI INU ADC fault	See [ABB]	See [ABB]	T1	1
556	PECFMI INU power supply fault	See [ABB]	See [ABB]	T1	1
557	PECFMI INU LEM supply fault	See [ABB]	See [ABB]	T1	1
558	GUSP ARU positive half power fail	See [ABB]	See [ABB]	T1	1
559	GUSP ARU negative half power fail	See [ABB]	See [ABB]	T1	1
560	GUSP INU positive	See [ABB]	See [ABB]	T1	1

	half power fail				
561	GUSP INU negative half power fail	See [ABB]	See [ABB]	T1	1
562	UltraLink 2 frame error (PECMI ARU -> PEC)	See [ABB]	See [ABB]	T1	1
563	UltraLink 2 watchdog (PECMI ARU -> PEC)	See [ABB]	See [ABB]	T1	1
564	UltraLink 3 frame error (PECMI INU -> PEC)	See [ABB]	See [ABB]	T1	1
565	UltraLink 3 watchdog (PECMI INU -> PEC)	See [ABB]	See [ABB]	T1	1
566	MultiLink 1 frame error (PEC -> PECMI LFU)	See [ABB]	See [ABB]	T1	1
567	MultiLink 1 watchdog (PEC -> PECMI LFU)	See [ABB]	See [ABB]	T1	1
568	MultiLink 1 frame error (PECMI LFU -> PEC)	See [ABB]	See [ABB]	T1	1
569	MultiLink 1 watchdog (PECMI LFU -> PEC)	See [ABB]	See [ABB]	T1	1
570	MultiLink 2 frame error (PEC -> PECMI ARU)	See [ABB]	See [ABB]	T1	1
571	MultiLink 2 watchdog (PEC -> PECMI ARU)	See [ABB]	See [ABB]	T1	1
572	MultiLink 2 frame error (PECMI ARU -> PEC)	See [ABB]	See [ABB]	T1	1
573	MultiLink 2 watchdog (PECMI ARU -> PEC)	See [ABB]	See [ABB]	T1	1
574	MultiLink 3 frame error (PEC -> PECMI INU)	See [ABB]	See [ABB]	T1	1
575	MultiLink 3 watchdog (PEC -> PECMI INU)	See [ABB]	See [ABB]	T1	1
576	MultiLink 3 frame error (PECMI INU -> PEC)	See [ABB]	See [ABB]	T1	1
577	MultiLink 3 watchdog (PECMI INU -> PEC)	See [ABB]	See [ABB]	T1	1
578	VLSCD fault INU	See [ABB]	See [ABB]	T1	1

	positive half				
579	VLSCD fault INU negative half	See [ABB]	See [ABB]	T1	1
580	VLSCD fault ARU positive half	See [ABB]	See [ABB]	T1	1
581	VLSCD fault ARU negative half	See [ABB]	See [ABB]	T1	1
582	VLSCD fault VLU positive half	See [ABB]	See [ABB]	T1	1
583	VLSCD fault VLU negative half	See [ABB]	See [ABB]	T1	1
584	Remote download to PECMI LFU failed	See [ABB]	See [ABB]	T1	1
585	Remote download to PECMI ARU failed	See [ABB]	See [ABB]	T1	1
586	Remote download to PECMI INU failed	See [ABB]	See [ABB]	T1	1
587	ASE 1 link frame error (ASE -> PEC)	See [ABB]	See [ABB]	T1	1
588	ASE 1 link watchdog (ASE -> PEC)	See [ABB]	See [ABB]	T1	1
589	ASE 2 link frame error (ASE -> PEC)	See [ABB]	See [ABB]	T1	1
590	ASE 2 link watchdog (ASE -> PEC)	See [ABB]	See [ABB]	T1	1
591	PEC power supply fault	See [ABB]	See [ABB]	T1	1
592	PEC CPU watchdog fault	See [ABB]	See [ABB]	T1	1
593	Emergency push button local	See [ABB]	See [ABB]	T1	1
594	Emergency push button remote	See [ABB]	See [ABB]	T1	1
602	S800 link error	See [ABB]	See [ABB]	T1	1
603	CAN heartbeat fail	See [ABB]	See [ABB]	T1	1
607	Critical switches in control cabinet off	See [ABB]	See [ABB]	T1	1
608	Uncritical switches in control cabinet off	See [ABB]	See [ABB]	T1	1
650	P const mode is active	See section 11.7		I	N.A.
651	Pn mode is active	See section 11.7		I	N.A.
652	Voltage control is on	See section 8.6		I	N.A.
653	Voltage control is off	See section 8.6		I	N.A.
654	Converter test mode on	The converter test mode has been switched on from screen		I	N.A.

		Zoom/Converter/Converter Test. See also section 9.9			
655	Converter test mode off	The converter test mode has been switched off from screen Zoom/Converter/Converter Test. See also section 9.9		I	N.A.
656	Generator power reduced	The generator power setpoint has been reduced due to high transformer temperature, high cooling water temperature, or low grid voltage	Check cooling system	I	N.A.

Appendix D. Configuration Data

The following list shows the most important configurable values of the control system and their actual settings:

Configuration parameter	Value	Unit	Description	Relation
SDTCnvOnTmOut	T#5m	[60s]	Converter on command timeout	Event 022
CDGenSpdHH	28	[rpm]	Generator speed very high	Event 027
CDNomPower	2000	[kW]	Maximum grid power : 5s > 2000*1.10 kW or 10m2000*1.05	Event 028
CDPower5sMxSp	110	[%]		
CDPower10mMxSp	105	[%]		
SDTCnvOffTmOut	T#1m	[60s]	Converter off command timeout	Event 032
SDHeartBeat	T#60s	[s]	Communication fault PLC-RIU	Event 036
SDTRot_HH	50	[deg C]	Rotor temperature very high	Event 365
SDTRot_H	45	[deg C]	Rotor temperature high	
SDTRotCab_LL	-5	[deg C]	Rotor cabinet temperature very low	Event 363
SDTRotCab_L	0	[deg C]	Rotor cabinet temperature low	Event 389
SDTRotCab_HH	50	[deg C]	Rotor cabinet temperature very high	Event 363
SDTRotCab_H	45	[deg C]	Rotor cabinet temperature high	Event 389
CDWacTimeout	T#300ms	[ms]	Watchdog time	
CDOvrCurr5s	22	[A]	Pitch motor current limits	Event 367, 368 and 369
CDOvrCurr60s	20	[A]		
CDOvrCurr10m	18	[A]		
SDNacVibR_HH	1.75	[m/s2]	Tower vibration very high - backup	Event 236
SDGnStaTmp_HH	150	[deg C]	Generator stator temperature very high	Event 244, ...
SDGnStaTmp_H	140	[deg C]	Generator stator temperature high	Event 264, ...
SDOutsideTmp_LL	-25	[deg C]	Outside temperature very low	Event 234
SDOutsideTmp_HH	45	[deg C]	Outside temperature very high	Event 234
SDNacCabTmp_LL	-5	[deg C]	Nacelle cabinet temperature very low	Event 256
SDNacCabTmp_L	0	[deg C]	Nacelle cabinet temperature low	
SDNacCabTmp_HH	50	[deg C]	Nacelle cabinet temperature very high	Event 256
SDNacCabTmp_H	45	[deg C]	Nacelle cabinet temperature high	
SDNacTmp_HH	50	[deg C]	Nacelle temperature very high	
SDNacTmp_H	45	[deg C]	Nacelle cabinet temperature high	Event 273
CDTauPT100	T#10s	[s]	Filter time constant temperature filters	
Lock_H	72	[%]	Locking pin sensor locked position	
Lock_L	11	[%]	Locking pin sensor unlocked position	
SDHyPres_H	190	[bar]	Reset condition for hydraulic pressure low events	
SDHyPres_L	170	[bar]	Hydraulic pressure low	Event 260
SDHyPres_LL	150	[bar]	Hydraulic pressure very low	Event 228
SDGnBrgTmp_H	85	[deg C]	Bearing temperature high	Event 240
SDGnBrgTmp_HH	95	[deg C]	Bearing temperature very high	Event 241
SDNacSmkDetDebTm	T#10s	[s]	Smoke detector event delay time	Event 258
CDTempOn	1	[deg C]	Deactivate meteo heating	

Configuration parameter	Value	Unit	Description	Relation
CDTempOff	0	[deg C]	Activate meteo heating	
SDVwind_hi_3s	35	[m/s]	Windspeed high - 3s	
SDVwind_hi_10m	25	[m/s]	Windspeed high - 10 min.	
SDVwind_OK_10m	20	[m/s]	Reset windspeed high - 10 min.	
SDVwind_Storm_10m	27	[m/s]	Windspeed very high - 10 min.	
SDVwind_Storm_3s	35	[m/s]	Windspeed very high - 3s	
SDVwind_NoStorm_10m	20	[m/s]	Reset windspeed very high - 3s	
SDVw5sWndVnEn	3.5	[m/s]	Windspeed condition to enable auto yaw - 5s	
SDVw60sWndVnEn	3	[m/s]	Windspeed condition to enable auto yaw - 60s	
SDVw60sWndVnDiscr	4.5	[m/s]	Windspeed condition to enable wind vane discrepancy	
SDWSpd_H	12	[m/s]	Windspeed high condition for "Hub access prohibited" indicator	
SDWSpd_OK	11	[m/s]	Reset windspeed high condition for "Hub access prohibited" indicator	
SDHysterease	5	[deg C]	Hysteresis for temperature control of nacelle cooling fan	
SDYwHyBrkTm	T#30s	[s]	Event delay time for hydraulic pressure low	
SDWspdMinRange	0.5	[m/s]	Wind speed fault detection - minimum speed	Event 230
SDRotSpdMinRange	6.5	[m/s]	Wind speed fault detection - minimum rotor speed	Event 230
SDWndVnFltTm	T#60s	[s]	Event delay time for wind vane fault	Event 231, 232
SDWndVnMaxDscrp	15	[deg]	Maximum discrepancy between wind vane 1 and wind vane 2	Event 233
SDWndVnDiscrTm	T#60s	[s]	Event delay time for wind vane discrepancy	Event 233
SDWndMtrFltTm	T#60s	[s]	Event delay time for anemometer fault	Event 230
SDYwAlgnAutoOff	2	[deg]	Auto yaw alignment stop conditions - 5s	
SDYwAlgnAutoOn	15	[deg]	Auto yaw alignment start conditions - 25s	
SDHyBrkLft	90	[bar]	Yaw brakes lifted condition	
SDYawTmOutFact	5	[-]	Yaw timeout factor	Event 223
SDYawTmOutOffs	0	[s]	Yaw timeout offset	Event 223
SDTwistHiHiHi	2	[turns]	Cablewindup very high	
SDTwistHiHi	1.5	[turns]	Cable windup high	
SDTwistHi	1	[turns]	Cable windup moderate	
funInitCTRG.p.v.GcvDsgn.PeTb[1]	0	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvDsgn.PeTb[2]	0	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvDsgn.PeTb[3]	0	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD	0	[kW]	Pn table - power table	

Configuration parameter	Value	Unit	Description	Relation
sgn.PeTb[4]				
funInitCTRG.p.v.GcvD sgn.PeTb[5]	0	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[6]	2.7	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[7]	82.2	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[8]	109.4	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[9]	160.6	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[10]	225.6	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[11]	306.1	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[12]	403.9	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[13]	520.5	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[14]	659.1	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[15]	1162.3	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[16]	1710.3	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[17]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[18]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[19]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[20]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[21]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.PeTb[22]	2072	[kW]	Pn table - power table	
funInitCTRG.p.v.GcvD sgn.RpmTb[1]	0	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[2]	3	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[3]	4.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[4]	6	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[5]	7.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[6]	9	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD	10	[rpm]	Pn table - rotor speed table	

Configuration parameter	Value	Unit	Description	Relation
sgn.RpmTb[7]				
funInitCTRG.p.v.GcvD sgn.RpmTb[8]	11	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[9]	12.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[10]	14	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[11]	15.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[12]	17	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[13]	18.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[14]	20	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[15]	21	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[16]	22	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[17]	22.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[18]	23	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[19]	24	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[20]	25	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[21]	27.5	[rpm]	Pn table - rotor speed table	
funInitCTRG.p.v.GcvD sgn.RpmTb[22]	30	[rpm]	Pn table - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[1]	-45	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[2]	9	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[3]	10.9286	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[4]	12.8571	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[5]	14.7857	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[6]	16.7143	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[7]	19.5	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[8]	21.5	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf g.NrpmPartList[9]	22.5	[rpm]	Finepitch - rotor speed table	
funInitCTRP.p.v.oprCf	45	[rpm]	Finepitch - rotor speed table	

Configuration parameter	Value	Unit	Description	Relation
g.NrpmPartList[10]				
funInitCTRP.p.v.oprCf g.ThdgPartList[1]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[2]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[3]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[4]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[5]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[6]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[7]	ThMin*	[deg]	Finepitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[8]	ThMin*+4.0 000000000 E+000	[deg]	Fine pitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[9]	ThMin*+4.0 000000000 E+000	[deg]	Fine pitch - pitch angle table	
funInitCTRP.p.v.oprCf g.ThdgPartList[10]	ThMin*+4.0 000000000 E+000	[deg]	Fine pitch - pitch angle table	
CDVecmOk	20	[V]	Minimum voltage battery pack system 24 VDC UPS supply part	Event 353
CDVecmOk	20	[V]	Minimum voltage battery pack system 24 VDC UPS supply part	Event 354
CDVecmOk	20	[V]	Minimum voltage battery pack system 24 VDC UPS supply part	Event 355
SDEmPitchSpeed	5	[deg/s]	Minimum emergency pitch speed	Event 356,357,3 58
SDEmgTstDly	T#3s	[s]	Emergency pitch test event inhibit time to allow for initial acceleration	Event 356, 357, 358
CDMaxTstTime	T#30s	[s]	Maximum allowed time for overspeed after start of test	Event 303
CDPBGPTMaxCycle	T#2m	[min.]	Maximum pitch bearing grease flow piston cycle time	Event 384
SDVaneLimitTolerance	5	[deg]	Maximum allowed deviation from blade encoder range[0-90 deg]	Event 326, 327, 328

* Thmin = Appendix B, parameter Blade work position.